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PROJECT REPORT

Course: Information System Analysis and Design

BURN ASSESSMENT AND FIRST-AID GUIDANCE INFORMATION SYSTEM

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TABLE OF CONTENTS

CHAPTER 1. PROJECT INTRODUCTION 6
1.1. Background 6
1.2. Problem Statement 6
1.3. Proposed Solution 7
1.4. Constraints and Assumptions 7
CHAPTER 2. LITERATURE REVIEW 9
2.1 Introduction 9
2.3 Deep Learning Approaches for Burn Classification 9
2.4 Segmentation with Advanced Neural Networks 10
2.5 Clinical Deployment and Mobile Applications 10
2.6 Summary of Literature and Identified Gaps 11
CHAPTER 3. REQUIREMENTS MODELING 12
3.1. OVERVIEW AND PURPOSES 12
3.1.1. Existing situation 12
3.1.2. Solution 12
3.2. STAKEHOLDERS 13
3.3. USER ROLES AND REQUIREMENTS 14
3.4. SCOPE OF WORK 15
3.4.1. In - scope 15
3.4.2. Out - of - scope 15
3.5.1. Functional requirements 15
3.5.2. Non-functional requirements 19
CHAPTER 4. DATA PROCESS MODELING 21
4.1. OVERVIEW AND PURPOSES 21
4.2. DATA FLOW DIAGRAM (DFD) 21
4.2.1. Context Diagram 21
4.2.2. DFD Level 1 22
4.2.3. DFD Level 2 23
CHAPTER 5. OBJECT MODELING 27
5.1. SCENARIOS, MODELS AND FUNCTIONALITY 27
5.1.1. Use Case Diagram 27
5.1.2. Use Case Descriptions 27
5.2. ACTIVITY DIAGRAMS 47
5.2.1. Sign In and Create Account (AD-01) 47

5.2.2. View Assessment History (AD-02) 49
5.2.3. Receive Emergency Warning (AD-03) 50
5.2.4. View Assessment Result (AD-04) 51
5.2.5. Submit Feedback (AD-05) 52
5.2.6. Perform Burn Assessment (AD-06) 53
5.2.7. Review User Feedback (AD-07) 54
5.2.8. Manage First-Aid Guidelines (AD-08) 55
5.2.9. Manage AI Models (AD-09) 56
5.2.10. Review Low-Confidence Cases (AD-10) 57
5.2.11. Manage User Account Status (AD-11) 58
5.2.12. Manage Personal Profile (AD-12) 59
5.2.13. Ask First-Aid Assistant (AD-13) 60
CHAPTER 6. DATABASE DESIGN 61
6.1. Overview and Purposes 61
6.2. Deliverables 61
6.3. Data dictionary 62
6.3.1. Account 62
6.3.2. Burn_case 62
6.3.3. Burn_area_assessment 64
6.3.5. First_aid_guideline 67
6.3.6. Assessment_guideline 68
6.3.7. AI_model 69
6.3.8. Feedback_review_log 70
6.3.10. Relationship Table 71
6.4. Entity Relationship Diagram (ERD) 74
6.4.1. Account Management Process: 74
6.4.2. Clinical Survey and Emergency Pre-screen Process: 75
6.4.3. Body Region Selection and Photo Capture Process: 75
6.4.4. AI Assessment and TBSA Calculation Process: 75
6.4.5. Case Scoring and Doctor Referral Process: 76
6.4.6. First Aid Guidance Process: 76
6.4.7. Feedback and Clinical Review Process: 76
6.5. Physical database in SQL Server 77
CHAPTER 7. MODEL DEVELOPMENT 79
7.1. AI Problem Definition 79

7.2. Burn Depth Classification Model (Classification Model)....80
7.2.1. Dataset Collecting & Preprocessing....80
7.2.2. Preparation and Training....81
7.3. Segmentation Model....84
7.3.1. Dataset Collecting & Preprocessing....84
7.3.2. Preparation and Training....85
7.4. Experimental Model Selection & Architecture Decisions....89
7.4.1. Objectives and Experimental Setup on NVIDIA A100 GPU....89
7.4.2. Classification Architecture Selection....90
7.4.3. Segmentation Architecture Selection....91
7.4.4. Direct Comparison of Two Segmentation Models on Independent Test Set....92
7.4.5. Final Model Decision and Deployment Limitations....93
7.5. Large Language Model & RAG....93
CHAPTER 8. SYSTEM ARCHITECTURE & USER INTERFACE DESIGN....95
8.1. THREE-TIER ARCHITECTURE DIAGRAM FOR PROTOTYPE....95
8.1.1. Overview....95
8.1.2. Architecture components....96
8.1.3. External Service - Groq Cloud LLM....97
8.1.4. Communication paths....97
8.2. NAVIGATION FLOW CHART FOR PROTOTYPE....97
8.3. INTERFACE PROTOTYPE....98
8.3.1. Launch, Introduction, and User Authentication....98
8.3.2. Home Navigation and Burn Wound Data Collection....100
8.3.3. Diagnostic Results Triage, Safety Warnings, and First Aid....102
8.3.4. Profile Management, History, and System Settings....104
8.4. LOCAL SYSTEM EXPERIMENTAL DEPLOYMENT....107
8.4.1. Local Deployment Rationale....107
8.4.2. Local Project Directory Tree....107
CHAPTER 9. TESTING, EVALUATION & CONCLUSION....110
9.1. REQUIREMENTS ACHIEVED....110
9.1.1. Burn Severity Classification and Injury Segmentation Using Computer Vision....110
9.1.2. Digital Clinical Rules, Burn Area Calculation, and Risk Classification....110
9.1.3. Medical Question-and-Answer Assistant and Technical Safety Checks....110
9.2. SYSTEM LIMITATIONS....111
9.2.1. Limitations in Clinical Scope and Types of Injury....111

9.2.2. Limitations in Features and System Architecture 111
REFERENCES 114
English materials 114
Vietnamese materials 114
APPENDIX 1: SOURCE CODE REPOSITORY 115
APPENDIX 2: TRAINING LOGS 115
APPENDIX 4: TABLES 117
APPENDIX 5: FIGURES 118

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CHAPTER 1. PROJECT INTRODUCTION

1.1. BACKGROUND

Burn injuries are a public health concern that can occur at home, in the workplace, and in other everyday settings. According to the World Health Organization (WHO), burns cause an estimated 180,000 deaths each year, with most occurring in low- and middle-income countries. Non-fatal burns can also lead to long hospital stays, disability, scarring, and emotional difficulties. These effects highlight the importance of burn prevention and access to reliable first-aid information.

Immediately after a burn occurs, an injured person or family member may need to provide first aid before medical help is available. However, identifying the seriousness of an injury can be difficult without medical knowledge. Burn assessment involves several factors, including wound depth, the percentage of total body surface area affected (%TBSA), the location of the injury, and signs of smoke inhalation. Therefore, the appearance of a wound alone does not provide enough information to determine the appropriate response. Clear first-aid instructions and guidance on when to seek medical attention are important in this situation.

Artificial intelligence (AI) and computer vision offer ways to support burn assessment by identifying wound boundaries and classifying burn depth from digital photographs. In the ABC-YOLO study, Şevik and Mutlu (2026) evaluated YOLO-based segmentation models, with YOLOv11x-seg achieving an F1 score of 0.87 and an mAP@0.5 of 0.91. These results suggest that AI-supported information systems could improve assessment consistency and support clinical triage and treatment planning. However, the model sometimes confuses neighbouring burn grades, and its performance may vary with lighting and imaging devices. Other research has also identified limitations related to data diversity and differences in performance across skin types. Therefore, AI outputs require careful evaluation and broader validation before they can be used reliably in practice.

Against this background, the Burn Assessment and First-Aid Guidance Information System is proposed as a prototype for people who experience a burn and family members assisting them before reaching a medical facility. The project brings together burn photographs, user-provided injury information, AI-supported preliminary assessment, and guidance based on trusted medical sources. Its intended role is to present useful information clearly, flag uncertain results, and help users recognize situations requiring medical attention. This direction reflects the project's focus on supporting early action while keeping the limitations of automated assessment visible to users.

1.2. PROBLEM STATEMENT

People who experience a burn at home may need to make immediate decisions before medical help is available. Without medical knowledge, patients and family members may find it difficult to judge the seriousness of the injury, choose appropriate first-aid actions, and recognize when medical attention is needed. Burn assessment requires information about wound depth, affected body surface area, injury location, symptoms, and possible smoke inhalation. The visible appearance of a wound alone cannot provide all the information needed for a preliminary risk assessment.

AI-based image classification and segmentation can support burn-depth prediction and localization of suspected burned tissue. However, these tasks have distinct limitations. Image-level classification does not establish the depth of every part of a wound, while a binary segmentation mask identifies suspected burn regions without distinguishing their depth. Neither output directly determines the percentage of the person's total body surface affected. Poor lighting, unsuitable camera angles, ambiguous wound appearance, and differences between training images and real-world photographs may also produce uncertain or conflicting outputs. Users may receive false reassurance if these predictions are presented as confirmed diagnoses.

The problem is therefore how to provide understandable preliminary burn assessment while communicating the limitations of image-based AI. The system needs to combine classification and segmentation with user-provided injury information, body-region-based Estimated TBSA calculation, and explicit risk rules. It must identify uncertainty, distinguish unconfirmed predictions from accepted results, and provide appropriate first-aid guidance and escalation advice without replacing professional medical evaluation.

1.3. PROPOSED SOLUTION

To address the identified problems, the project proposes the Burn Assessment and First-Aid Guidance Information System, a prototype that combines AI-supported image analysis, structured injury information, and guidance from trusted medical sources. Users upload burn photographs and complete a short survey covering age, injury cause, affected body regions, time of injury, and accompanying symptoms. The system checks image quality, identifies burn regions, and provides a preliminary burn-depth classification. It then combines these results with the survey responses and an estimated percentage of total body surface area affected (Estimated TBSA) to assess potential risks through explicit, reviewable rules.

The system presents the assessment results, first-aid instructions, and warnings about situations requiring medical attention in a clear interface. When an image is unsuitable or a prediction is uncertain, it displays an inconclusive result and explains the next steps. A source-grounded assistant helps users understand the approved guidance, while assessment records allow users to review previous entries.

The general objective is to develop and evaluate an integrated prototype that supports early burn assessment and access to reliable first-aid information. Specific objectives are to measure classification and segmentation performance, verify Estimated TBSA calculations, evaluate the handling of uncertain inputs, and check that guidance and risk warnings follow the selected medical references. These evaluations will establish the prototype’s capabilities and limitations within its defined scope.

1.4. CONSTRAINTS AND ASSUMPTIONS

Several assumptions and constraints guide the development of the Burn Assessment and First-Aid Guidance Information System. First, users are assumed to have a smartphone or another device with a camera and sufficient internet access to upload photographs and use the system. Patients who are conscious and able to operate the device may enter information themselves, while family members may assist adult patients who need help. Users are expected

to provide accurate injury details, although stress, incomplete information, and poor image quality may affect assessment reliability.

The project utilizes raw image samples collected from Kaggle repositories (including the Skin Burn Detection dataset by Fares Abbas and non-burn skin benchmarks). These images were uploaded to the Roboflow platform for manual cropping, quality verification, and label re-annotation across four target classes: healthy skin (None), first-degree, second-degree, and third-degree burns. Images selected from the iToBoS 2024 Skin Lesion Detection dataset are used as non-burn examples for the No Burn class. In this project, No Burn indicates the absence of a burn injury rather than the absence of other skin conditions. The burn examples come from the Skin Burn Detection notebook by Fares Abbas, which contains 543 files for first-degree burns, 488 for second-degree burns, and 326 for third-degree burns. Data preparation requires checking image suitability, label consistency, duplicate files, and usage conditions. Training, validation, and test sets must be separated carefully to avoid overlap. These classification labels do not establish the availability of pixel-level masks, so the data used for segmentation must be documented separately.

The system is developed as an academic prototype for preliminary assessment and first-aid information. Its operation depends on available computing resources, correctly loaded model files, network connectivity, and any external services used. Unsuitable images, uncertain predictions, or service failures must produce an inconclusive result. Emergency calling and location features depend on device capabilities and service availability; direct transmission of assessment data to emergency personnel requires a separate integration. Within the current prototype scope, assessment and Estimated TBSA calculation are limited to patients aged 18 years and above. Patients under 18 are excluded from these functions, even when a guardian operates the device.

Finally, development and evaluation are constrained by the course schedule, team capacity, and access to suitable data and medical review. Priority is given to the core assessment, uncertainty handling, and first-aid guidance functions. Clinical validation, wider deployment, and advanced emergency-service integration remain outside the current prototype scope.

CHAPTER 2. LITERATURE REVIEW

2.1 INTRODUCTION

Burn injuries are among the most painful and complex trauma cases in clinical medicine, demanding precise assessment of depth and affected area to prevent complications, guide surgical intervention, and optimize patient outcomes. Traditionally, burn evaluation relies on visual inspection and palpation by experienced clinicians. However, clinical assessment alone exhibits variable accuracy, particularly in distinguishing between deep partial-thickness and full-thickness burns during the early post-injury phase.

To address these limitations, computer vision and machine learning techniques have been increasingly investigated. Computerized visual analysis offers non-invasive, objective, and reproducible evaluation of skin lesions. With recent advancements in deep learning, deep convolutional neural networks (CNNs) and real-time object detection models have shown substantial promise in automating both burn depth classification and lesion segmentation.

This chapter provides a comprehensive review of existing literature on automated burn assessment. Section 2.2 reviews traditional machine learning approaches. Section 2.3 discusses deep learning classification methodologies. Section 2.4 focuses on image segmentation using modern object detection architectures. Section 2.5 examines existing mobile and clinical deployment frameworks. Finally, Section 2.6 summarizes the key research gaps addressed by this work.

2.2 Traditional Machine Learning Methods

Early efforts in automated burn assessment relied on conventional image processing techniques paired with classical machine learning classifiers. These methods generally involved handcrafted feature extraction-combining color space transformations (e.g., HSV, L*a*b*) and texture descriptors (e.g., Gray-Level Co-occurrence Matrix, LBP), followed by classification using algorithms such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN), or Random Forests.

While these foundational studies demonstrated proof-of-concept for automated assessment, handcrafted feature extraction is sensitive to variations in lighting conditions, skin tone, and camera specifications. Furthermore, classical pipeline models often require separate manual pre-processing steps, limiting their utility in fast-paced clinical or field environments.

2.3 DEEP LEARNING APPROACHES FOR BURN CLASSIFICATION

The emergence of Convolutional Neural Networks (CNNs) marked a major shift in medical image analysis. Pretrained convolutional architectures, fine-tuned via transfer learning, consistently outperform traditional handcrafted feature pipelines by automatically learning hierarchical visual representations.

Several pre-trained architectures, including VGG-16, GoogLeNet, ResNet variants, and EfficientNet, have been evaluated on burn image datasets. These studies typically focus on multi-class classification (e.g., distinguishing healthy skin, superficial, partial-thickness, and full-thickness burns).

Table 2.1: Key Studies in Deep Learning Burn Classification

Author(s)	Architecture, Method	Key Findings
Cirillo et al. (2019)	VGG-16, GoogLeNet, ResNet-50, ResNet-101	Evaluated time-independent burn depth classification; ResNet-101 achieved top performance.
Tran et al. (2020)	DenseNet-121, EfficientNet	Demonstrated high accuracy on small clinical datasets using domain-specific data augmentation.
Abubakar et al. (2021)	MobileNetV2	Emphasized lightweight models for portable and mobile-assisted triage scenarios.

Despite encouraging classification metrics, standalone classification models do not explicitly delineate burn boundaries or isolate lesion areas from surrounding healthy tissue and background artifacts.

2.4 SEGMENTATION WITH ADVANCED NEURAL NETWORKS

To provide precise spatial localization alongside diagnostic assessment, semantic segmentation and object detection frameworks have been adapted for burn care. Real-time object detection models have gained traction due to their high processing speeds and balanced performance.

UNet and Mask R-CNN frameworks have been deployed for fine-grained pixel-level segmentation. These unified models allow real-time boundary delineation, which is critical for accurate Total Body Surface Area (TBSA) calculation. While UNet variants offer detailed boundary preservation, lightweight architectures are increasingly favored for real-time edge computing applications.

2.5 CLINICAL DEPLOYMENT AND MOBILE APPLICATIONS

Translating deep learning models into clinical practice requires integration into accessible platforms. Mobile devices present an ideal medium for field triage, emergency medical services, and point-of-care evaluation in remote areas.

Deploying deep learning models on mobile platforms involves architectural optimization to balance inference speed, memory efficiency, and diagnostic accuracy. Model quantization, pruning, and lightweight backbones (e.g., MobileNet) enable real-time analysis directly on mobile processors without relying on persistent cloud connectivity.

Recent frameworks emphasize user-centric mobile interfaces that provide immediate visual feedback, overlaying segmentation masks directly on captured images to support healthcare providers during initial assessment.

2.6 SUMMARY OF LITERATURE AND IDENTIFIED GAPS

Table 2.2. Summary of the existing studies reviewed

Study	Task	Data and models	Main result
Cirillo et al. (2019)	Time-independent burn depth classification	VGG-16, GoogLeNet, ResNet-50, ResNet-101, pretrained	ResNet-101 best: mean accuracy 81.66% (min 72.06%, max 88.06%) in 10-fold cross-validation
Abubakar et al. (2021)	Burn image classification and depth estimation	MobileNetV2, custom transfer learning pipeline	Achieved >88% classification accuracy on clinical test subsets with lightweight edge footprint
Tran et al. (2020)	Automated burn severity scoring	DenseNet-121, EfficientNet-B0	Demonstrated robust performance under varying illumination using advanced color normalization

Despite the progress documented in existing literature, several critical gaps remain:

1. **Integrated depth and boundary assessment:** Most prior studies treat classification and segmentation as decoupled tasks, whereas clinical utility requires unified, single-pass assessment.
2. **Edge optimization:** High-performing models often rely on large parameter footprints, making them difficult to deploy on standard mobile hardware without latency penalties.
3. **Robustness to environmental variability:** Limited existing public datasets lead to overfitting on specific lighting conditions, background clutter, and skin pigmentation variations.

The research presented in this thesis aims to bridge these gaps by developing an optimized, light-weight end-to-end framework capable of real-time segmentation and classification directly on mobile devices.

CHAPTER 3. REQUIREMENTS MODELING

3.1. OVERVIEW AND PURPOSES

3.1.1. Existing situation

The current process considered in this project begins when a person experiences a burn at home or at the scene of an accident, before professional medical assessment is available. The injured person or a family member must decide what first aid to provide and whether to seek medical attention. They may rely on personal experience, advice from others, or online searches. In this process, finding information, assessing the injury, and deciding on the next action are separate activities, with the user responsible for connecting them.

A major difficulty is judging the seriousness of the injury. Users without medical training may focus on redness, blisters, or pain while overlooking other relevant factors. Burn assessment also considers wound depth, affected body surface area, injury location, and possible inhalation injury. Even a small burn may require specialist evaluation when it affects a critical body area. These factors make visual self-assessment difficult and create a risk of underestimating an injury.

Accessing suitable first-aid information presents another challenge. Users must find a reliable source, understand its instructions, and determine which advice applies to their situation. For example, NHS guidance provides specific instructions on cooling and covering a burn, as well as actions to avoid. However, reading general guidance does not automatically help users organize their injury details or identify which warning signs are relevant to them.

Information may also remain scattered across photographs, search results, and verbal descriptions. Without a structured record, users may find it difficult to review what they entered or explain the injury clearly when seeking help. The project therefore addresses a fragmented self-assessment and information-seeking process. This description represents the project's assumed usage context; it does not establish measured waiting times, error rates, or user behaviour from a field study.

3.1.2. Solution

The proposed solution is the Burn Assessment and First-Aid Guidance Information System, an academic prototype that brings injury information, AI-supported image analysis, and first-aid guidance into one workflow. Its primary users are adult patients who are conscious and able to operate the device, and family members assisting an adult patient. Assessment and Estimated TBSA functions are limited to patients aged 18 years and above within the current prototype scope.

For image-based assessment, users upload photographs of the affected regions. The system checks image quality and applies classification and segmentation models to suggest burn depth and highlight the identified wound area. The classification component distinguishes between No Burn, first-degree, second-degree, and third-degree burn images. No Burn means

that the model has not identified a burn; it does not confirm that the skin is healthy or cancel warning signs reported by the user.

Estimated TBSA is calculated separately using the adult body-region reference and the proportion of each region entered by the user. The segmentation overlay supports visual review of the photographed wound; it is not treated as a direct measurement of whole-body burn percentage. The system combines the available assessment results, Estimated TBSA, and survey responses through explicit, reviewable risk rules.

The outputs include a preliminary classification, a wound overlay where available, Estimated TBSA, identified risk factors, and guidance on the next action. Unsuitable images, uncertain predictions, or processing failures produce an inconclusive result with an explanation. First-aid instructions come from a reviewed knowledge base, while the AI assistant helps users understand that content. The interface also provides access to the emergency-call function when indicated. Direct transmission to emergency services or hospital systems remains a separate integration.

Assessment records link the submitted information, available results, and guidance for later review. The project is expected to make the process easier to follow, improve access to consistent first-aid information, and reduce the risk of users overlooking warning signs. These expectations will be evaluated through model performance, uncertainty handling, calculation checks, guidance consistency, and workflow testing; they are not presented as proven clinical outcomes.

3.2. STAKEHOLDERS

Table 3.1. System Stakeholders

Stakeholders	Needs	Influence Level	Methodology
Department of Health, public health organizations	Reduce the burden on hospital emergency departments from minor burn cases and promote standardized public first-aid practices.	High	Newspapers, medical journals, and reference materials
Patient and Family members	Receive appropriate assessment and first-aid guidance for minor burns and timely emergency assistance for severe burns.	High	Surveys and behavioral observation
Medical advisor (Burn specialists)	Ensure that first-aid guidance follows medical standards and validate the reliability of the AI model.	High	Newspapers, medical journals, and reference materials

IT Staff	Ensure stable system performance and protect users' personal image data.	Medium	Technical design reviews
Emergency services	Receive emergency phone calls initiated by users via the device's native dialer when recommended by the app.	Low	Document analysis

3.3. USER ROLES AND REQUIREMENTS

This *Burn Assessment and First-Aid Guidance Information System* consists of three user groups with different functions and access permissions: Patient and Family members, Medical Advisors, and IT Staff. Each role has specific responsibilities within the system. This role-based access structure helps ensure clear system operations, data security, and alignment with the system's intended purpose.

Table 3.2. User roles and requirements

User roles	Description	Responsibility
User (Patient and Family member)	This is the primary user group of the system. They use it when they need a quick preliminary assessment of a burn, suspect a burn injury, or require appropriate first-aid guidance.	- Create and manage personal accounts and profiles. - Upload burn images and provide relevant information for preliminary assessment. - View burn assessment results and corresponding first-aid guidance. - Receive warnings when the situation may require urgent medical attention. - Browse first-aid and burn-related guidance without performing an assessment.
Medical Advisor	Medical advisors are responsible for reviewing and managing professional first-aid content within the system. They help ensure that first-aid procedures are appropriate, easy to understand, and consistent with medical standards.	- Review and update burn-related medical information. - Manage first-aid guidance and recommendations. - Review the content associated with assessment results and emergency warnings. - Ensure that medical information is consistent with professional guidance.

IT Staff	IT Staff are responsible for managing the AI-based assessment models and maintaining the system. They ensure that the system functions, user accounts, and model-generated assessments operate appropriately and effectively.	- Manage and maintain user accounts and access status. - Monitor system performance and technical errors. - Maintain and update the burn assessment model. - Manage system security and data protection. - Support system maintenance and troubleshooting.

3.4. SCOPE OF WORK

3.4.1. In - scope

Within the project scope, the system is designed for adult users aged 18 and above. It supports the preliminary assessment of burns and provides appropriate first-aid guidance for each situation. However, it should be noted that this is only a support system intended to help users access information and determine initial actions. It does not completely replace the role of doctors and medical professionals.

The functions of the system for users include burn assessment based on images and information provided by users. The system then displays the results and provides corresponding first-aid guidance, along with warnings when the assessment indicates a severe case that may require urgent medical assistance. Users can also view guidance information without submitting an assessment request and view their personal profiles. Medical advisors are granted access to review and update professional first-aid content. Finally, tasks such as account management, system monitoring, and AI model improvement are assigned to IT Staff.

3.4.2. Out - of - scope

The project is clearly defined as a support system and cannot replace a doctor's diagnosis, consultation, or treatment decisions. Any medical intervention beyond first aid requires advice from qualified professionals at healthcare facilities. In addition, the target users are limited to adults aged 18 and above, and the system does not currently apply to infants or children. Activities outside the project scope, such as providing actual burn treatment, prescribing medication, monitoring and advising on patient care, sharing location information with emergency services, or coordinating ambulances, will not be performed in this project.

3.5. SYSTEMS REQUIREMENTS DOCUMENTS

3.5.1. Functional requirements

Table 3.3. Functional requirements

ID	Functional Description	User	Collection Method	Priority
FR-01	Account Registration (email or phone number, password, Google or Facebook)	Patient and Family members	User Survey	Must have
FR-02	Log in to the system	Patient and Family members	User Survey	Must have
FR-03	Forgot Password	Patient and Family members	Document Analysis (Common App UX Standards)	Should have
FR-04	View and agree to the terms of use and medical disclaimer before first use	Patient and Family members	Document Analysis (Legal Requirements)	Must have
FR-05	Manage personal profile (full name, age, and emergency contact phone number)	Patient and Family members	User Survey	Must have
FR-06	Take or upload a burn image to the system	Patient and Family members	Usage Behavior Observation	Must have
FR-07	Complete a quick survey (patient's age, gender, smoke inhalation)	Patient and Family members	Survey	Must have
FR-08	Ask AI Chatbot	Patient and Family members	Survey	Should Have
FR-09	The model segments the burned area in the image	System Processing	Document analysis	Must have

FR-10	Calculate %TBSA (total body surface area affected by burns) using the 1-3-6-9-18 rule.	System Processing	Document analysis	Must have
FR-11	Burn-severity classification by model	System Processing	Document analysis	Must have
FR-12	The model classifies burn severity from minor to severe	Patient and Family members (receive results)	Document Analysis (Paper 1.4)	Must have
FR-13	Aggregate internal processing results (burned areas, %TBSA, internal ABSI for clinician referral, and burn severity...)	System Processing	Document Analysis	Must have
FR-14	Calculate the ABSI value	System	Document analysis	Must have
FR-15	Display step-by-step first-aid guidance corresponding to the assessment severity	System	Medical Advisor Interview	Must have
FR-16	Display a warning and recommend calling emergency services immediately when a severe case is assessed	System	Document analysis	Must have
FR-17	Provide a button that redirects users to the phone app to make an emergency call	Patient and Family members	User Interview	Must have
FR-18	Save the history of self-assessments, including segmented images	System	Survey	Must have

FR-19	Send reminders to monitor the wound after first aid	System	User Interview	Should have
FR-20	Submit feedback and rate the usefulness of the guidance after use	Patient and Family members	Survey	Must have
FR-21	Log out of the account	Patient and Family members	Document analysis	Must have
FR-22	Medical advisors log in to the administration portal to draft or approve first-aid guidance content	Medical advisor	Medical Advisor Interview	Should have
FR-23	IT Staff manage and update classification and segmentation model versions (retraining and changing versions when new datasets become available)	IT Staff	Technical Design Review	Must have
FR-24	Compose new first-aid guidance content for each burn severity level	Medical advisor	Medical Advisor Interview	Must have
FR-25	Edit and update existing guidance when medical guidelines change	Medical advisor	Medical Advisor Interview	Should have
FR-26	View user feedback on the usefulness of the guidance to improve content	Medical advisor	Survey and Focus Group	Should have
FR-27	View a list of cases assessed by the system as having low confidence for manual review	Medical advisor	Document analysis	Should have
FR-28	Monitor system performance (uptime, response time, and runtime errors)	IT Staff	Technical Design Review	Must have

FR-29	Manage user accounts (suspend or delete accounts upon request while complying with data privacy requirements)	IT Staff	Document Analysis (Personal Data Protection Regulations)	Should have

3.5.2. Non-functional requirements

Table 3.4. Non-functional requirements

ID	Type	Description	Relevant Stakeholders	Priority
NFR-01	Performance	Burn severity classification results must be returned within 3 minutes of image upload	Patient, family member	Must have
NFR-02	Performance	The emergency call button for 115 must respond immediately, within 1 second, regardless of network speed	Patient, family member	Must have
NFR-03	Reliability	The emergency call function and display of basic first-aid guidance must operate in offline mode without an internet connection	Patient, family member	Must have
NFR-04	Usability	The interface must be simple, use large text, and require minimal interaction, with no more than 2 to 3 steps, so it can be used even when users are distressed	Patient, family member	Must have
NFR-05	Security	Encrypt users' image data and personal information during storage and transmission	Patient, family member, IT Staff	Must have
NFR-06	Compliance	Medical guidance content must be approved by a medical specialist before publication and must comply with regulations for health-support applications	Medical advisor	Must have

NFR-07	Availability	The system must achieve a minimum uptime of 99%, especially for emergency functions	IT Staff	Must have
NFR-08	Compatibility	The application must run on both Android and iOS and be compatible with low-specification devices	Patient, family member	Should have
NFR-09	Scalability	The system must withstand sudden increases in traffic, such as during holidays when more burn accidents may occur due to fireworks or fires	IT Staff	Should have
NFR-10	Maintainability	First-aid guidance content and AI models must be updateable without releasing the entire application again	IT Staff	Should have
NFR-11	Localization	Vietnamese and English must be supported as the primary languages, with the possibility of adding other languages later	Patient and Family members	Should have
NFR-12	Accuracy (Segmentation Model)	The burn-area segmentation model must meet the minimum performance metrics	IT Staff	Must have
NFR-13	Accuracy (Classification Model)	The burn-depth classification model must achieve a minimum average accuracy of approximately 70%	IT Staff	Must have

CHAPTER 4. DATA PROCESS MODELING

4.1. OVERVIEW AND PURPOSES

Based on the requirements analysis, the primary objective of this chapter is to establish a structured data-process framework for the *Burn Assessment and First-Aid Guidance Information System*. Data Flow Diagrams (DFDs) are designed to visually model the movement, transformation, and storage of information across the system. In this architecture, data serves as the critical backbone connecting three external entities (Patient and Family members, Medical Advisors, and IT Staff) with six primary processes and five central data stores (D1 to D5). The DFD model explicitly details how raw user inputs, such as burn images and clinical survey responses, are processed through AI-driven segmentation and classification algorithms to compute burn metrics. By mapping these data interactions from high-level Context Diagrams to detailed Level 2 DFDs, this chapter provides a clear technical blueprint for reliable guidance delivery, emergency response, and clinical case reviews.

4.2. DATA FLOW DIAGRAM (DFD)

4.2.1. Context Diagram

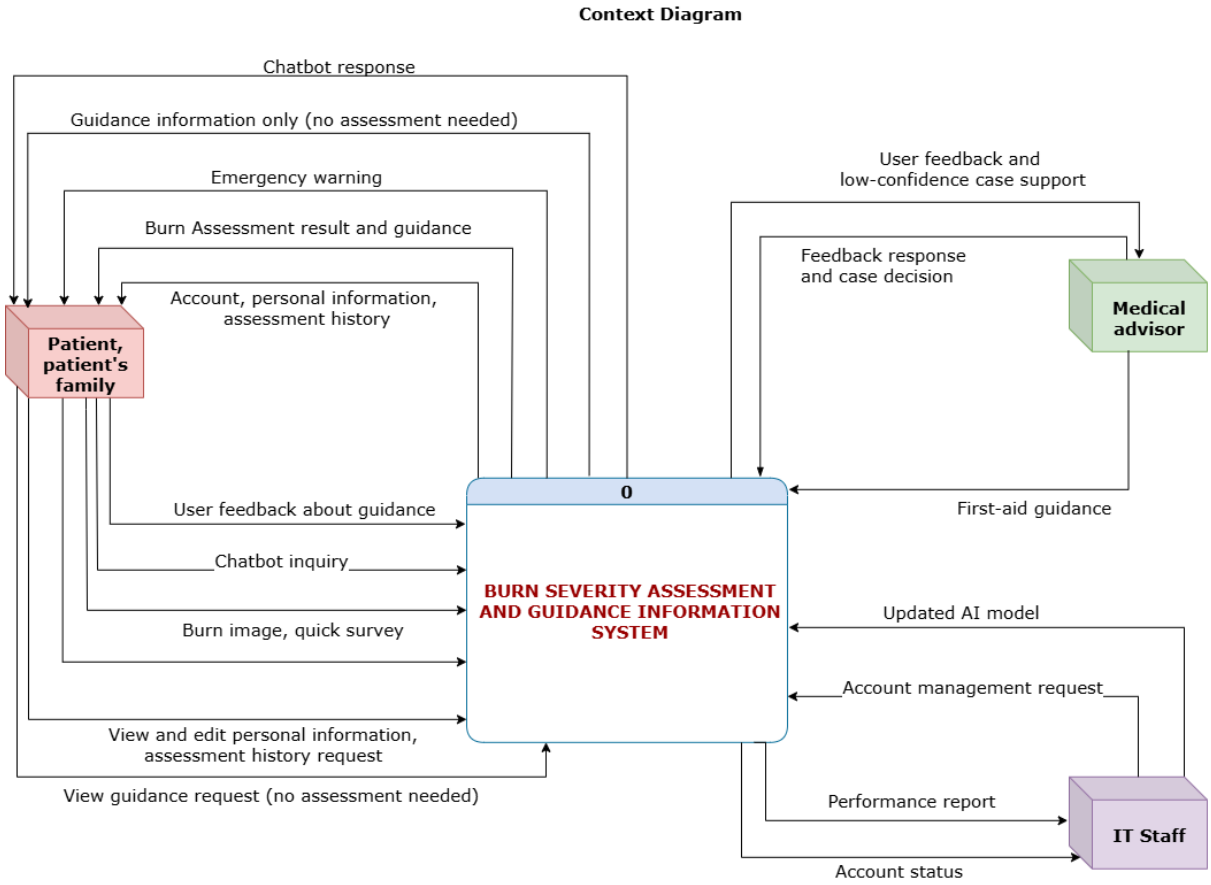


Figure 4.1. Context Diagram

The context diagram establishes the overall boundary of the *Burn Assessment and First-Aid Guidance Information System* by representing the system as a single central process and

showing its interactions with external entities. The system interacts with three main external entities: Patient and Family members, Medical advisor, and IT Staff.

The diagram illustrates the major data flows exchanged between the system and its external environment. Patient and Family members provide account information, personal information, burn images, survey responses, assessment requests, guidance requests, and feedback. In return, the system provides account information, assessment history, burn assessment results, emergency warnings, first-aid guidance, and responses to guidance-related inquiries. Medical Advisor provides and updates first-aid guidance content and receives user feedback and information related to cases requiring review. IT Staff provides account management information and AI model updates to the system.

By representing the system as a single process, the context diagram defines the system boundary and identifies the external entities and major data flows connected to the system. The internal processes, data stores, and detailed data transformations are excluded from this level and are further decomposed in the subsequent DFD levels.

4.2.2. DFD Level 1

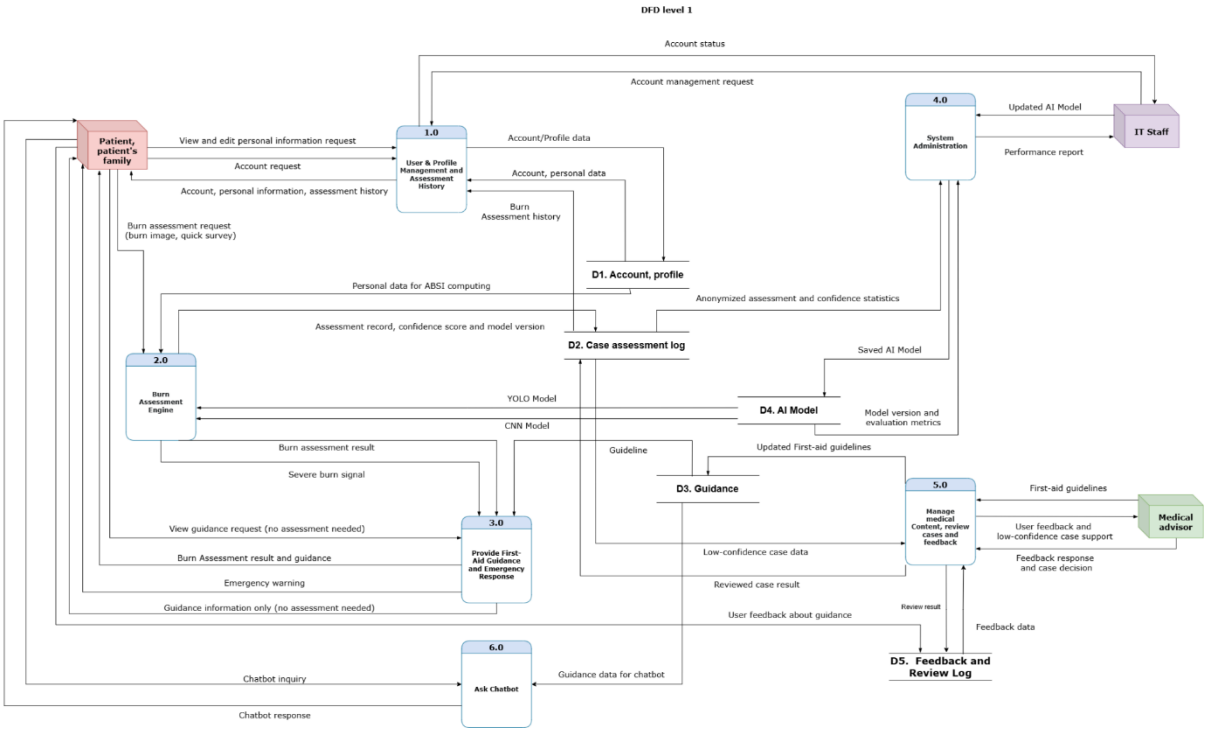


Figure 4.2. DFD Level 1

DFD Level 1 provides a more detailed view of the system represented in the Context Diagram by presenting six main business processes, including “User & Profile management and Assessment history” (Process 1.0), “Burn assessment engine” (Process 2.0), “Provide first-aid guidance and Emergency response” (Process 3.0), “System administration” (Process 4.0), “Manage medical content, Review cases and feedback” (Process 5.0), and finally, “Ask chatbot” (Process 6.0). Together, these processes show how data is received, transformed, stored, and exchanged across the system’s main functional areas.

At this level, five data stores, D1 to D5, are introduced to represent the main data resources supporting the system. D1 stores account and profile information, D2 stores burn assessment records, D3 stores first-aid guidance content, D4 stores AI models, and D5 stores feedback and review records. These data stores provide shared data resources for related processes and connect different functional areas through the storage and retrieval of relevant data.

Among the six processes, Process 2.0 serves as the main data-processing function for burn assessment. It receives burn images and quick survey responses from the user, uses the AI models stored in D4 to process the assessment inputs, and generates burn assessment results. These results are then transferred to Process 3.0, which retrieves relevant first-aid guidance from D3 and provides first-aid guidance or emergency warnings to the user. Moreover, the main user can send inquiries to the chatbot, and then Process 6.0 will deliver a response to them in return.

Meanwhile, Process 1.0 manages account and profile information and retrieves assessment records from the relevant data stores. Process 4.0 manages system-related information and AI model data, while Process 5.0 maintains first-aid guidance content provided by the medical advisor and updates the corresponding information in D3. In addition, this process also retrieves relevant assessment and feedback data from D2 and D5 to support case review and records the resulting review information.

Overall, DFD Level 1 presents the system's main data-processing architecture by showing the relationships among the six business processes, three external entities, and five data stores. It therefore provides the structural basis for further decomposition of selected processes into more detailed DFD Level 2 diagrams.

4.2.3. DFD Level 2

To provide a more detailed representation of the data flow within the system, three of the six Level 1 processes are further decomposed into Level 2: Process 2.0, Process 3.0, and Process 5.0. The other processes are already the lowest level of data processing in the system and therefore are not further decomposed.

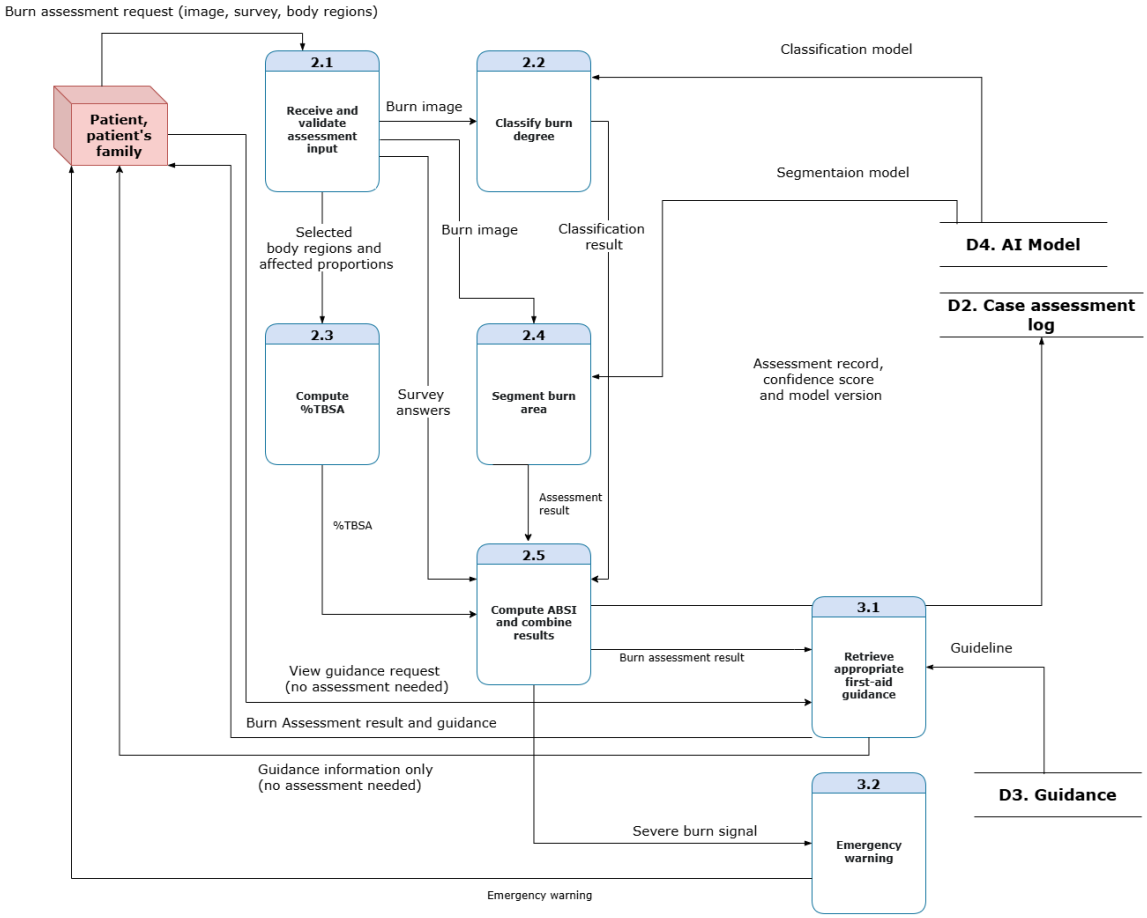


Figure 4.3. DFD Level 2 of Process 2.0 (Burn Assessment Engine) and Process 3.0 (Provide First-Aid Guidance and Enemergency Response)

For Process 2.0, the assessment flow is decomposed into five subprocesses, from Receive and Validate assessment input (2.1) to Compute ABSI and Combine results (2.5). The process receives burn images, selected body regions and affected proportions, and survey answers from patient and family members. Process 2.1 validates the assessment input before the data is passed to the subsequent assessment subprocesses. The classification and segmentation models are retrieved from AI model storage (D4) and applied to implement the burn classification and segmented burn area. The burn body proportions that the user selected at first are then used to compute %TBSA, while the relevant assessment indicators are combined to produce the final assessment result with its confidence score. The assessment record, including the model version and confidence score, is stored in the Case assessment log (D2) for subsequent retrieval and review.

In Figure 4.3 (DFD Level 2), Process 2.2 (Classify burn degree) and Process 2.4 (Segment burn area) are modeled as receiving the validated burn image independently from Process 2.1, reflecting the strict DFD convention of representing data transformations rather than execution sequence. At the system implementation level, however, the application coordinates these two processes via a gated execution mechanism, if Process 2.2 classifies the image as healthy skin (None), the system bypasses Process 2.4 to avoid false-positive segmentation masks and conserve computational resources.

For Process 3.0, the process is decomposed into Retrieve appropriate first-aid guidance (3.1) and Emergency warning (3.2). The process receives the burn assessment result generated by Process 2.0 and uses the assessment information to retrieve the corresponding guidance information from the guidance data store (D3). Based on the assessment result, the system provides first-aid guidance to the user. When the assessment indicates a severe burn condition, the relevant severe burn signal is routed to Process 3.2, which generates an emergency warning. This decomposition shows how assessment data is transformed into guidance and emergency-response outputs.

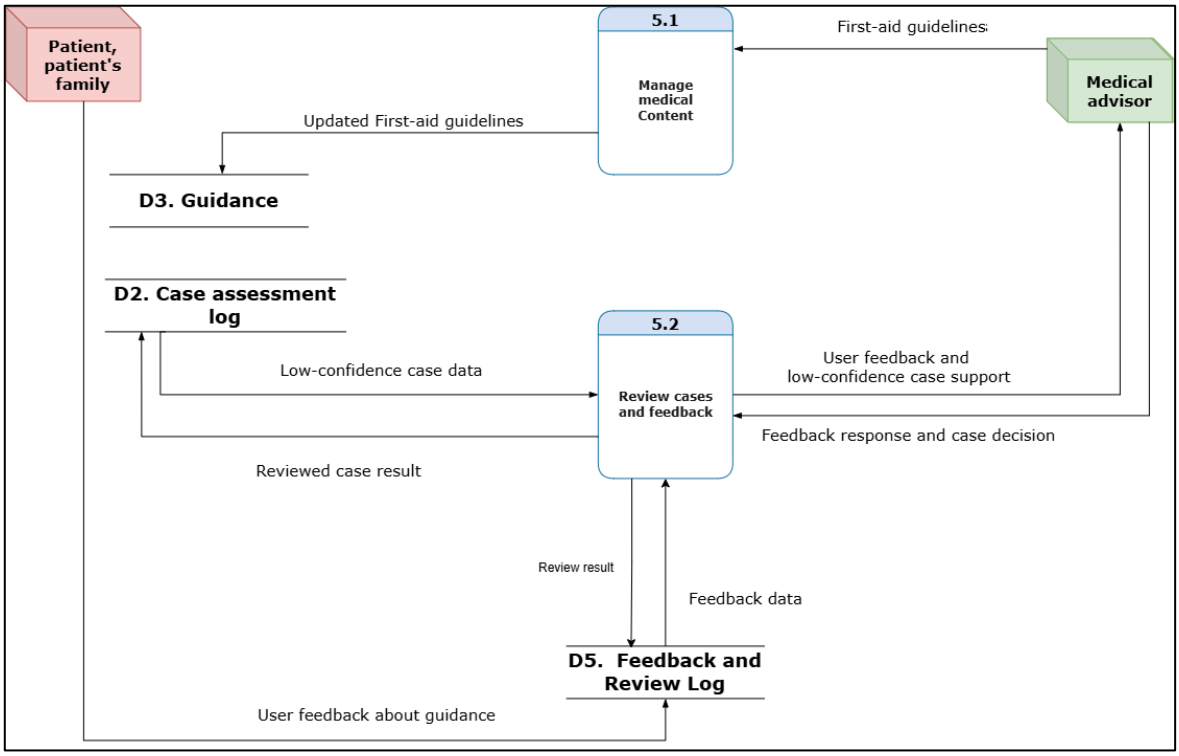


Figure 4.4. DFD Level 2 of Process 5.0 (Manage medical content, review cases, and feedback)

Process 5.0 is decomposed into two subprocesses: Manage medical content (5.1) and Review cases and feedback (5.2). Process 5.1 manages first-aid guidance content by receiving medical content from the medical advisor, validating and updating the relevant information, and storing the resulting guidance in D3. This ensures that the guidance data used by the system can be maintained and updated through the corresponding data store.

Process 5.2 handles case review and feedback by retrieving relevant assessment records and user feedback from D2 and D5. The advisor reviews the available case and feedback information and provides review results, which are then recorded in D5. Through these data flows, Process 5.0 supports the management of medical content and the review of assessment-related information within the system.

Overall, the decomposition of Process 5.0 illustrates how medical content, assessment records, and feedback are retrieved, processed, updated, and stored through the relevant subprocesses and data stores.

CHAPTER 5. OBJECT MODELING

5.1. SCENARIOS, MODELS AND FUNCTIONALITY

5.1.1. Use Case Diagram

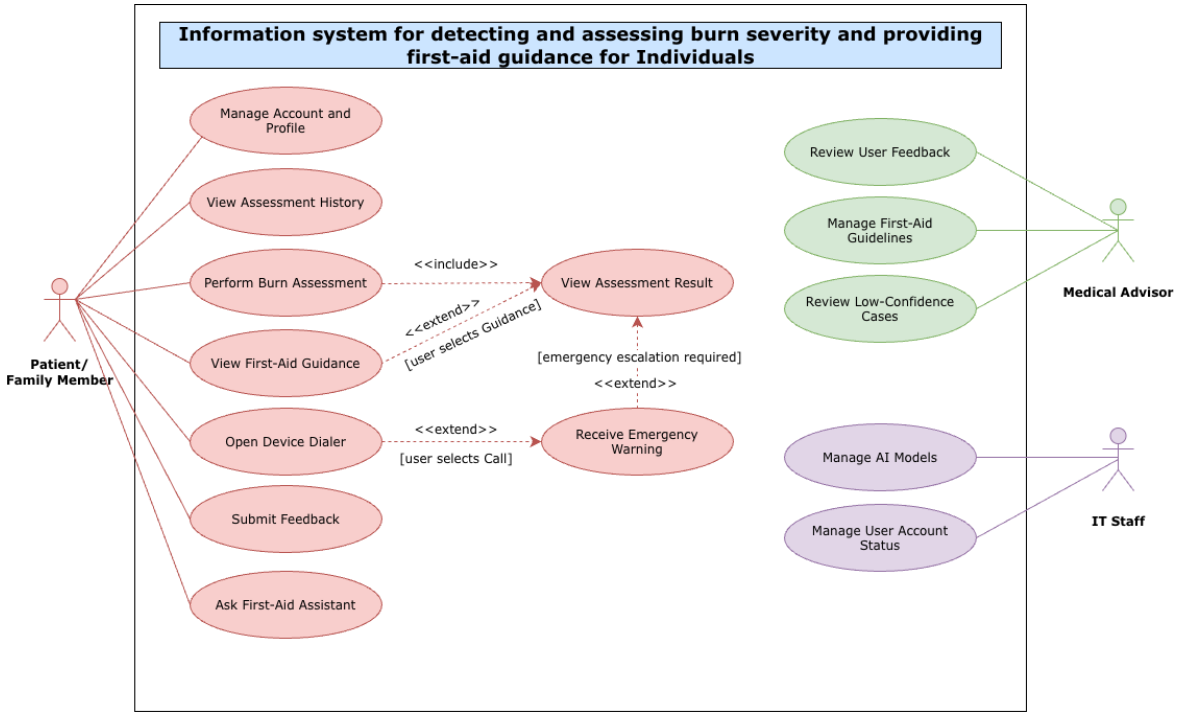


Figure 5.1. Detecting and assessing burn severity and providing first-aid guidance for Individuals System Use Case diagram

5.1.2. Use Case Descriptions

Table 5.1. List of Use Cases

ID	Use Case
UC-01	Manage Account and Profile
UC-02	View Assessment History
UC-03	Perform Burn Assessment
UC-04	View Assessment Result
UC-05	View First-Aid Guidance
UC-06	Receive Emergency Warning
UC-07	Open Device Dialer
UC-08	Submit Feedback
UC-09	Review User Feedback

UC-10	Manage First-Aid Guidelines
UC-11	Review Low-Confidence Cases
UC-12	Manage AI Models
UC-13	Manage User Account Status
UC-14	Ask First-Aid Assistant

Table 5.2. UC-01 Manage Account and Profile

1. Name, ID	UC-01 - Manage Account and Profile
2. Importance Level	High
3. Primary Actor	Patient, Family Member
4. Use Case Type	User-goal; system-level
5. Stakeholders & Interests	- Patient and Family Member: wants secure account access and accurate personal information. - System Owner: requires valid profile data for assessment and ABSI calculation. - IT Staff: requires controlled account status and access management.
6. Brief Description	Allows a user to create an account, sign in, view their profile, and update personal information. Profile viewing and editing require an authenticated session.
7. Trigger	The user selects Sign Up, Sign In, or Account and Profile.
8. Relationships	Association: Patient and Family Member. Related to UC-03 and UC-02 because profile identity links assessments to the correct user.
9. Normal Flow of Events	1. The user opens the account or profile function. 2. The system displays the available options according to the user's authentication status. 3. The user selects Sign Up, Sign In, or View and Edit Profile. 4. The system executes the corresponding subflow. 5. The system displays the outcome.

10. Subflows & Exceptional Flows	- S1. Sign up 1. The user enters the required registration information. 2. The system validates the information and checks that the email is unique. 3. The system hashes the password and creates the account with role patient, account status active, and a creation timestamp. 4. The system establishes an authenticated session for the new account. 5. The system confirms successful registration and displays the Home screen. - S2. Sign in 1. The user enters their email and password. 2. The system verifies the credentials and checks that the account status is active. 3. The system establishes an authenticated session and displays the home screen. - S3. View and Edit profile 1. The system verifies the user's authenticated session and retrieves their profile. 2. The system displays the profile. 3. If the user chooses to edit, the user updates the permitted personal information. 4. The system validates and saves the changes and updates updated_at. 5. The system displays the updated profile. - E1. Required information is missing or invalid: the system identifies the affected fields and requests correction. - E2. Credentials are incorrect: the system denies access and offers another attempt or password recovery. - E3. The account is locked or deactivated: The system blocks sign-in and displays an appropriate account-status message. - E4. The account already exists: the system directs the user to sign in. - E5. Saving fails: the system retains the previous profile and reports the error.

Table 5.3. UC-02 View Assessment History

1. Name, ID	UC-02 - View Assessment History
2. Importance Level	Medium
3. Primary Actor	Patient, Family Member
4. Use Case Type	User-goal; system-level
5. Stakeholders & Interests	- Patient, Family Member: wants to review earlier assessments and guidance. - System Owner: must show only records belonging to the signed-in account.
6. Brief Description	Allows a signed-in user to view their saved burn cases and open a selected case to review its case-level results and associated burned-area assessments.
7. Trigger	The signed-in user selects Assessment History.
8. Relationships	Association: Patient, Family Member. Opening a selected burn case provides access to UC-04 View Assessment Result, including the associated burned-area assessments.
9. Normal Flow of Events	1. The user opens Assessment History. 2. The system verifies that the user has a valid authenticated session and an active account. 3. The system retrieves burn cases whose account_id matches the authenticated account. 4. The system displays the burn cases in reverse chronological order, showing each case's creation date and time, case status, and overall risk level when available. 5. The user selects a burn case. 6. The system verifies that the selected case belongs to the authenticated account. 7. The system retrieves the selected case and its associated burned-area assessments. 8. The system presents the case through UC-04 View Assessment Result, including case-level results and available results for each burned area. 9. The user may return to the burn case list.

10. Subflows & Exceptional Flows	- E1. No saved burn cases: The system displays an empty-history message and offers access to UC-03 Perform Burn Assessment. - E2. Session expired or account not active: The system denies access to assessment history and requests sign-in or displays an appropriate account-status message. - E3. Selected case is unavailable: The system explains that the case cannot be retrieved and returns the user to the history list. - E4. Unauthorized case access: The system denies access if the selected case does not belong to the authenticated account and does not disclose its contents. - E5. Incomplete or inconclusive case: The system displays the case and area statuses, identifies unavailable results, and does not present partial results as a confirmed case-wide assessment. - E6. Guest assessment: Cases created without an owning account are not included in the signed-in user's account history.

Table 5.4. UC-03 Perform Burn Assessment

1. Name, ID	UC-03 - Perform Burn Assessment
2. Importance Level	Critical
3. Primary Actor	Patient, Family Member
4. Use Case Type	User-goal; core system use case
5. Stakeholders & Interests	- Patient, Family Member: wants a timely and understandable preliminary burn assessment. - Medical Advisor: wants low-confidence cases identified for professional review. - System Owner: requires traceable AI and rule-based calculations.
6. Brief Description	Uses burn images and survey information to assess an adult's burn injury, calculate Estimated TBSA from selected body regions and affected proportions, calculate ABSI when applicable, determine risk, and save the assessment. ABSI is available only to authorized Medical Advisors.
7. Trigger	The user selects Perform Burn Assessment.
8. Relationships	Association: Patient, Family Member. Includes UC-04 View Assessment Result. UC-06 extends UC-04 under [emergency escalation required].

9. Normal Flow of Events	1. The user captures or uploads a burn image. 2. The system validates the image format and quality. If acceptable, it displays the questionnaire and body-area map. 3. The user provides or confirms the injured person's age and sex, enters incident information and symptoms, and selects the affected body region and proportion. Profile information may be used when available. 4. The system validates the required assessment information and confirms that the injured person is aged 18 or older. 5. The segmentation component identifies the burn region in the image when supported. 6. The classification component predicts burn degree I, II, III, or None and produces a confidence score. 7. The system calculates preliminary regional Estimated TBSA from the selected body region and affected proportion, excluding Degree I burns, and then checks whether the prediction meets the configured confidence threshold. Low-confidence results follow E3. 8. The user may repeat the image capture and regional assessment steps for additional burned areas. Overlapping regions are not counted twice. 9. When no further regions are added, the system finalizes total Estimated TBSA when sufficient information is available and calculates case-wide ABSI when applicable. ABSI is available only to authorized Medical Advisors. 10. The system evaluates available assessment results, incident information, symptoms, and uncertainty to determine the overall risk and recommended action. 11. The system saves the assessment inputs, available outputs, model references, and prediction acceptance or rejection information. 12. The system invokes UC-04 View Assessment Result, retaining uncertainty warnings for low-confidence results, and initiates UC-06 if emergency escalation is required.
10. Subflows & Exceptional Flows	- E1. Missing, blurred, or unsupported image: The system asks the user to provide another image. - E2. Required survey data is missing: The system highlights the missing fields and requests completion.

	- E3. Low-confidence prediction: The system displays an uncertainty notice and offers another image attempt. If the user continues without retaking the image, the system flags the assessment as low-confidence, retains available raw AI output until it is successfully replaced by a conclusive Medical Advisor-reviewed outcome, and creates an entry for Medical Advisor review. Available AI predictions are displayed only as unconfirmed results with an explicit warning, not as accepted predictions. - E4. AI inference fails: The system reports that image assessment is temporarily unavailable and does not fabricate a result. Safety evaluation continues using other available information. - E5. Assessment cannot be saved: The system may display available results but clearly reports that the assessment was not saved to history. - E6. Incomplete or inconclusive assessment: The system identifies unavailable or unconfirmed results. It does not present a partial estimate as a confirmed total Estimated TBSA. ABSI remains unavailable when it cannot be calculated, while safety evaluation continues using available information.

Table 5.5. UC-04 View Assessment Result

1. Name, ID	UC-04 - View Assessment Result
2. Importance Level	Critical
3. Primary Actor	Patient, Family Member
4. Use Case Type	User-goal; included by UC-03 and accessible from assessment history.
5. Stakeholders & Interests	- Patient/ Family Member: needs a clear result and an explanation of limitations. - Medical Advisor: requires uncertainty and model version to remain traceable.
6. Brief Description	Displays available burn-assessment results, including burn-depth predictions, localization, confidence information, Estimated TBSA, overall risk, and recommended action. Accepted results are distinguished from low-confidence predictions, which may be viewed only with an explicit uncertainty warning. Unavailable results are identified. ABSI scores and prognosis are not displayed to the Patient, Family Member.

7. Trigger	UC-03 produces an assessment outcome, the user selects a saved assessment from UC-02 View Assessment History, or the user opens an in-app notification about an updated assessment.
8. Relationships	Association: Patient, Family Member. Included by UC-03 Perform Burn Assessment. Accessible from UC-02 View Assessment History. Extended by UC-05 View First-Aid Guidance. Extended by UC-06 Receive Emergency Warning.
9. Normal Flow of Events	1. The system obtains the current assessment outcome or retrieves the saved assessment selected by the user. 2. For a saved assessment, the system verifies that the user is authorized to access it. 3. The system displays the assessment date and indicates whether the assessment is complete or inconclusive. 4. The system displays the overall risk level and recommended action prominently. If emergency escalation is required, it immediately initiates UC-06. 5. For each assessed body region, the system displays available burn-depth predictions, confidence information, and localization. Low-confidence predictions are clearly labelled as unconfirmed and are not presented as accepted results. 6. When a Medical Advisor-reviewed outcome is available, the system displays it as the current result, labelled Medical Advisor-reviewed with the review date. The superseded AI prediction and confidence are no longer displayed. 7. The system displays each region's Estimated TBSA and the total Estimated TBSA when sufficient information is available. 8. It explains that the estimate comes from selected body regions and affected proportions, rather than the proportion of burned pixels in a close-up image. 9. The system identifies any unavailable or inconclusive results and explains how these limitations affect the assessment. 10. The system states that the assessment is preliminary, may be incorrect, and does not replace professional medical evaluation. 11. Confidence information is not presented as a guarantee of correctness. 12. The system offers access to UC-05 View First-Aid Guidance. 13. The user may view the guidance, return to assessment history when available, or leave the result screen.

10. Subflows & Exceptional Flows	- E1. Low-confidence prediction: The system displays an uncertainty notice and offers another image attempt or professional assessment. If the user continues, available predictions remain clearly labelled as low-confidence and unconfirmed. Continuing does not change the prediction's acceptance status, and the assessment is retained for Medical Advisor review. - E2. Incomplete Estimated TBSA: If one or more affected regions remain unresolved, the system indicates that total Estimated TBSA is unavailable or incomplete. Available regional estimates are clearly labelled and are not presented as the confirmed total for the whole injury. - E3. Emergency escalation: The system displays the emergency warning and recommended action before optional result details. The user may select Call through UC-06 and UC-07. Uncertain or incomplete AI results do not suppress warnings supported by other risk information. - E4. Burn-region localization unavailable: The system states that localization is unavailable and does not display a fabricated mask or overlay. Other results are shown only if they remain valid under the assessment's acceptance criteria. - E5. Stored image unavailable: The system indicates that the image is unavailable and displays other saved results that can still be retrieved. - E6. Unauthorized access or expired session: The system does not disclose the saved assessment. It requests sign-in when appropriate or displays an access-denied message. - E7. Assessment unavailable: The system explains that the selected assessment cannot be retrieved and offers a return to assessment history or the main menu. - E8. Assessment not saved: The system displays available results and safety advice but clearly states that the assessment was not saved to history.

Table 5.6. UC-05 View First-Aid Guidance

1. Name, ID	UC-05 - View First-Aid Guidance
2. Importance Level	Critical
3. Primary Actor	Patient, Family Member
4. Use Case Type	User-goal; standalone or extending use case

5. Stakeholders & Interests	- Patient, Family Member: wants simple, approved, step-by-step first-aid instructions. - Medical Advisor: requires only approved and current guidance to be displayed.
6. Brief Description	Displays approved first-aid guidance either independently from the main menu or in the context of an assessment result.
7. Trigger	The user selects First-Aid Guidance from the main menu or selects Guidance while viewing an assessment result.
8. Relationships	Association: Patient, Family Member. Extends UC-04 under [user selects Guidance]. It can also execute independently without UC-03 or UC-04.
9. Normal Flow of Events	1. The user opens First-Aid Guidance. 2. The system determines whether the request comes from an assessment result. 3. If assessment context exists, the system uses relevant incident information, affected body regions, and accepted assessment results when available to select appropriate guidance. 4. If no assessment context exists, the user selects a burn cause or guidance topic. 5. The system retrieves the current approved guidance. 6. The system displays step-by-step first-aid instructions, cautions, and escalation advice. 7. The user may return to the topic list or close the guidance.
10. Subflows & Exceptional Flows	- E1. No matching approved guidance: Display approved general guidance when available and explain the limitation. - E2. Guidance cannot be retrieved: Report temporary unavailability. - E3. Emergency escalation applies: Prioritize the emergency warning and recommended action. - E4. Assessment is inconclusive: Do not use rejected predictions to select degree-specific guidance.

Table 5.7. UC-06 Receive Emergency Warning

1. Name, ID	UC-06 - Receive Emergency Warning
2. Importance Level	Critical

3. Primary Actor	Patient, Family Member
4. Use Case Type	System-initiated; extending use case
5. Stakeholders & Interests	- Patient, Family Member: must be alerted clearly when the available information indicates high risk. - System Owner: requires deterministic and auditable emergency rules.
6. Brief Description	Displays an emergency warning and recommended action when the available assessment outcome meets configured emergency-escalation criteria.
7. Trigger	The system identifies a need for emergency escalation during risk evaluation following the image-assessment attempt.
8. Relationships	Extends UC-04 View Assessment Result under [emergency escalation required]. May be extended by UC-07 Open Device Dialer under [user selects Call].
9. Normal Flow of Events	1. The system evaluates available image-assessment results, Estimated TBSA, incident information, symptoms, and uncertainty against configured emergency rules. 2. The system identifies one or more conditions requiring emergency escalation. 3. The system displays a prominent emergency warning. 4. The system explains the detected risk factors in plain language. 5. The system advises the user to seek urgent professional medical care. 6. The system offers a Call action that may invoke UC-07.
10. Subflows & Exceptional Flows	- E1. Some risk inputs are missing: the system states that the risk evaluation is incomplete and recommends a cautious next step. - E2. The user dismisses the warning: The system returns to the originating screen while keeping the emergency alert accessible. Dismissing the warning does not clear the identified risk. - E3. No emergency-escalation condition exists: This use case is not executed.

Table 5.8. UC-07 Open Device Dialer

1. Name, ID	UC-07 - Open Device Dialer
2. Importance Level	High

3. Primary Actor	Patient, Family Member
4. Use Case Type	Optional extending use case
5. Stakeholders & Interests	- Patient, Family Member: wants quick access to the device calling interface after receiving a warning. - System Owner: must not imply that the application itself places or completes a call.
6. Brief Description	Requests the device to open its native dialer after the user selects Call from an emergency warning. The application does not contact a healthcare facility itself.
7. Trigger	The user selects Call while UC-06 is displayed.
8. Relationships	Association: Patient, Family Member. Extends UC-06 under [user selects Call].
9. Normal Flow of Events	1. The user selects Call on the emergency warning. 2. The system asks for confirmation if required. 3. If confirmation is requested, the user confirms the action. 4. The application sends a request to open the device's native dialer. 5. The device displays the dialer with the configured emergency number when supported. 6. The user decides whether to place the call outside the application.
10. Subflows & Exceptional Flows	- E1. The device does not support a dialer: the system displays the emergency number for manual use. - E2. Permission or operating-system handling fails: the system explains that the dialer could not be opened. - E3. The user cancels: the system returns to the emergency warning without initiating a call.

Table 5.9. UC-08 Submit Feedback

1. Name, ID	UC-08 - Submit Feedback
2. Importance Level	Medium
3. Primary Actor	Patient, Family Member
4. Use Case Type	User goal: supporting use case

5. Stakeholders & Interests	- Patient, Family Member: wants to submit feedback about the application under an appropriate category. - Medical Advisor: wants actionable feedback for review and improvement. - System Owner: needs traceable feedback without automatically changing medical content or assessment results.
6. Brief Description	Allows the user to independently submit feedback about the application by selecting a category and providing a rating, comments, or both.
7. Trigger	The user selects Submit Feedback.
8. Relationships	Association: Patient, Family Member. Related to UC-09 Review User Feedback, no include or extend relationship is required in the submitted diagram.
9. Normal Flow of Events	1. The system displays the feedback form. 2. The user selects a feedback category. 3. The user provides a rating, comments, or both. 4. The system validates the selected category and submitted information. 5. The system records the feedback, selected category, and submission time. 6. The system places the feedback in the Medical Advisor review queue. 7. The system confirms successful submission.
10. Subflows & Exceptional Flows	- E1. Required feedback fields are missing: the system requests completion. - E2. The comment exceeds an allowed limit or contains unsupported content: the system requests revision. - E3. Submission fails: the system preserves the entered text where possible and offers another attempt.

Table 5.10. UC-09 Review User Feedback

1. Name, ID	UC-09 - Review User Feedback
2. Importance Level	Medium
3. Primary Actor	Medical Advisor

4. Use Case Type	Administrative; supporting use case
5. Stakeholders & Interests	- Medical Advisor: wants to identify guidance issues and improvement opportunities. - Patient, Family Member: expects submitted concerns to inform safer content. - System Owner: requires review status and decisions to be auditable.
6. Brief Description	Allows the medical advisor to inspect, categorize, comment on, and close feedback submitted by users.
7. Trigger	The Medical Advisor opens the pending feedback queue.
8. Relationships	Association: Medical Advisor. Receives information originating from UC-08 Submit Feedback.
9. Normal Flow of Events	1. The Medical Advisor opens the feedback queue. 2. The system displays pending feedback with relevant non-excessive context. 3. The Medical Advisor selects a feedback item. 4. The system displays the feedback, related guidance version, and assessment reference when available. 5. The Medical Advisor examines the issue and records review notes. 6. The Medical Advisor selects a disposition, such as no change, investigation, or guideline update required. 7. The system saves the decision and marks the item as reviewed.
10. Subflows & Exceptional Flows	- E1. Referenced data is unavailable: the system shows the feedback without the missing context. - E2. The item was updated by another reviewer: the system refreshes the latest status before saving. - E3. The Medical Advisor lacks permission: The system denies access to the feedback review function.

Table 5.11. UC-10 Manage First-Aid Guidelines

1. Name, ID	UC-10 - Manage First-Aid Guidelines

2. Importance Level	Critical
3. Primary Actor	Medical Advisor
4. Use Case Type	Administrative content-management use case
5. Stakeholders & Interests	- Medical Advisor: wants to maintain accurate and approved medical guidance. - Patient, Family Member: must receive only current approved guidance. - System Owner: requires versioning, approval status, and an audit trail.
6. Brief Description	Allows an authorized Medical Advisor to create, edit, approve, publish, archive, and version first-aid guidance.
7. Trigger	The Medical Advisor selects Manage First-Aid Guidelines or chooses to update guidance after a review.
8. Relationships	Association: Medical Advisor. Supplies the approved content used by UC-05 and UC-06.
9. Normal Flow of Events	1. The Medical Advisor opens the guideline management workspace. 2. The system displays existing guidelines with their statuses and versions. 3. The Medical Advisor adds a new guideline or updates an existing one, providing completed content, applicability conditions, warnings, and sources. 4. The system validates the required information and applicability conditions. 5. The Medical Advisor reviews the content and confirms approval. 6. The system saves and makes the approved version available. When replacing an existing version, it archives the superseded version while preserving its past assessment references.
10. Subflows & Exceptional Flows	- S1. Archive a guideline: The Medical Advisor selects a guideline to archive. The system excludes it from new guidance selections while preserving its previous versions and assessment references. - E1. Required medical or source information is missing: the system prevents publication. - E2. Invalid or conflicting applicability conditions: The system identifies the affected conditions and requests correction before approval.

	- E3. Another advisor has changed the guideline: the system requests review of the latest version. - E4. Saving or publishing fails: the current approved version remains active.

Table 5.12. UC-11 Review Low-Confidence Cases

1. Name, ID	UC-11 - Review Low-Confidence Cases
2. Importance Level	High
3. Primary Actor	Medical Advisor
4. Use Case Type	Administrative clinical review use case
5. Stakeholders & Interests	- Medical Advisor: wants to examine uncertain AI outputs using professional judgment. - Patient, Family Member: needs uncertainty to be handled transparently. - AI Development Team: benefits from reviewed labels for later controlled evaluation or retraining.
6. Brief Description	Allows a Medical Advisor to review cases whose AI confidence falls below the configured threshold and record a professional review decision.
7. Trigger	The Medical Advisor opens the low-confidence case queue, or the system reports a new low-confidence case.
8. Relationships	Association: Medical Advisor. Cases originate from UC-03, but no UML include or extend relationship is required.
9. Normal Flow of Events	1. The system displays the low-confidence review queue. 2. The Medical Advisor selects a low-confidence assessment entry for a burned area. 3. The system displays the available burn image, relevant survey information, segmentation output, raw AI prediction and confidence, and the reason for rejection. Accepted results, regional Estimated TBSA, and case-wide Estimated TBSA and ABSI are displayed only when available. 4. The Medical Advisor reviews the available information. 5. The Medical Advisor submits a review decision and notes, including a corrected classification when appropriate.

	6. The system updates the existing area assessment with the Medical Advisor's reviewed outcome, reviewer identity, notes, and review time, retaining the same assessment ID. 7. The system removes the superseded raw AI prediction, confidence score, and rejection reason, marks the review entry as reviewed, and updates affected case-level calculations and guidance when applicable. 8. The system makes the updated assessment available in the user's history, sends an in-app notification to the owning user when available, and confirms completion to the Medical Advisor.
10. Subflows & Exceptional Flows	- E1. Insufficient information for review: The Medical Advisor records the limitation without assigning an unsupported classification. The assessment remains inconclusive, and its existing AI outputs are not replaced or deleted. - E2. The review entry has changed since it was opened: The system requests a refresh and review of the latest information before saving. - E3. Updating fails: The system retains the existing assessment and its AI outputs, leaves the review entry pending, and reports the failure. Old outputs are removed only after the replacement outcome is successfully saved. - E4. A reviewed label is available: the system does not automatically retrain or deploy a model without the authorized model-management process.

Table 5.13. UC-12 Manage AI Models

1. Name, ID	UC-12 - Manage AI Models
2. Importance Level	Critical
3. Primary Actor	IT Staff
4. Use Case Type	Administrative and technical management use case
5. Stakeholders & Interests	- IT Staff: wants to monitor and safely manage model versions. - Medical Advisor: expects reviewed cases and medical decisions not to be overwritten by technical actions. - System Owner: requires reproducible comparison, controlled deployment, and rollback.
6. Brief Description	Allows IT Staff to review model metrics and confidence trends, compare trained versions, register an updated model, deploy an approved version, and roll back when necessary.

7. Trigger	IT Staff opens AI Model Management.
8. Relationships	Association: IT Staff. Uses assessment logs and model metadata. It does not authorize IT staff to make medical review decisions in UC-11.
9. Normal Flow of Events	1. The system verifies IT staff authorization. 2. The system displays registered model versions, their task types, activation statuses, and available evaluation summaries. 3. IT Staff reviews evaluation metrics, confidence distribution, low-confidence rate, errors, and relevant operational logs. 4. IT Staff selects existing versions for comparison, including pretrained and from-scratch experiments when available. 5. IT Staff uploads or registers a candidate model with its task type, architecture, version, dataset version, applicable decision threshold, and evaluation information. 6. The system validates file integrity, compatibility, class mapping, and required evaluation information. 7. IT Staff approves activation of the validated candidate. 8. The system activates the validated candidate for the selected task, deactivates the version it replaces, and preserves previous versions and historical assessment references. 9. The system reports the activation status and displays the registered evaluation information.
10. Subflows & Exceptional Flows	- E1. Model validation fails: the system rejects deployment and displays the reasons. - E2. Required model metadata or metrics are missing: the system prevents activation. - E3. Deployment fails: the system keeps or restores the previous stable model. - E4. Rollback required: IT Staff selects a previously validated model version for the same task, and the system restores it as the active version.

Table 5.14. UC-13 Manage User Account Status

1. Name, ID	UC-13 - Manage User Account Status
2. Importance Level	High

3. Primary Actor	IT Staff
4. Use Case Type	Administrative security use case
5. Stakeholders & Interests	- IT Staff: wants to maintain valid account access and resolve account-status issues. - Patient, Family Member: expects account information and assessment records to remain protected. - System Owner: requires least-privilege access and controlled account-status changes.
6. Brief Description	Allows authorized IT Staff to search accounts, inspect their roles and current statuses, and change account statuses according to permitted transitions among active, locked, and deactivated.
7. Trigger	IT Staff selects Manage User Account Status.
8. Relationships	Association: IT Staff. Related to UC-01 account data but does not allow IT Staff to alter medical assessment results.
9. Normal Flow of Events	1. The system verifies IT Staff authorization. 2. The system retrieves and displays the account list. 3. IT Staff searches for and selects an account. 4. The system displays the selected account's role, current status, and permitted administrative information. 5. IT Staff selects a permitted target account status. 6. The system validates permission for the selected account and the requested status transition. 7. If valid, the system updates the account status. 8. The system displays the updated account status and confirms success.
10. Subflows & Exceptional Flows	- E1. The account does not exist: the system reports that no matching account was found. - E2. IT Staff lacks permission for the selected action: the system denies the request. - E3. Invalid status transition: The system rejects the requested change, displays the reason, and retains the current account status. - E4. The update fails: the previous account status remains unchanged, and the system reports the error.

Table 5.15. UC-14 Ask First-Aid Assistant

1. Name, ID	UC-14 - Ask First-Aid Assistant
2. Importance Level	High
3. Primary Actor	Patient, Family Member
4. Use Case Type	User-goal; supporting use case
5. Stakeholders & Interests	- Patient, Family Member: wants understandable answers to questions about burn first aid. - Medical Advisor: requires explanations to remain consistent with approved guidance and medical safety boundaries. - System Owner: requires appropriate handling of uncertainty, unsupported questions, and emergency warning signs.
6. Brief Description	Allows the user to ask questions about burn first aid and receive explanations grounded in medically approved guidance. Available assessment context may support the response, but rejected predictions are not treated as confirmed findings. The assistant does not replace professional medical assessment or provide inpatient treatment instructions.
7. Trigger	The user opens the First-Aid Assistant and submits a question.
8. Relationships	Association: Patient, Family Member. Uses approved guidance associated with UC-05 View First-Aid Guidance. No include or extended relationship with UC-05 is required.
9. Normal Flow of Events	1. The user opens the First-Aid Assistant. 2. The system displays the assistant's purpose and limitations. 3. The user enters a first-aid question. 4. The system checks whether the question is clear and within the assistant's scope. 5. The system retrieves relevant approved guidance and considers available assessment context when applicable. 6. The system provides an understandable explanation, relevant cautions, and source references when available. 7. The user may ask a follow-up question or close the assistant.
10. Subflows & Exceptional Flows	- E1. Reported emergency symptoms: The assistant advises the user to seek immediate medical assistance rather than relying on chatbot guidance. - E2. Unclear question: The system asks the user to clarify the question before providing specific guidance.

	- E3. Question outside scope: The system explains its limitations and recommends an appropriate healthcare professional. It does not provide invasive procedures, intravenous-fluid calculations, or other inpatient treatment instructions. - E4. No suitable approved guidance: The system states that it cannot provide a supported answer and offers applicable approved general safety advice when available. - E5. Incomplete or inconclusive assessment: The system explains the uncertainty and does not use rejected AI predictions as confirmed findings. - E6. Service unavailable: The system reports that the assistant is temporarily unavailable and offers access to available first-aid guidance and emergency information.

5.2. ACTIVITY DIAGRAMS

5.2.1. Sign In and Create Account (AD-01)

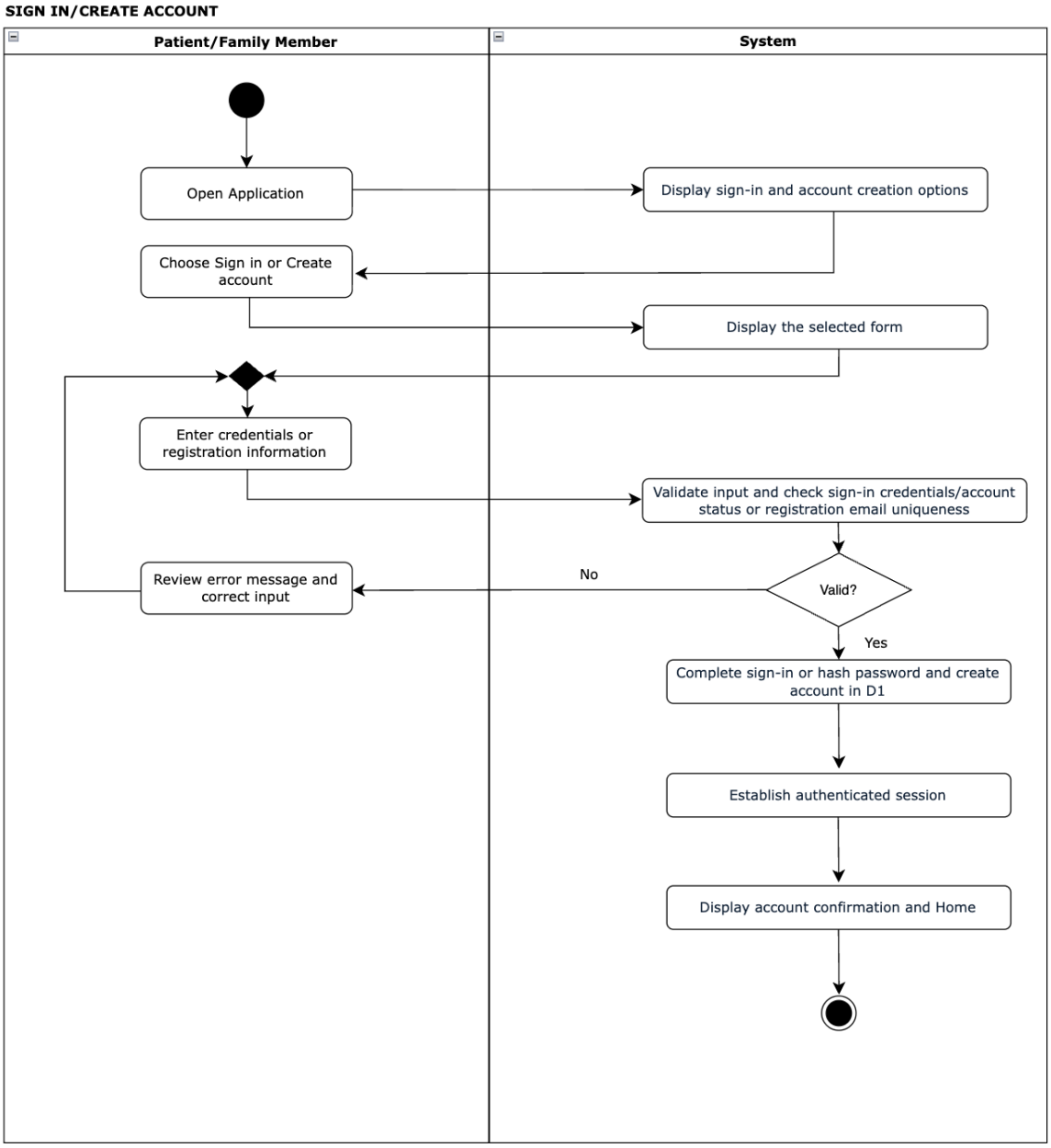


Figure 5.2. Sign In/Create Account Activity Diagram

The Sign In/Create Account activity diagram illustrates how users access the application through an existing or newly registered account. After selecting an option, the user completes the corresponding form. The system validates the input, checking credentials and account status for sign-in or email uniqueness for registration. Unsuccessful validation prompts the user to review the error and retry. When validation succeeds, the system completes sign-in or hashes the password and creates the account in D1. Both successful paths establish an authenticated session and direct the user to the Home screen.

5.2.2. View Assessment History (AD-02)

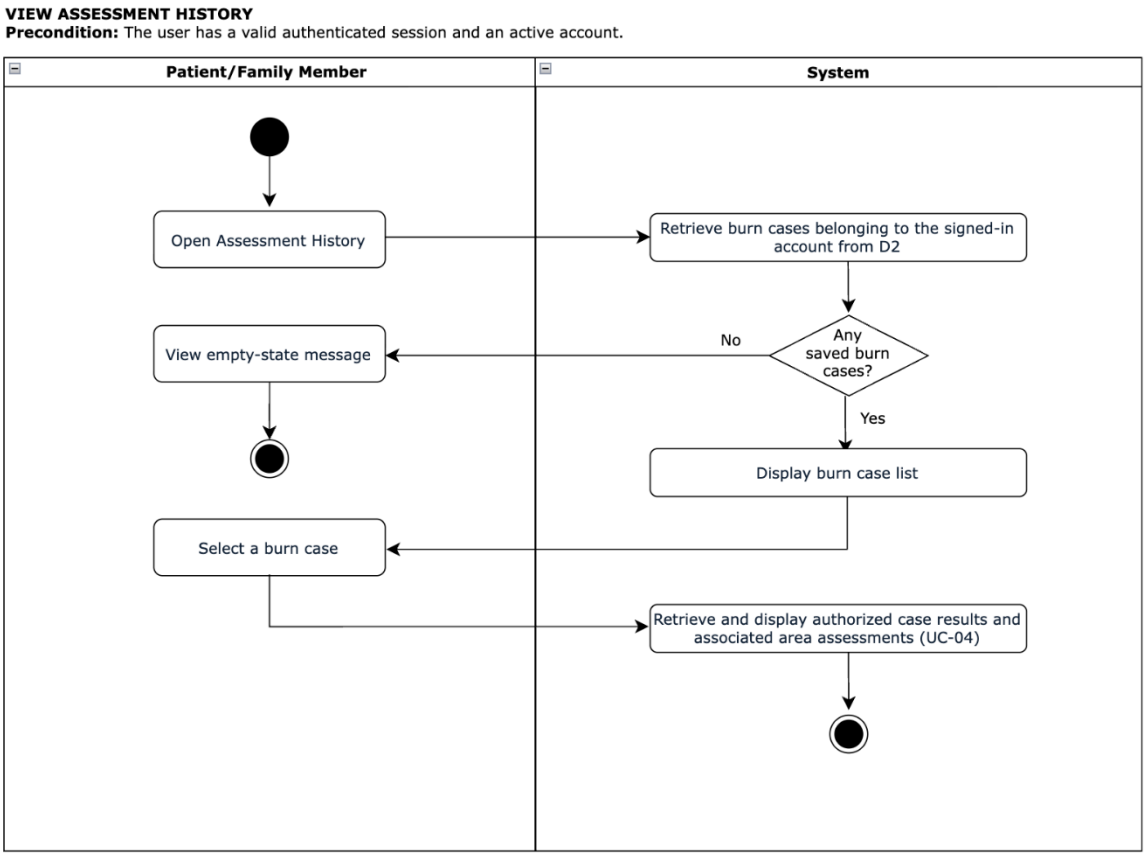


Figure 5.3. View Assessment History Activity Diagram

The View Assessment History activity diagram shows how a signed-in user accesses previously saved burn cases. The system retrieves cases belonging to the authenticated account and checks whether any records exist. If none are found, the user sees an empty-history message. Otherwise, the system displays the case list, allowing the user to select a case and view its authorized results and associated burned-area assessments through UC-04. This workflow keeps case-level and regional results connected while restricting access to the user's own records.

5.2.3. Receive Emergency Warning (AD-03)

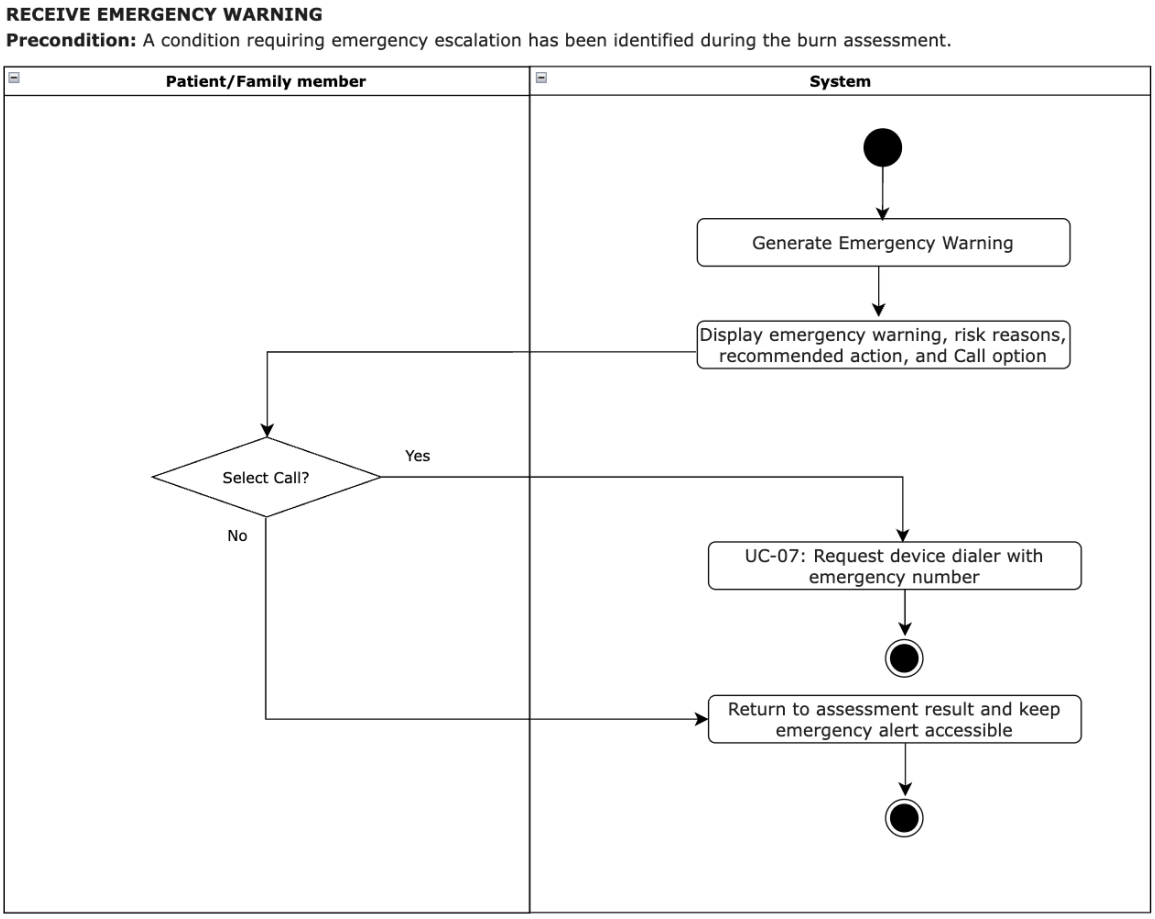


Figure 5.4. Receive Emergency Warning Activity Diagram

The Receive Emergency Warning activity diagram shows how the system alerts users when a burn assessment identifies a condition requiring emergency escalation. The system displays a warning, explains the risk factors, and provides recommended actions with a Call option. If the user selects Call, UC-07 requests the device dialer with the emergency number; the user decides whether to place the call. Otherwise, the system returns to the assessment result while keeping the emergency alert accessible. This workflow communicates urgent risks without automatically contacting emergency services.

5.2.4. View Assessment Result (AD-04)

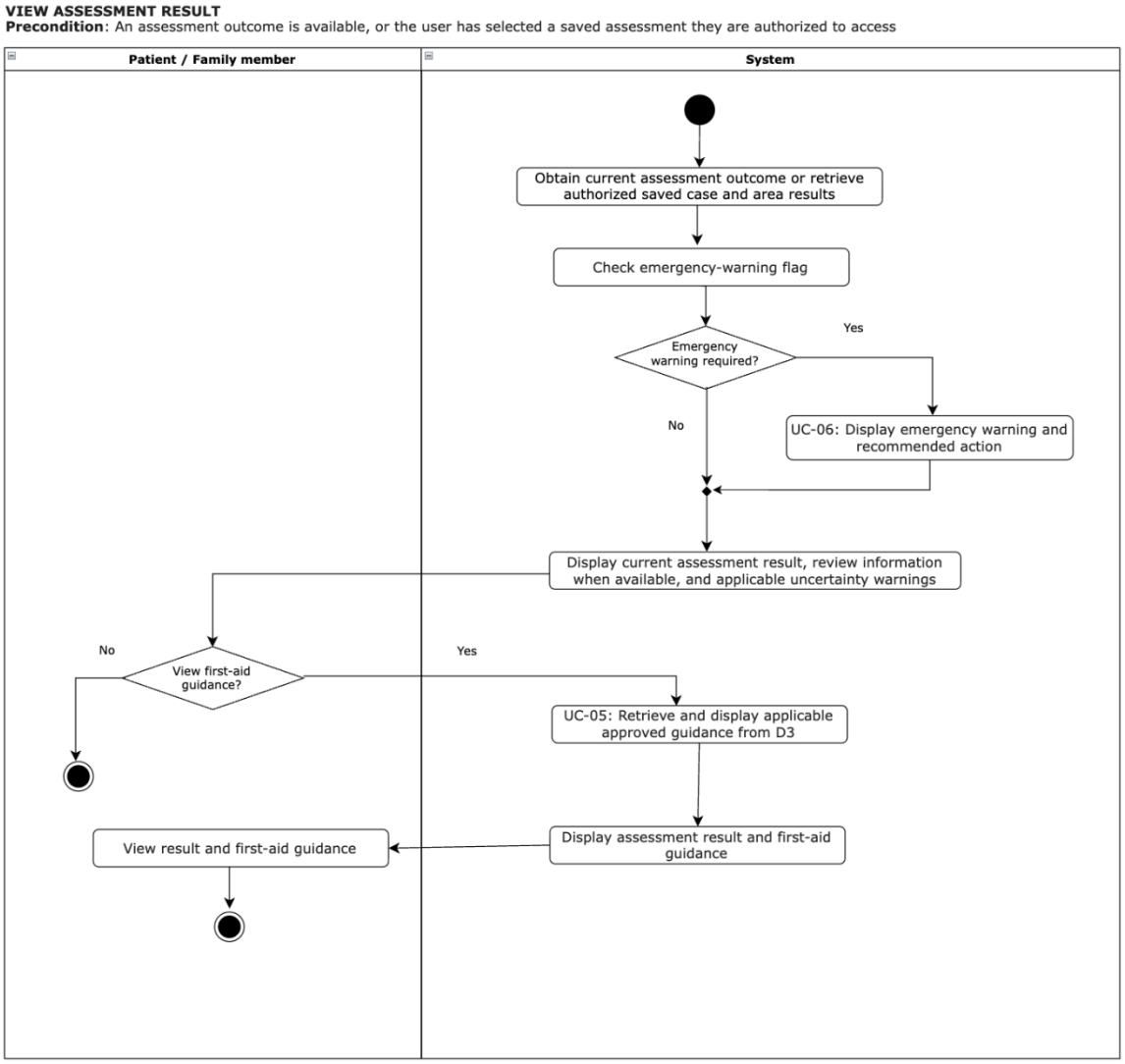


Figure 5.5. View Assessment Result Activity Diagram

The View Assessment Result activity diagram shows how users access a current or authorized saved burn assessment. The system checks whether an emergency warning is required and displays the warning and recommended action before presenting result details. Available case and regional results are shown with confidence information, Estimated TBSA, risk, and assessment limitations. Inconclusive results are identified, while ABSI remains hidden from users. The user may then view applicable approved first-aid guidance or leave the result screen. This workflow prioritizes urgent advice and communicates the preliminary nature of the assessment. When a reviewed outcome is available, it replaces the superseded AI prediction on the result screen and is labelled Medical Advisor-reviewed with the review date.

5.2.5. Submit Feedback (AD-05)

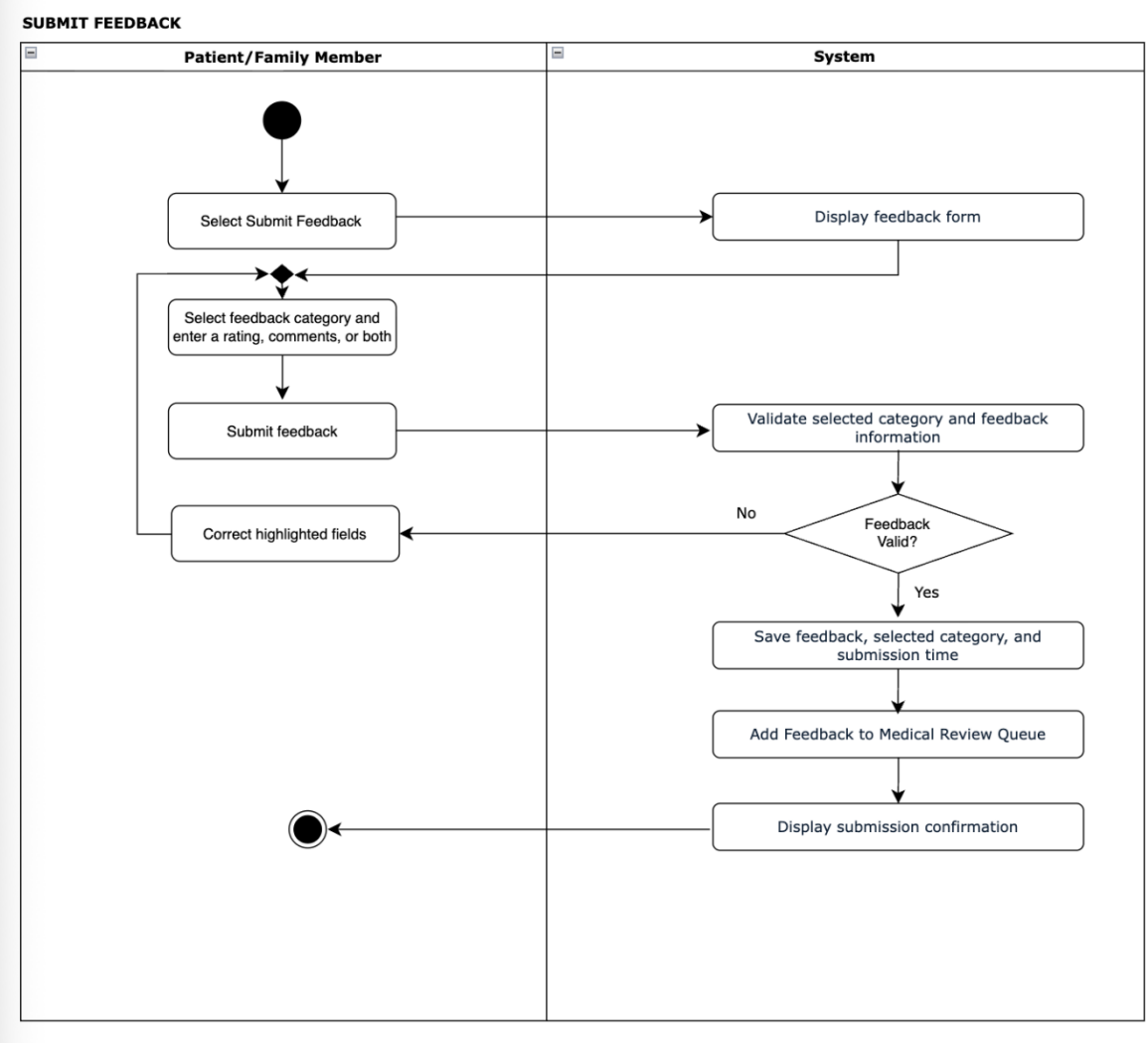


Figure 5.6. Submit Feedback Activity Diagram

The Submit Feedback activity diagram shows how users independently submit feedback by selecting a category and providing a rating, comments, or both. The system validates the submission and requests corrections when necessary. Valid feedback is saved with its category and submission time, added to the Medical Advisor review queue, and acknowledged with a confirmation message. This workflow organizes user feedback for subsequent review without automatically changing medical guidance or assessment results.

5.2.6. Perform Burn Assessment (AD-06)

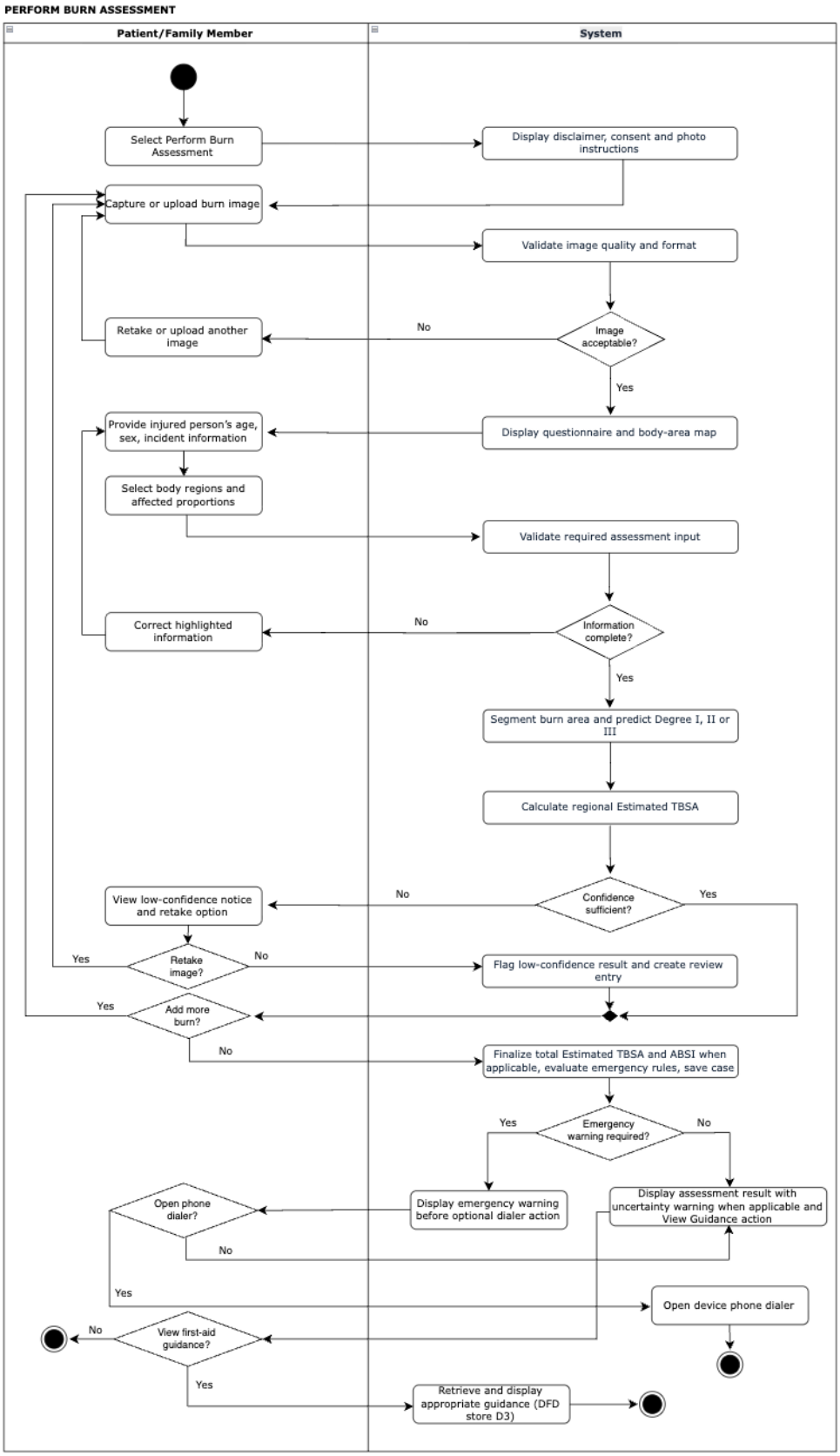


Figure 5.7. Perform Burn Assessment Activity Diagram

The Perform Burn Assessment activity diagram illustrates how users submit burn images and assessment information for preliminary evaluation. The system validates image quality and required inputs, then performs burn segmentation and depth prediction and calculates regional Estimated TBSA. Low-confidence predictions prompt a retake option; if the user continues, the system flags the result and creates a review entry. Users may assess additional regions before the system finalizes total Estimated TBSA and ABSI when applicable, evaluates emergency rules, and saves the case. Results retain uncertainty warnings where needed, while emergency cases receive prominent advice and an optional device-dialer action. Users may also view approved first-aid guidance.

5.2.7. Review User Feedback (AD-07)

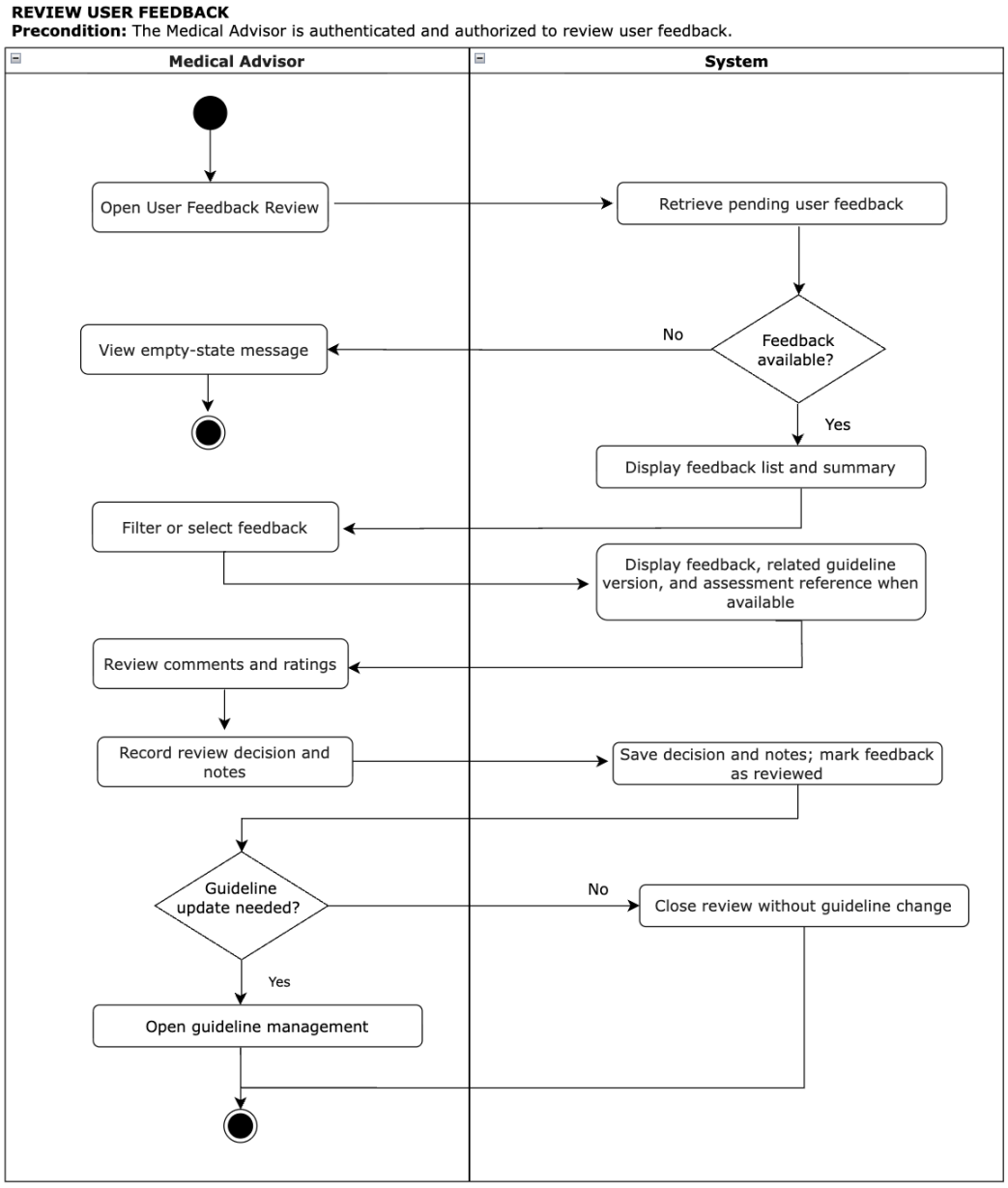


Figure 5.8. Review User Feedback Activity Diagram

The Review User Feedback activity diagram illustrates how an authorized Medical Advisor examines submitted feedback. The system retrieves pending entries and displays either an empty-state message or a feedback list. After selecting an entry, the advisor reviews its comments, ratings, and available guideline or assessment references, then records a decision and notes. The system saves the review and marks the feedback as reviewed. If a guideline update is needed, the advisor opens guideline management; otherwise, the review closes without changing the guidance. This workflow preserves review decisions while keeping feedback review separate from content updates.

5.2.8. Manage First-Aid Guidelines (AD-08)

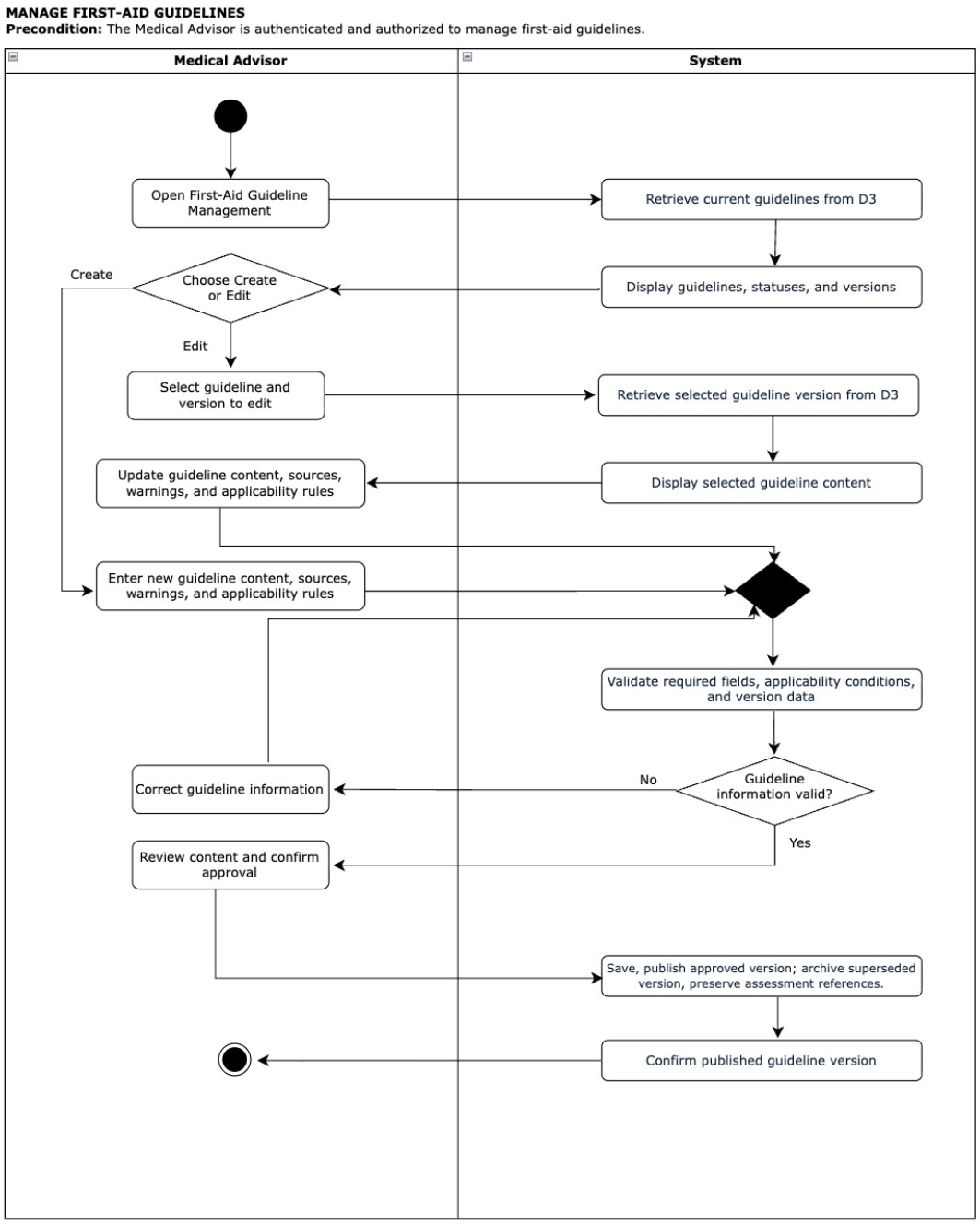


Figure 5.9. Manage First-Aid Guidelines Activity Diagram

The Manage First-Aid Guidelines activity diagram illustrates how an authorized Medical Advisor creates or updates guidance. The system retrieves existing guidelines and displays their statuses and versions. After the advisor enters or revises the content, sources, warnings, and applicability conditions, the system validates the information and requests corrections when necessary. Valid content is reviewed and approved by the advisor before the system saves and publishes the approved version. Superseded versions and their assessment references are preserved, supporting traceability while making updated guidance available to users.

5.2.9. Manage AI Models (AD-09)

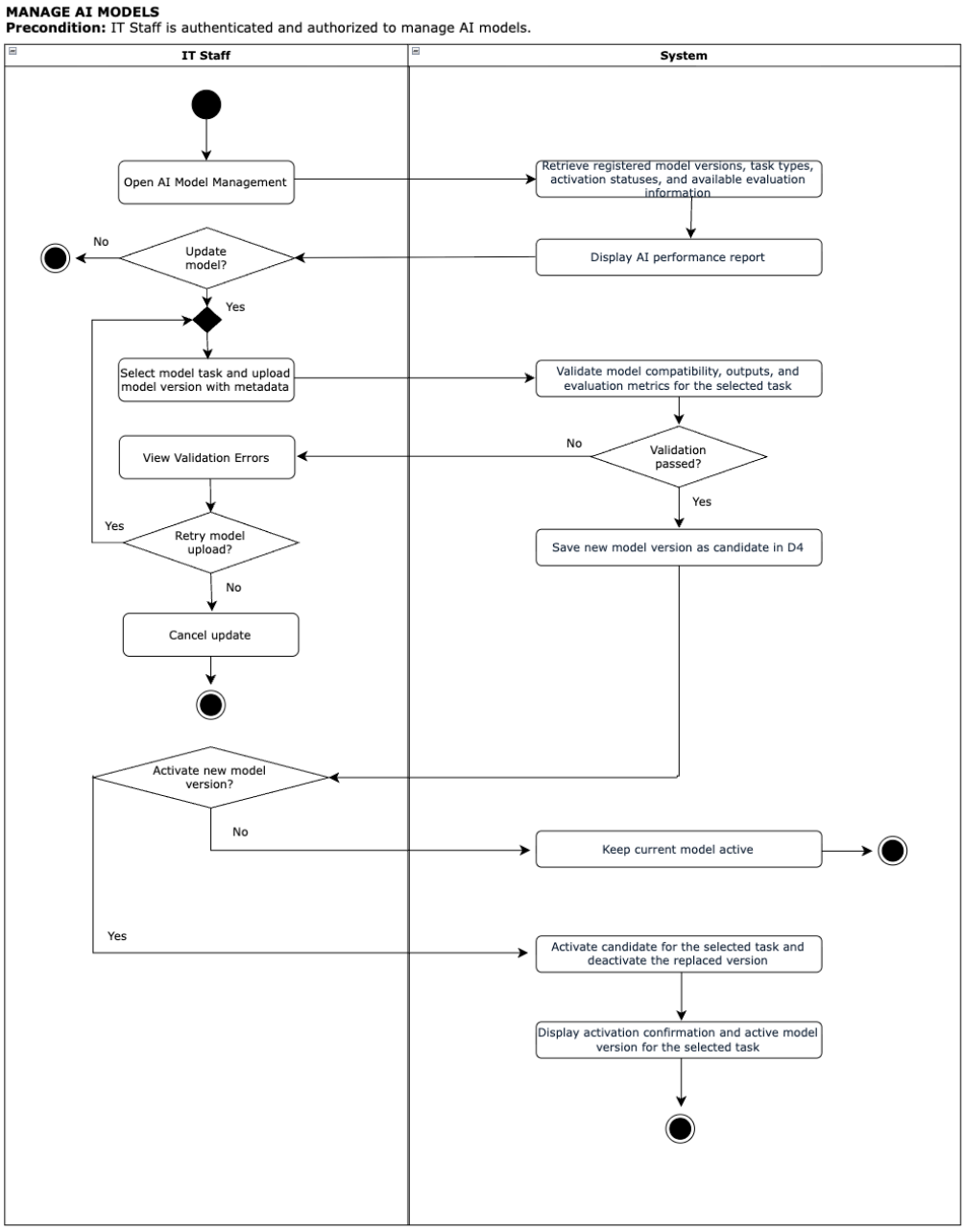


Figure 5.10. Manage AI Models Activity Diagram

The Manage AI Models activity diagram shows how authorized IT Staff review model performance and manage classification or segmentation model versions. When an update is needed, staff select the task and upload a model with its metadata. The system validates compatibility, outputs, and evaluation metrics, allowing staff to retry or cancel if validation fails. A validated model is saved as a candidate for optional activation. If activated, it replaces the active version for the selected task; otherwise, the current model remains active. The system then confirms the activation and identifies the active version.

5.2.10. Review Low-Confidence Cases (AD-10)

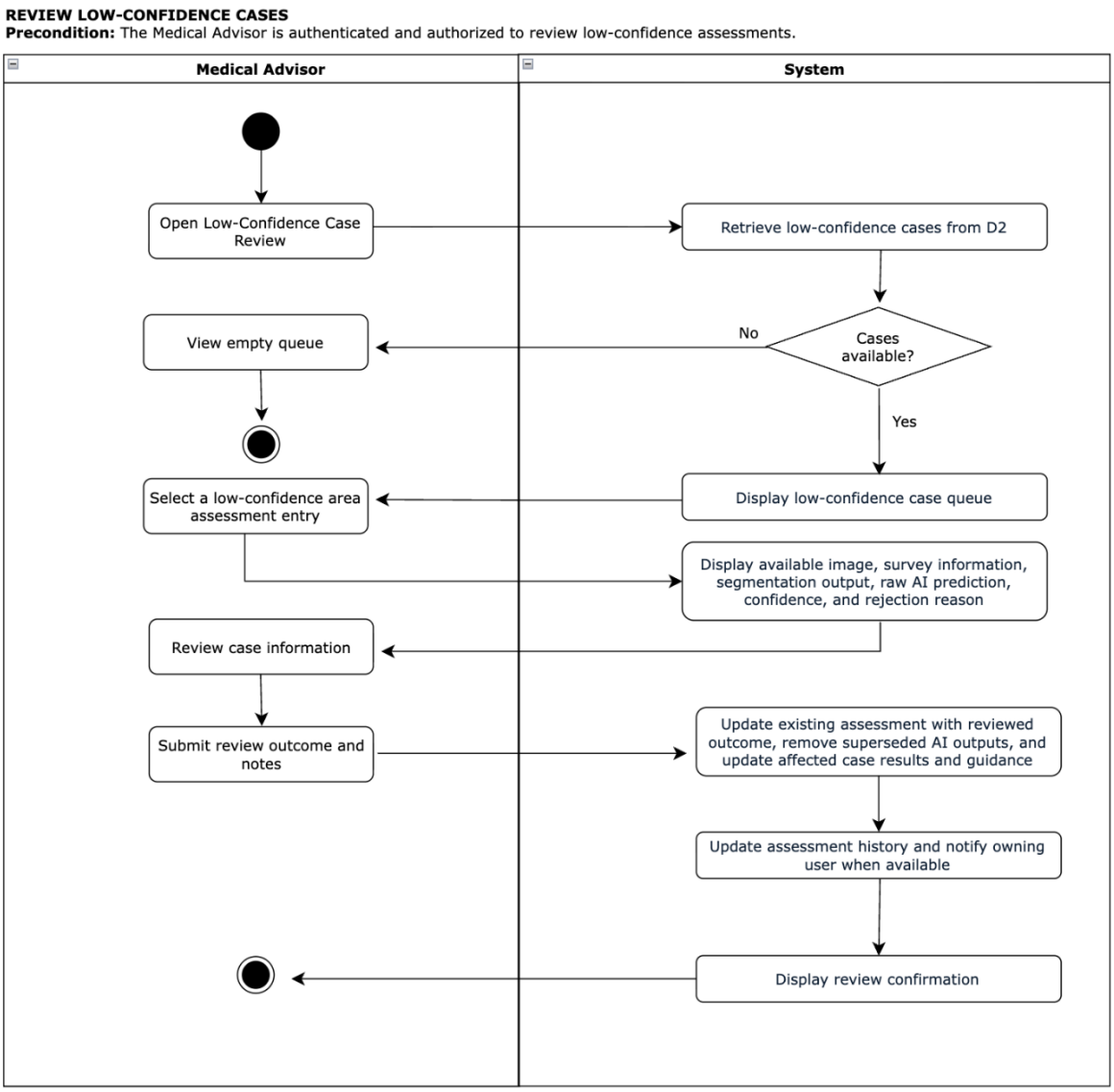


Figure 5.11. Review Low-confidence Cases Activity Diagram

The Review Low-Confidence Cases activity diagram shows how an authorized Medical Advisor reviews uncertain assessments. After selecting an entry, the advisor examines the available image, survey information, AI outputs, and rejection reason, then submits a review outcome and notes. When a conclusive outcome is successfully saved, the system updates the existing assessment while retaining its ID, removes superseded AI outputs, and updates

affected case results and guidance. It marks the review entry as reviewed, makes the updated result available in assessment history, notifies the owning user when available, and confirms completion to the advisor.

5.2.11. Manage User Account Status (AD-11)

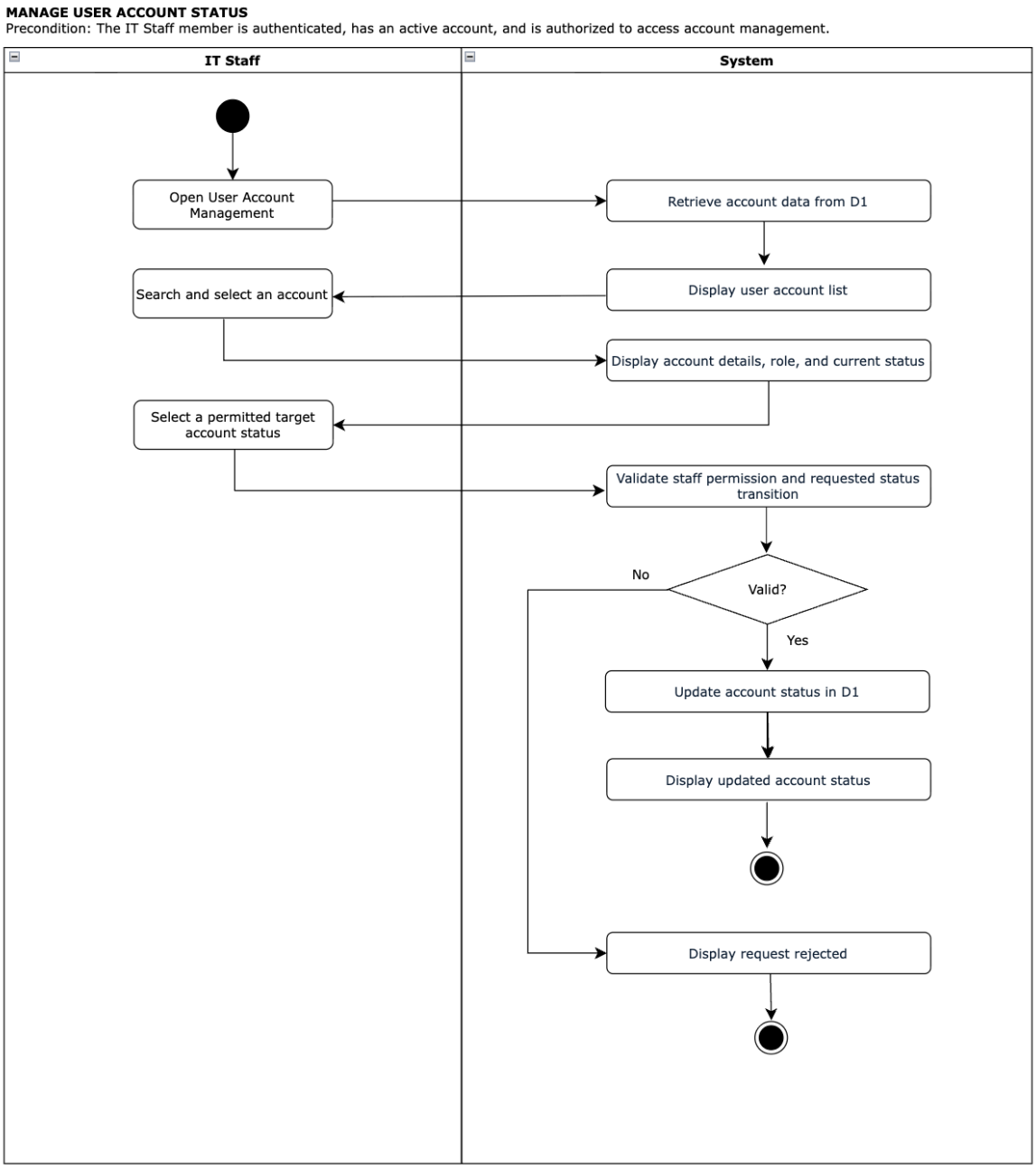


Figure 5.12. Manage User Account Status Activity Diagram

AD-11 describes how authorized IT Staff manage user account statuses. After opening account management, the staff member searches the account list and selects an account to inspect its details, role, and current status. The staff member then chooses a permitted target status. Before applying the change, the system checks both the staff member's permission and the validity of the requested transition. A valid request updates the account record in D1 and displays the new

status; an invalid request is rejected without changing the existing status. This workflow supports controlled account access and prevents unauthorized or disallowed status changes.

5.2.12. Manage Personal Profile (AD-12)

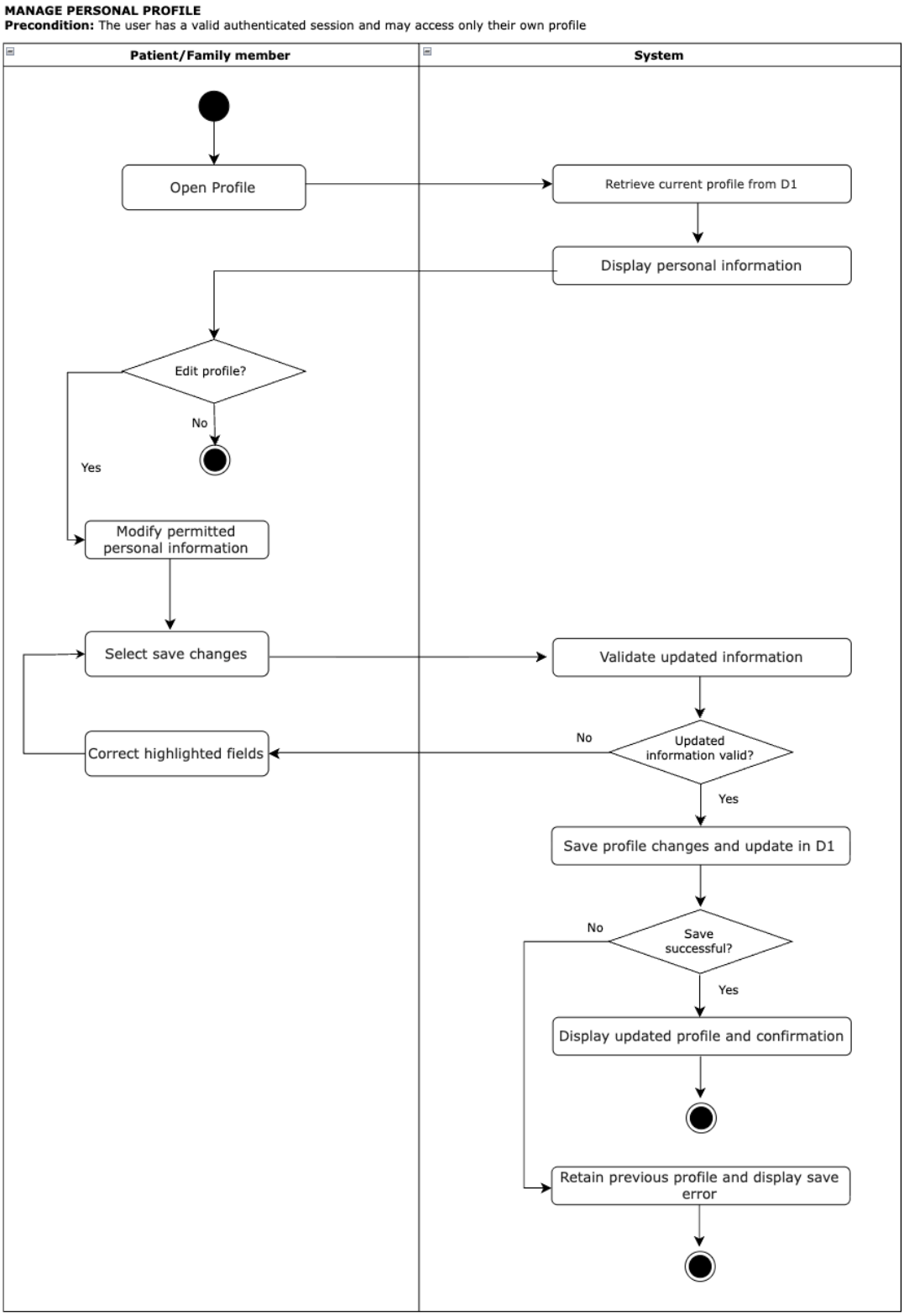


Figure 5.13. Manage Personal Profile Activity Diagram

The Manage Personal Profile activity diagram shows how an authenticated user views and edits their own personal information. The system retrieves the current profile from D1, allowing the user to review it or modify permitted details. Submitted changes are validated, and invalid fields are highlighted for correction. Once the information is valid, the system attempts to save the changes. A successful update displays the revised profile and confirmation, while a failed save preserves the previous profile and displays an error. This workflow helps maintain accurate profile information through controlled updates and validation.

5.2.13. Ask First-Aid Assistant (AD-13)

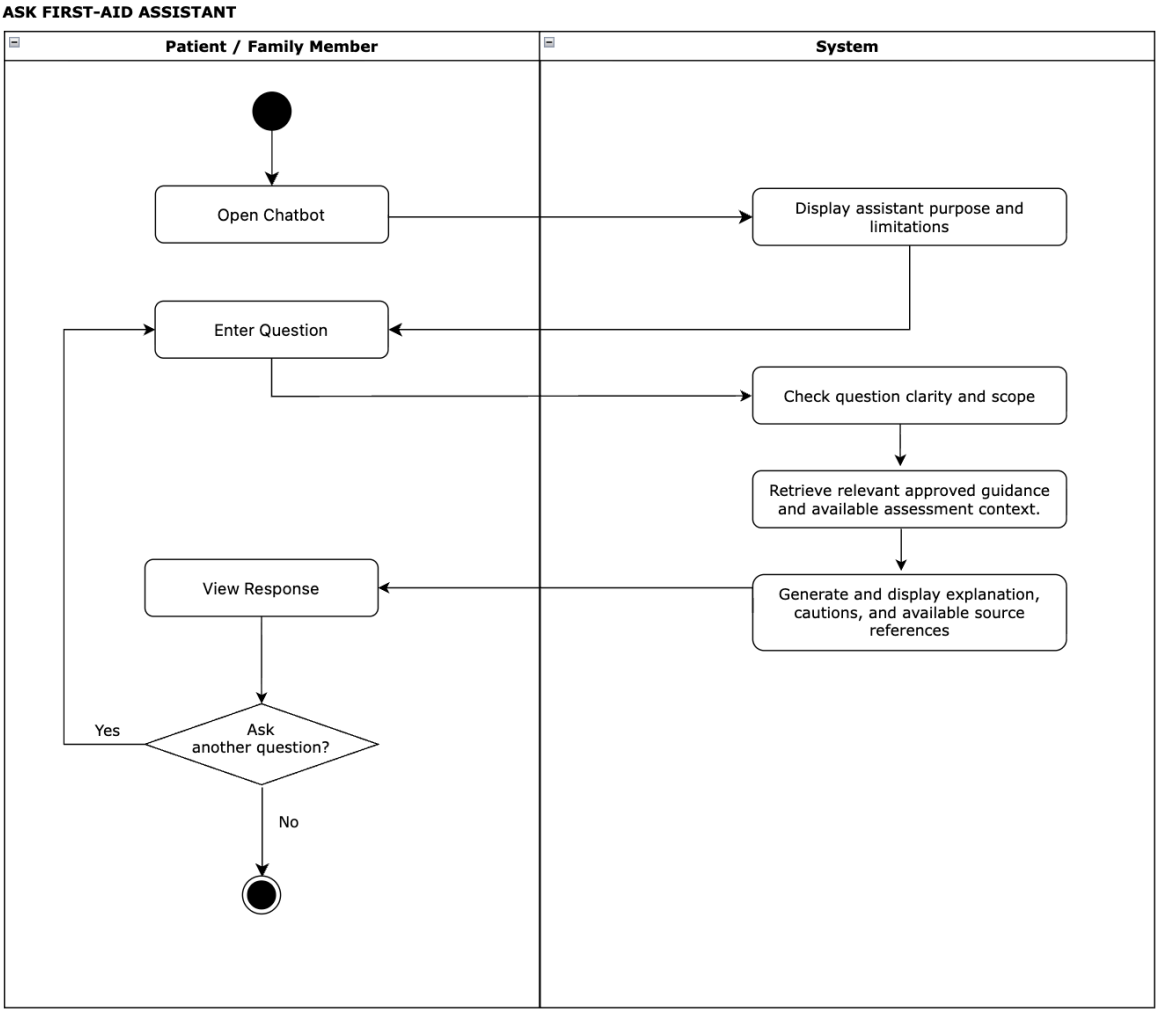


Figure 5.14. Ask First-Aid Assistant Activity Diagram

The Ask First-Aid Assistant activity diagram illustrates how users ask questions about burn first aid. After displaying the assistant's purpose and limitations, the system receives the question and checks its clarity and scope. It then retrieves relevant approved guidance and available assessment context to generate an explanation with cautions and source references when available. The user reads the response and may ask another question or end the interaction. This workflow supports access to first-aid information while communicating the assistant's limitations.

CHAPTER 6. DATABASE DESIGN

6.1. OVERVIEW AND PURPOSES

This section provides an overview of the database designed for BurnAid, a burn assessment system that helps adults check a burn injury from a phone. The primary purpose of this database is to establish a structured, consistent, and reliable repository for all system data, including burn cases, burned body regions, photos, AI results, first aid guidance, and clinician reviews.

The database is designed with a strong emphasis on data integrity through the use of Primary Keys, Foreign Keys, unique constraints, and validation rules. Reference data such as body regions and their standard %TBSA values is stored once in its own table, and the raw output of the AI models is kept even when a prediction is rejected. These mechanisms prevent data redundancy, maintain consistency across related entities, and keep an audit trail that clinicians can check later.

In addition, the database is organized in modules, such as accounts, burn cases, burned areas, guidelines, AI models, and feedback. A burn case can contain several burned areas, and guidelines and AI models are versioned, so the system can grow in the number of users, regions, and model versions without changing the structure.

By centralizing all operational data, the database serves as the single source of truth for the system. It directly supports the clinical survey, body region selection, AI assessment, the calculation of %TBSA and ABSI, first aid guidance, and the referral of each case to a doctor. The user only sees the total %TBSA, while the ABSI score and all other case information are sent to the doctor, and the database keeps these two views of the same case separate.

6.2. DELIVERABLES

This database design phase produces a set of key deliverables that define the complete data model of the system, from conceptual design to physical implementation.

- Identification of Entities, Relationships, and Attributes: A specification of all database entities is developed, including account, Burn_case, Burn_area_assessment, Body_region_reference, First_aid_guideline, Assessment_guideline, AI_model, Feedback_review_log, and Case_referral. For each entity, the attributes, data types, and constraints are defined, and the relationships between entities form the structural foundation of the database.
- Entity Relationship Diagram (ERD): A visual representation of the database structure illustrates the logical relationships between entities, and each table is explained through the process in which its attributes are created or updated.
- Data Normalization Analysis: The schema is analyzed against First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF) to remove redundant data and avoid insert, update, and delete anomalies.

- Physical Database Implementation Script (SQL): A SQL Server script creates all tables with their constraints and relationships, loads the standard body region data, and defines the views that separate what the user sees from what the doctor sees.

6.3. DATA DICTIONARY

6.3.1. Account

This table stores system accounts of patients, medical advisors, and IT staff.

Table 6.1. Entity Account

Attribute	Data Type	Constraint	Description
account_id	INT	PK, auto-increment	Unique account identifier
email	VARCHAR(150)	NOT NULL, UNIQUE	Login email
password_hash	VARCHAR(255)	NOT NULL	Hashed password
full_name	VARCHAR(100)	NOT NULL	Full name
phone_number	VARCHAR(20)	NULL	Contact phone number
date_of_birth	DATE	NULL	Date of birth
sex	VARCHAR(10)	NULL	Sex
role	VARCHAR(20)	NOT NULL, patient, medical_advisor, it_staff	User role
account_status	VARCHAR(20)	NOT NULL, active/ locked/deactivated	Account status
created_at	DATETIME	NOT NULL	Account creation timestamp
updated_at	DATETIME	NULL	Timestamp of the most recent profile update

6.3.2. Burn_case

This table atores one patient encounter, including the survey answers, the emergency pre-screen answers, and the case-wide results such as total %TBSA and ABSI.

Table 6.2. Entity Burn_case

Attribute	Data Type	Constraint	Description

case_id	INT	PK, auto-increment	Unique identifier for the whole burn case (one patient encounter)
account_id	INT	FK (account.account_id), NULL	Owning account; NULL for a guest user
age_at_assessment	INT	NOT NULL, >= 18	Patient age (>= 18)
sex_at_assessment	VARCHAR(10)	NOT NULL, male/female/other/unspecified	Biological sex
Burn_cause	VARCHAR(20)	NOT NULL, thermal	Cause of the burn (strictly thermal: hot liquid, flame, steam, hot surface)
injury_time	DATETIME	NULL	Time the accident occurred
smoke_exposure_reported	BOOLEAN	NOT NULL	Reported smoke/toxic-fume exposure
respiratory_symptoms	BOOLEAN	NOT NULL	Reported respiratory symptoms
suspected_inhalation_injury	BOOLEAN	NOT NULL	TRUE when smoke_exposure_reported or respiratory_symptoms is TRUE
total_tbsa_percent	DECIMAL(5,2)	NULL until completed, 0 to 100	Sum of tbsa_percent of all areas in the case; the only score shown to the user
absi_score	INT	NULL, 2 to 18	ABSI score of the whole case; sent to the doctor only
absi_risk_category	VARCHAR(50)	NULL	ABSI risk category; sent to the doctor only
absi_not_applicable_reason	VARCHAR(50)	NULL, case_incomplete,	Reason why ABSI was not computed

		tbsa_zero_superficial_only, sex_not_binary	
overall_risk_level	VARCHAR(20)	NOT NULL, DEFAULT 'undetermined', undetermined, low, moderate, high, emergency	Overall risk level for the case
medical_escalation_level	VARCHAR(50)	NOT NULL, DEFAULT 'undetermined', undetermined, none, routine, urgent, emergency	Case-wide escalation routing, medical alert level
show_emergency_call_prompt	BOOLEAN	NOT NULL, DEFAULT FALSE	Flag to display the call-115 prompt
emergency_reasons	JSON	NULL	Array of risk-flag codes explaining the escalation decision
case_status	VARCHAR(20)	NOT NULL, DEFAULT 'draft', draft, in_progress, completed, inconclusive	Case status, derived from the status of its areas
finalized_at	DATETIME	NULL	Timestamp when case-level results (ABSI, overall risk) were finalized
created_at	DATETIME	NOT NULL	Case creation timestamp

6.3.3. Burn_area_assessment

This table stores one burned region of a case, with its photo, AI results, and estimated TBSA. One case has one or more areas.

Table 6.3. Entity Burn_area_assessment

Attribute	Data Type	Constraint	Description

area_assessment_id	INT	PK, auto-increment	Unique identifier for this region's assessment
case_id	INT	FK (Burn_case.case_id), NOT NULL	Parent case this area belongs to
region_code	VARCHAR(30)	NOT NULL, FK (Body_region_reference.region_code)	Selected body region
affected_proportion_percent	DECIMAL(5, 2)	NOT NULL, 1 to 100	Percentage of the region that is burned
Burn_image_path	VARCHAR(255)	NULL	Path of the captured photo; NULL until captured or after deletion
Burn_mask_path	VARCHAR(255)	NULL	URL of the segmentation mask image
image_deleted_at	DATETIME	NULL	Timestamp the image was deleted
image_retention_status	VARCHAR(20)	NOT NULL, DEFAULT 'pending', pending, active, deleted, archived	Image retention status
quality_status	VARCHAR(20)	NOT NULL, pending, acceptable, rejected	Image quality status
quality_issues	JSON	NULL	Detailed reasons for image rejection
quality_check_model_id	INT	FK (AI_model.model_id), NULL	Image quality checker model used

area_status	VARCHAR(20)	NOT NULL, DEFAULT 'draft', draft, processing, completed, inconclusive	This area's own lifecycle status
model_predicted_degree	VARCHAR(10)	NULL, degree_1, degree_2, degree_3	Raw AI burn depth, kept even when rejected
model_confidence_score	DECIMAL(5, 4)	NULL, 0 to 1	Raw AI confidence, kept even when rejected
classification_model_id	INT	FK (AI_model.model_id), NULL	Classification model used
segmentation_model_id	INT	FK (AI_model.model_id), NULL	Segmentation model used
threshold_used	DECIMAL(5, 4)	NULL	Confidence threshold applied at decision time
prediction_decision	VARCHAR(10)	NULL, accepted, rejected	Whether the raw AI result was accepted
rejection_reason	VARCHAR(100)	NULL	Reason for rejection, for example low_confidence
predicted_degree	VARCHAR(10)	NULL until accepted, degree_1, degree_2, degree_3, None	Burn depth shown to the user
confidence_score	DECIMAL(5, 4)	NULL until accepted, 0 to 1	Confidence shown with the result (0.8750 = 87.50%)
is_low_confidence	BOOLEAN	NOT NULL, DEFAULT FALSE	Flags a low-confidence image/AI result

tbsa_percent	DECIMAL(5, 2)	NULL until completed, 0 to 100	TBSA of this region (0 for degree_1)
clinical_review_status	VARCHAR(20)	NOT NULL, DEFAULT 'pending', pending, approved, flagged, rejected	Doctor review of the AI result for this area
assessed_at	DATETIME	NULL until completed	Time the area assessment was completed

6.3.4. Body_region_reference

This table stores the standard body regions and their reference %TBSA values.

Table 6.4. Entity Body_region_reference

Attribute	Data Type	Constraint	Description
region_code	VARCHAR(30)	PK	Body region code, for example ARM_TOTAL_L or CHEST
region_name	VARCHAR(50)	NOT NULL	Region name shown in the UI
side	VARCHAR(10)	NOT NULL, left, right, none	Body side, encoded in region_code
reference_tbsa_percent	DECIMAL(5,2)	NOT NULL	Standard %TBSA of the whole region (Le The Trung, 1999)
is_critical_priority	BOOLEAN	NOT NULL, DEFAULT FALSE	TRUE for face, hands, feet and major joints; always escalated

6.3.5. First_aid_guideline

This table stores versioned first aid guidelines written and approved by Medical Advisors.

Table 6.5. Entity First_aid_guideline

Attribute	Data Type	Constraint	Description
guideline_id	INT	PK	Unique identifier
guideline_code	VARCHAR(50)	NOT NULL	Stable identifier code

version	INT	NOT NULL	Version number
body_region_code	VARCHAR(30)	FK (Body_region_reference.region_code), NULL	Region the guideline applies to; NULL for a general guideline
Burn_degree	VARCHAR(10)	degree_1, degree_2, degree_3	Applicable burn degree
title	VARCHAR(150)	NOT NULL	Title
content	TEXT	NOT NULL	Full guideline text
warnings	TEXT	NULL	Warnings
source_reference	TEXT	NULL	Source reference
applicability_rules	TEXT	NULL	Activation conditions
status	VARCHAR(20)	draft, approved, archived	Guideline status
created_by	INT	FK (account)	Authoring clinician
approved_by	INT	FK (account)	Approving clinician
approved_at	DATETIME	NULL	Approval timestamp
updated_at	DATETIME	NULL	Last update timestamp

6.3.6. Assessment_guideline

This table is an associative entity which links burned areas with the guidelines shown for them.

Table 6.6. Entity Assessment_guideline

Attribute	Data Type	Constraint	Description
Assessment_guideline_id	INT	PK, auto-increment	Link identifier
area_assessment_id	INT	FK (Burn_area_assessment.area_assessment_id), NOT NULL	Area this guideline set applies to
guideline_id	INT	FK (First_aid_guideline.	Guideline version reference

		guideline_id), NOT NULL	
display_order	INT	NOT NULL	Display priority order
assigned_at	DATETIME	NOT NULL	Assignment timestamp

6.3.7. AI_model

This table stores the registered AI models, including classification, segmentation, and image quality models.

Table 6.7. Entity AI_model

Attribute	Data Type	Constraint	Description
model_id	INT	PK	Model identifier
task_type	VARCHAR(30)	NOT NULL, Burn_classification, Burn_segmentation, image_quality_check	Classification, segmentation or image quality check
architecture	VARCHAR(50)	NOT NULL	Model architecture, for example ResNet50 or YOLOv8
training_strategy	VARCHAR(100)	NULL	e.g. transfer learning + fine-tune, augmentation recipe used
dataset_version	VARCHAR(50)	NOT NULL	Version/tag of the training dataset used
version	VARCHAR(50)	NOT NULL	Model version/tag
decision_threshold	DECIMAL(5,4)	NULL	Confidence threshold used to accept or reject predictions
evaluation_metrics	JSON	NULL	Evaluation metrics
model_size	VARCHAR(20)	NULL	Model artifact size (e.g. '45MB') or parameter count

inference_time_ms	INT	NULL	Average inference time in milliseconds
file_path	VARCHAR(255)	NOT NULL	Stored file path
is_active	BOOLEAN	NOT NULL	Currently deployed flag
uploaded_by	INT	FK (account)	IT staff who registered the model
uploaded_at	DATETIME	NOT NULL	Registration timestamp
activated_at	DATETIME	NULL	Activation timestamp

6.3.8. Feedback_review_log

This table stores user feedback and low-confidence cases that need a review.

Table 6.8. Entity Feedback_review_log

Attribute	Data Type	Constraint	Description
review_entry_id	INT	PK, auto-increment	Entry identifier
account_id	INT	FK (account.account_id), NULL	Submitting user; NULL for a guest
area_assessment_id	INT	FK (Burn_area_assessment.area_assessment_id), NULL	Related burned area
guideline_id	INT	FK (First_aid_guideline.guideline_id), NULL	Related guideline
entry_type	VARCHAR(20)	NOT NULL, user_feedback, low_confidence_case	Entry type
rating	INT	NULL, 1 to 5	Star rating
comment	TEXT	NULL	Feedback text
reviewed_by	INT	FK (account.account_id), NULL	Reviewing doctor

review_decision	VARCHAR(100)	NULL	Clinician decision
review_notes	TEXT	NULL	Review notes
status	VARCHAR(20)	NOT NULL, DEFAULT 'pending', pending, reviewed	Status of the feedback entry
submitted_at	DATETIME	NOT NULL	Entry creation timestamp
reviewed_at	DATETIME	NULL	Review completion timestamp

6.3.9. Case_referral

This table stores the case package sent to a doctor, which includes the ABSI score that the user does not see.

Table 6.9. Entity Case_referral

Attribute	Data Type	Constraint	Description
referral_id	INT	PK, auto-increment	Referral identifier
case_id	INT	FK (Burn_case.case_id), NOT NULL, UNIQUE	Case sent to the doctor
assigned_to	INT	FK (account.account_id), NULL	Doctor who takes the case; NULL until assigned
sent_at	DATETIME	NOT NULL	Time the case was sent, created when the case is finalized
referral_status	VARCHAR(20)	NOT NULL, DEFAULT 'sent', sent, viewed, reviewed	Doctor handling status of the whole case
viewed_at	DATETIME	NULL	Time the doctor first opened the case
doctor_notes	TEXT	NULL	Doctor's comments on the case
reviewed_at	DATETIME	NULL	Time the doctor finished the review

6.3.10. Relationship Table

Table 6.10. Entity Relationship

Parent Entity	Child Entity	Relationship	Note
Account	Burn_case	0..1 - N	One account may own many cases. A guest case has no account.
Burn_case	Burn_area_assessment	1 - 1..N	One case contains one or more burned areas.
Burn_case	Case_referral	1 - 0..1	One finalized case is sent to a doctor once.
Account	Case_referral	0..1 - N	One Medical Advisor may take many referrals.
Body_region_reference	Burn_area_assessment	1 - N	One region may be used by many burned areas.
Body_region_reference	First_aid_guideline	0..1 - N	One region may have many guidelines. A general guideline has no region.
Burn_area_assessment	Assessment_guideline	1 - N	One burned area may receive many guidelines.
First_aid_guideline	Assessment_guideline	1 - N	One guideline may be assigned to many burned areas.
AI_model	Burn_area_assessment	1 - N	One model may process many areas as a classification, segmentation, or image quality model (three foreign keys).
Account	AI_model	1 - N	One IT Staff account may upload many models (uploaded_by).
Account	First_aid_guideline	1 - N	One Medical Advisor may write and approve many guidelines (created_by, approved_by).
Account	Feedback_review_log	1 - N	One account may submit or review many entries

			(account_id, reviewed_by).
Burn_area_assessment	Feedback_review_log	1 - N	One burned area may receive many feedback entries.
First_aid_guideline	Feedback_review_log	1 - N	One guideline may receive many feedback entries.

6.4. ENTITY RELATIONSHIP DIAGRAM (ERD)

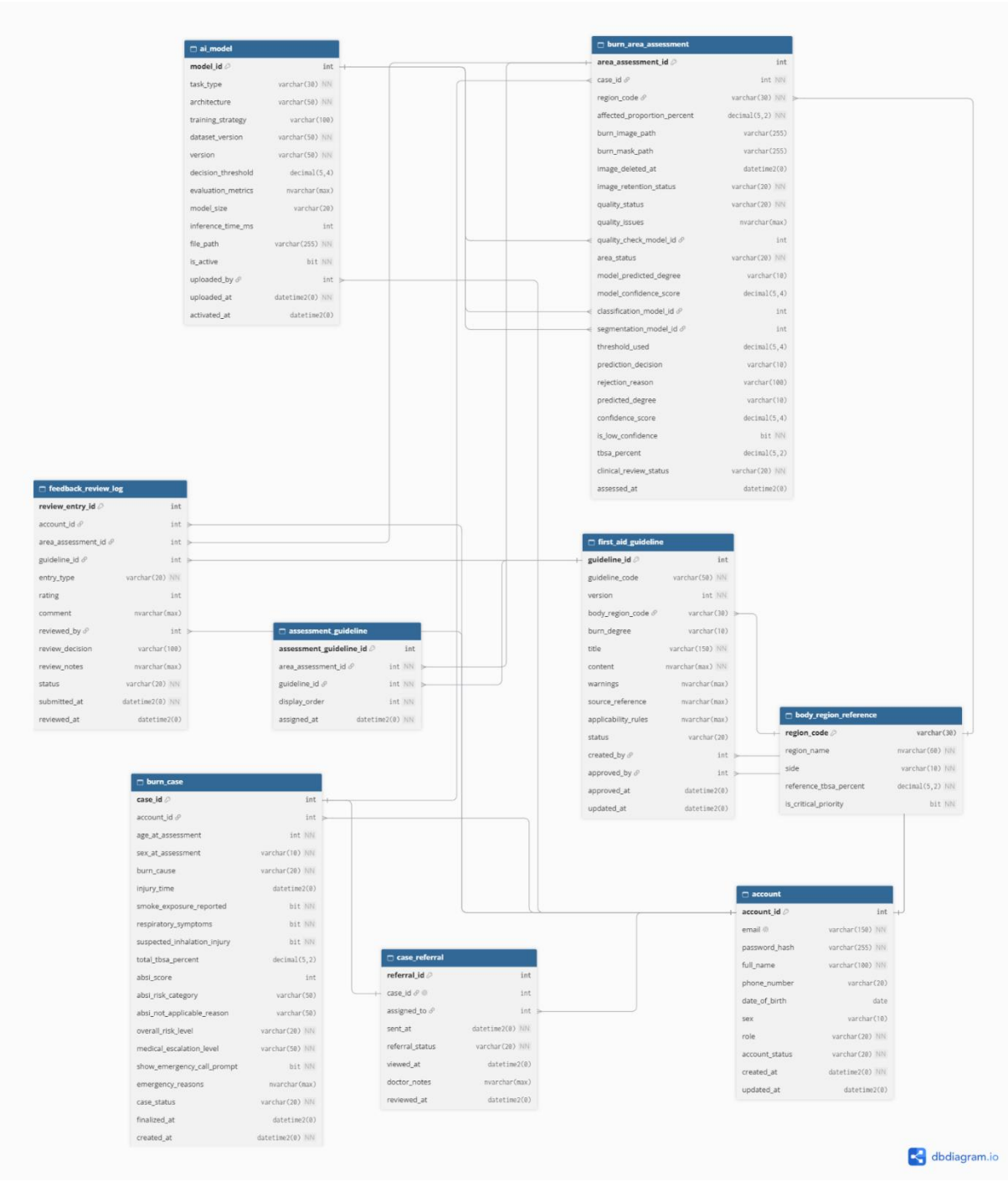


Figure 6.1. Entity Relationship Diagram (each box shows the keys and main attributes; the complete list is in Table 1 to Table 9)

6.4.1. Account Management Process:

This process manages the accounts of patients, Medical Advisors, and IT Staff. It involves the *Account* table. When a person registers, the system inserts *email, password_hash, full_name, phone_number, date_of_birth,* and *sex*, and sets *role*, *account_status,* and

created_at. Later changes to the profile update *updated_at.* A person may also use the app as a guest, so *account_id* in *Burn_case* is optional. The accounts of Medical Advisors and IT Staff appear again in other tables as *created_by* and *approved_by* in *First_aid_guideline, assigned_to* in *Case_referral, reviewed_by* in *Feedback_review_log*, and *uploaded_by* in *AI_model*.

6.4.2. Clinical Survey and Emergency Pre-screen Process:

This process starts every assessment and mainly involves the *Burn_case* table. When the user begins the survey, the system inserts a new record with *account_id* (if the user is logged in), and the answers fill *age_at_assessment, sex_at_assessment, Burn_cause, injury_time, smoke_exposure_reported,* and *respiratory_symptoms*. The system then sets *suspected_inhalation_injury* from the smoke and breathing answers. The clinical survey focuses strictly on thermal burns (scald, flame, hot contact) and records age, sex, burn cause, injury time, and inhalation injury signs (smoke exposure and respiratory symptoms) as structured fields, so the urgency rules do not depend on free text. The new case starts with *case_status* set to draft.

6.4.3. Body Region Selection and Photo Capture Process:

This process involves *Body_region_reference, Burn_area_assessment,* and *AI_model*. When the user touches the body map, the system reads *region_code, region_name,* and *side* from Body_region_reference. After the user confirms the region and moves the slider, the system inserts a record into Burn_area_assessment with *case_id, region_code*, and *affected_proportion_percent,* and changes *case_status* to in_progress. Scalp, buttocks, perineum, and genitalia are blocked at this step, so no record is created for them. If the user has several burns, the same step is repeated, and every new record uses the same *case_id*. After the photo is taken, the system stores *Burn_image_path* and *image_retention_status*, and the image quality check writes *quality_status, quality_issues,* and *quality_check_model_id*. The *area_status* then changes from draft to processing.

6.4.4. AI Assessment and TBSA Calculation Process:

This process involves *AI_model, Burn_area_assessment, Body_region_reference, and Feedback_review_log.* Before it can run, IT Staff register each model in AI_model with *task_type, architecture, dataset_version, version, decision_threshold,* and *is_active*. During the assessment, the segmentation model writes *segmentation_model_id* and *Burn_mask_path,* and the classification model writes *classification_model_id, model_predicted_degree, model_confidence_score, threshold_used,* and *prediction_decision*. These raw values are always kept. If the prediction is accepted, the system fills *predicted_degree* and *confidence_score,* reads *reference_tbsa_percent* from Body_region_reference, and calculates *tbsa_percent* as (*reference_tbsa_percent* × *affected_proportion_percent*)/100, with 0 for Degree I. It then sets *assessed_at* and changes *area_status* to completed. If the prediction is rejected, the system stores *rejection_reason* and *is_low_confidence*, changes area_*status* to inconclusive, leaves *predicted_degree* and *confidence_score* empty so that nothing wrong is

shown, and inserts a record into *Feedback_review_log* with *entry_type* set to low_confidence_case and the related *area_assessment_id*.

6.4.5. Case Scoring and Doctor Referral Process:

This process involves *Burn_case, Body_region_reference*, and *Case_referral*, and it runs after every area of the case has finished. The Safety Rule Engine adds up *tbsa_percent* of all areas into *total_tbsa_percent,* then calculates *absi_score* and *absi_risk_category*. When ABSI cannot be scored, *absi_not_applicable_reason* explains why: the case is incomplete, the total TBSA is 0, or the sex is other or unspecified. The engine also uses *is_critical_priority* from Body_region_reference to set *overall_risk_level, medical_escalation_level, show_emergency_call_prompt,* and *emergency_reasons,* and the case ends with *case_status* set to completed and *finalized_at* filled in. The user app only reads *total_tbsa_percent* (and the *tbsa_percent* of each area), the risk level, and the urgency prompt. The ABSI score is never shown to the user. Instead, the system inserts a record into Case_referral with *case_id, sent_at,* and *referral_status* set to sent, and the doctor receives the ABSI score together with all other case information, such as the survey answers, the burned regions, the photos, and the raw and accepted AI results. When the doctor opens the case, the system sets *assigned_to, viewed_at*, and *referral_status* to viewed, and after the review, the doctor writes *doctor_notes* and *reviewed_at,* and *referral_status* becomes reviewed. This design keeps the ABSI prognosis, which needs clinical interpretation, in the hands of a doctor.

6.4.6. First Aid Guidance Process:

This process involves *First_aid_guideline* and *Assessment_guideline*. Medical Advisors write guidelines with *guideline_code*, *version*, *title*, *content*, *warnings*, *source_reference*, *applicability_rules*, *body_region_code*, and *Burn_degree*, and the system saves *created_by*. A guideline starts with *status* set to draft and becomes approved when a Medical Advisor approves it, which fills *approved_by* and *approved_at*. Only approved guidelines are shown. When a burned area is completed, the system matches approved guidelines by *body_region_code* and *Burn_degree*, and inserts one record into Assessment_guideline for each match with *area_assessment_id*, *guideline_id*, *display_order*, and *assigned_at*. The same guideline cannot be assigned twice to the same area.

6.4.7. Feedback and Clinical Review Process:

This process involves *Feedback_review_log* and *Burn_area_assessment*. After seeing a result or a guideline, the user may submit feedback, and the system inserts a record with *entry_type* set to *user_feedback, rating, comment, submitted_at*, the related *area_assessment_id* or *guideline_id*, and *account_id* when the user is logged in. A doctor reviews low-confidence cases and feedback entries by writing *reviewed_by, review_decision, review_notes*, and *reviewed_at,* and changing the status from pending to reviewed. The doctor can also judge the AI result of a single area, which updates *clinical_review_status* in *Burn_area_assessment*. Therefore, status tracks a feedback entry, *clinical_review_status*

tracks the review of one AI result, and *referral_status* in *Case_referral tracks* the doctor's handling of the whole case.

6.5. PHYSICAL DATABASE IN SQL SERVER

The physical database of BurnAid was implemented in SQL Server, in a database named BurnAidDB, based on the final ERD and the data dictionary. A single script, run in SQL Server Management Studio (SSMS), creates the nine tables in the order of their foreign keys, so every foreign key points to a table that already exists. The script defines the Primary Keys, Foreign Keys, unique constraints, CHECK constraints, default values, and indexes for the common joins, and it loads the 22 standard body regions into *Body_region_reference*.

The data types of the data dictionary were mapped to SQL Server types as follows: BOOLEAN to BIT, JSON and TEXT to NVARCHAR(MAX), DATETIME to DATETIME2(0), and auto-increment keys to INT IDENTITY(1,1). JSON columns also have a CHECK with ISJSON(), so only valid JSON can be stored.

Most business rules are enforced directly by CHECK constraints, so wrong data is rejected even if the application has a bug. Status and category columns accept only their listed values, and *age_at_assessment* must be at least 18. The value of *suspected_inhalation_injury* must match *smoke_exposure_reported* and *respiratory_symptoms*. A completed burn area must have predicted_degree, confidence_score, tbsa_percent, assessed_at, and a classification_model_id, while these fields must stay empty until completion. The raw AI values are always kept, and predicted_degree and *confidence_score* must equal them when the prediction is accepted. The ABSI score, its category, and the reason why it is missing must agree with each other. Rules that depend on several rows, such as the value of *case_status*, the sum of *total_tbsa_percent*, the ABSI calculation, and the escalation rules, are handled by the application and the Safety Rule Engine.

The script also creates three views to separate the two audiences of a case. The views *v_case_user_result* and *v_area_user_result* expose the total and per-area %TBSA, the burn depth, and the urgency information without any ABSI column, and the user app reads only these views. The view *v_case_clinician_summary* joins *Burn_case* with *Case_referral*, so the doctor receives the ABSI score and all other case information.

Table 6.11 summarizes the logical entities used by the prototype, and the relationships between them are listed in the Relationship Table in section 6.3.3. The complete script is provided in the file *BurnAid_Database_v9.1_Full_Production.sql.*

Table 6.11. Logical entities of the prototype

Entity	Role in the prototype	Primary key	Foreign keys
account	Users and staff who log in	account_id	None

Burn_case	One patient encounter and its case-wide results	case_id	account_id
Burn_area_assessment	One burned region with its photo and AI result	area_assessment_id	case_id, region_code, quality_check_model_id, classification_model_id, segmentation_model_id
Body_region_reference	Standard regions and %TBSA values	region_code	None
First_aid_guideline	Approved first aid guidance with versions	guideline_id	body_region_code, created_by, approved_by
Assessment_guideline	Guidelines shown for each burned area	Assessment_guideline_id	area_assessment_id, guideline_id
AI_model	Registered AI models and their settings	model_id	uploaded_by
Feedback_review_log	User feedback and low-confidence cases	review_entry_id	account_id, area_assessment_id, guideline_id, reviewed_by
Case_referral	Case package sent to a doctor, with the ABSI score	referral_id	case_id, assigned_to

CHAPTER 7. MODEL DEVELOPMENT

7.1. AI PROBLEM DEFINITION

In the BurnAid AI project, the goal is to support pre-hospital first aid and classify thermal burn injuries into one of four labels: degree_1 (first-degree burn), degree_2 (second-degree burn), degree_3 (third-degree burn), and None (healthy skin, no burn), as well as to locate suspected injured regions on images. The system consists of two specialized computer vision steps combined with an independent pre-hospital clinical rule engine. These two deep learning tasks are designed in parallel under an independent dual-branch pipeline, where each branch handles a specific technical objective:

For the burn depth classification branch, the team uses a modern convolutional neural network, ConvNeXtV2 (Nano version). The model takes an RGB image of size 224 × 224 pixels and outputs a probability distribution across four classes: First-degree burn (degree_1), Second-degree burn (degree_2), Third-degree burn (degree_3), and Healthy skin without burns (None). This single-label, image-level classification guides the assessment of how deep the burn injury is.

For the lesion segmentation branch, the team uses the U-Net++ (Nested U-Net) architecture combined with a ResNet-34 feature extractor (encoder) pretrained on ImageNet. The model takes an RGB image of size 320 × 320 pixels and predicts a single-channel probability map. A threshold of 0.50 is applied to generate a binary mask, where a value of 1 represents suspected burn tissue and 0 represents the background. The model does not predict burn depth for individual pixels; instead, it focuses on finding the exact boundary of the burned region.

This design combines two independently executed neural networks. The classification branch plays the core role in identifying burn depth, while the segmentation branch provides visual bounding and helps detect conflicting signals. The system does not use a rigid rule like "an empty mask means no burn" which prevents missing superficial or small burn injuries in real situations. In terms of deep learning development, the two models were trained independently following a specialized dual-branch architecture. However, in the actual production system workflow, the execution flow operates via a conditional sequential mechanism: Process 2.2 in Chapter 4 (Classify burn degree) executes first to determine injury depth. If healthy non-burned skin (None) is detected, the system proactively bypasses the downstream segmentation stage to eliminate false-positive mask generation on normal skin while conserving computational resources. Once a burn injury is verified, the data flow proceeds to Process 2.4 (Segment burn area) to delineate lesion boundaries, estimate affected area proportions, and apply visual color-coded overlays.

In addition, there is a clear separation between machine learning and clinical rule calculations. In the overall architecture, deep learning models are strictly limited to visual feature extraction (recognizing burn depth and the proportion of the affected area on the image). All subsequent clinical interpretation steps, including estimating Total Body Surface Area (%

TBSA) using Professor Le The Trung's 1-3-6-9-18 rule, risk stratification (Emergency, High, Moderate, Mild), screening for inhalation burn signs, and automatically triggering emergency 115 calls when a Third-Degree burn is detected, are handled completely independently by the Clinical Rule Engine. This design prevents the AI models from making clinical judgments beyond their training data, ensuring maximum safety for pre-hospital medical applications.

7.2. BURN DEPTH CLASSIFICATION MODEL (CLASSIFICATION MODEL)

7.2.1. Dataset Collecting & Preprocessing

The classification dataset was constructed by collecting raw burn images from open-source Kaggle benchmark repositories (primarily the Kaggle Skin Burn Detection dataset by Fares Abbas, supplemented with healthy skin samples). These raw images were then imported into the Roboflow workspace, where the team performed image cropping, visual quality filtering, and manual re-annotation to standardize the labels into four distinct target classes: degree_1, degree_2, degree_3, and healthy skin (None). The None class was supplemented with a separately gathered set of non-burned skin images (estimated at about 987 images in the initial stage; this estimate is not identical to the total count of the None class after all final splits). Because the Internet images were collected from various websites, full URLs and licenses for every image are not available. The team acknowledges this traceability limitation and does not assume all images have open licenses. After merging all sources, the dataset reached a total of 4,686 RGB images across four target classes: 1,259 First-Degree burn images (26.87%), 1,201 Second-Degree burn images (25.63%), 1,055 Third-Degree burn images (22.52%), and 1,171 Healthy skin images (24.99%). This relatively balanced distribution helps reduce prediction bias during model training.

After merging, the team detected duplicate and near-duplicate images using Perceptual Hashing (pHash) and Hamming distance, and continued filtering out unsuitable or low-quality images. pHash helps detect visual similarity even if files were resized or compressed; however, it does not guarantee that every duplicate is removed. Since no record was kept of the exact number of images removed at each filtering step, this report presents the final counts. The dataset used in the notebook contains 4,686 images, consisting of 3,908 training images, 462 validation images, and 316 test images. Images are loaded in RGB format. The training set applies RandomResizedCrop to 224 × 224 pixels, horizontal/vertical flips, rotation, and ColorJitter. The validation and test sets use a fixed resize to 224 × 224 pixels. All three sets are converted to tensors and normalized using ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).

Table 7.1. Class distribution of the classification dataset after preprocessing

Burn Class	Training Set (Train)	Validation Set (Val)	Test Set (Test)	Total Images	Proportion (%)

First-Degree Burn (degree_1)	1,076	110	73	1,259	26.87%
Second-Degree Burn (degree_2)	1,021	97	83	1,201	25.63%
Third-Degree Burn (degree_3)	834	140	81	1,055	22.52%
Healthy Skin (None)	977	115	79	1,171	24.99%
Total	3,908	462	316	4,686	100.00%

The actual proportions of the train, validation, and test sets are 83.40%, 9.86%, and 6.74%, respectively. These are the observed numbers after data aggregation and cleaning; they should not be described as an exact 80/10/10 split or a strictly stratified split without split creation logs. At the label level, each CSV row must contain exactly one value of 1, with the rest set to 0. Rows with missing image files or invalid labels were skipped or produced loading errors.

The Hamming distance between two perceptual image hashes (pHash) is defined as follows:

$$d_H(h_1, h_2) = \sum_{i=1}^{n} I(h_{1,i} \neq h_{2,i})$$

Here, n represents the hash bit-length, h1,i and h2,i denote the i-th bits of the two hashes, and I(·) is the indicator function evaluating to 1 if the bits differ and 0 if they match. A smaller distance indicates greater visual similarity between the two images. In the segmentation v8 notebook, near-duplicate checking between splits used a maximum threshold of 4 bits. Because the initial classification filtering did not have a recorded threshold log, the two screening setups are not identical.

7.2.2. Preparation and Training

The core classification model uses the ConvNeXtV2 Nano architecture via the timm library, initialized with pretrained weights convnextv2_nano.fcmae_ft_in22k_in1k. The classification head was customized with four outputs corresponding to the four burn classes. Fine-tuning uses layer-wise learning rates: the backbone learns more slowly than the head to preserve low-level visual features. The AdamW optimizer is combined with the OneCycleLR scheduler over 15 epochs. To prevent overfitting, the team applied data augmentation (RandomResizedCrop, flips, rotation, ColorJitter), dropout of 0.4, and Label Smoothing of 0.05 in the CrossEntropyLoss function. Checkpoints were selected based on validation Macro-F1 score, not test metrics.

Table 7.2. Training parameter settings for the ConvNeXtV2 Nano model

Technical Parameter	Configured Value	Purpose, Notes
Model Architecture	ConvNeXtV2 Nano (timm)	Modern convolutional network, 14.99M parameters
Initial Weights	convnextv2_nano.fcmae_ft_in22k_in1k	Pretrained FCMAE on ImageNet-22k and 1k
Input Dimensions	224 × 224 pixels, 3 RGB channels	Normalized using ImageNet mean and std
Batch Size	64	Optimizes GPU throughput
Maximum Epochs	15 epochs	Reduces degradation from overfitting
Optimizer	AdamW (Weight Decay = 0.05)	Stable weight updates with weight decay
Learning Rate	Backbone: 2×10^{-4}, Head: 2×10^{-3}	Layer-wise learning rate strategy
LR Scheduler	OneCycleLR (pct_start = 0.15)	Increases LR early then decays via cosine schedule
Regularization	Dropout = 0.4; Label Smoothing = 0.05	Prevents overconfidence on noisy labels
Data Augmentation	RandomResizedCrop(0.8-1.0), HFlip, VFlip, Rotation ±15°, ColorJitter	Simulates diverse camera angles and lighting
Checkpoint Criterion	Highest Validation Macro-F1	Saved at epoch 14 (convnextv2_Burn_4class_best.pth)

During training, the training loss decreased from 0.7878 (epoch 1) to 0.2285 (epoch 15), while the validation loss fluctuated without decreasing monotonically. The best model was saved at epoch 14, achieving a validation accuracy of 0.8528 and a Macro-F1 score of 0.8537. The gap between training loss and validation loss toward the end of training indicates that overfitting must be handled with care. Selecting checkpoints based on validation performance prevents using the final epoch model simply because its training loss was lower.

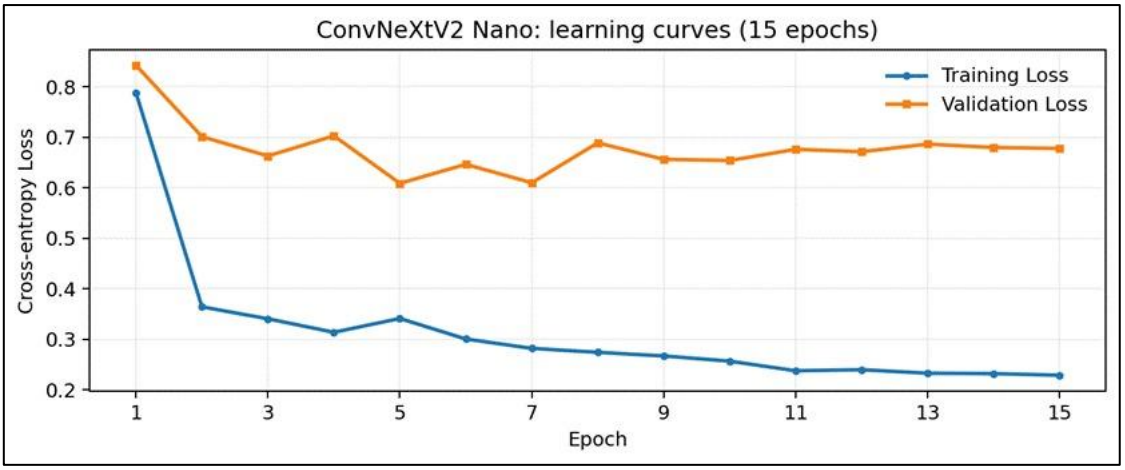


Figure 7.1. Training Loss and Validation Loss curves of the ConvNeXtV2 Nano model over 15 epochs

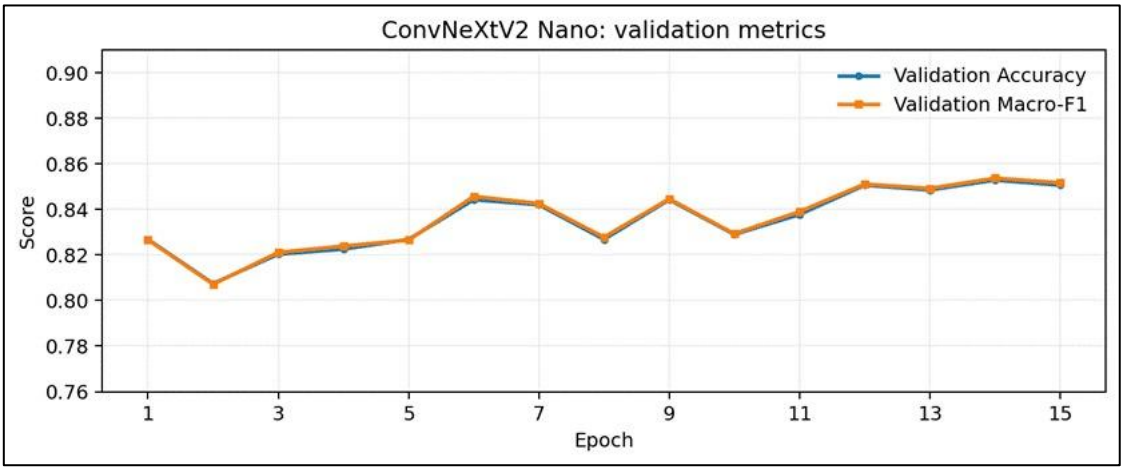


Figure 7.2. Validation Accuracy and Macro F1-score across epochs

After loading the best checkpoint, the training notebook evaluated the model on the 316 test split images, recording an Accuracy of 0.8576 and a Macro-F1 of 0.8559. This represents an internal test result on the available split, not external validation from independent data sources. In particular, the degree_1 class had a recall of 0.6986, which is lower than degree_2 (0.8193) and degree_3 (0.9012).

Table 7.3. Detailed class-by-class evaluation results on the independent test set (n = 316)

Target Class	Precision	Recall	F1-score	Test Images
First-Degree Burn (degree_1)	0.8500	0.6986	0.7669	73 images
Second-Degree Burn (degree_2)	0.8831	0.8193	0.8500	83 images
Third-Degree Burn (degree_3)	0.7300	0.9012	0.8066	81 images
Healthy Skin (None)	1.0000	1.0000	1.0000	79 images
Macro Average	0.8658	0.8548	0.8559	316 images

The None class achieved 1.0 on this test split, but this does not mean the model can perfectly rule out burns on all new images. In testing outside the dataset, the classifier occasionally assigned first-degree burn to sample images of normal skin. Therefore, the report clearly notes this limitation on out-of-distribution data.

7.3. SEGMENTATION MODEL

7.3.1. Dataset Collecting & Preprocessing

The segmentation data was built from images with manual polygon annotations and labeled images that lacked precise masks. The v8 pipeline converts polygons into binary masks, counting a mask as valid only if the polygon exists and its area is greater than 0. Empty annotations from skipped images are not interpreted as None images. For image sources without masks, a baseline U-Net++ model was used to generate pseudo-masks. These predicted masks underwent Auto-QC based on confidence scores, mask area plausibility, and output consistency. Unreliable pseudo-masks were rejected or isolated rather than included in training. Images labeled as None at the image level were used as examples of background masks, but they are not pixel-level negative masks verified by medical specialists.

The final segmentation manifest contains 2,198 training records, 21 validation records, and 13 test records with manual polygon masks. The training set consists of 300 manual polygon records, 921 accepted pseudo-masks, and 977 None images. The 300 manual training records come from approximately 100 original image groups due to augmentation variants. Before curriculum training, 98 None images were set aside to evaluate the false positive rate, leaving 2,100 source training records (300 manual, 921 pseudo, 879 None). Each epoch samples with replacement according to the source proportions; therefore, $80\ steps \times batch\ size\ 8 = 640$ sampled images per epoch does not mean exactly 640 unique images. Images are resized to 320 × 320 pixels and normalized with ImageNet statistics; masks are resized using nearest-neighbor interpolation to preserve binary values. The source code also checks SHA-256 hashes, pHash, original image groups, and prevents cross-split duplication where verifiable data exists.

Table 7.4. Segmentation dataset composition and the 98 holdout None images

Mask Source	Train in Manifest	Train after Holdout	Validation	Test
Manual Polygon	300	300	21	13
Auto-QC Pseudo-mask	921	921	0	0
None, Background Mask	977	879	0	0
TOTAL	2,198	2,100	21	13

Because the segmentation validation and test sets contain only manual masks, are small in size, and may include variants of original images, Dice and IoU metrics should be interpreted

cautiously. Polygon data describes suspected burn locations, not medical ground truth for burn depth.

7.3.2. Preparation and Training

The final segmentation model is U-Net++ (Nested U-Net) with a ResNet-34 encoder initialized from ImageNet weights, three input channels, and a single-channel logits output. The nested skip connections allow combining features across multiple resolution levels. Training uses AdamW and a combined loss function of Binary Cross-Entropy with Logits (BCE) and Soft Dice with equal weights: $L = 0.5 \times BCE + 0.5 \times (1 - SoftDice)$. Data augmentation includes random horizontal and vertical flips; validation evaluation does not use random augmentation.

The team evaluated four sampling strategies: Uniform, Easy-to-Hard, Hard-to-Easy, and Random Curriculum. To control confounding factors, all experiments used the same architecture, batch size, number of steps per epoch, and source sampling ratios of 25% manual, 55% pseudo, 20% None. In the curriculum setups, positive data eligible for sampling started at 30% and expanded gradually to 100% by roughly 75% of the training duration. Version v5 also examined ranking pseudo-mask difficulty; ablation experiment v5b showed that difficulty ranking did not provide a consistent advantage across random seeds.

Table 7.5. Training settings for the Curriculum U-Net++ ResNet-34 model

Technical Parameter	Configured Value	Purpose, Notes
Model Architecture	U-Net++ (Nested U-Net), Encoder ResNet-34	Optimizes boundaries via nested skip connections
Initial Weights	ImageNet Pretrained Weights	Accelerates convergence and stabilizes feature extraction
Input, Output Dimensions	320 × 320 RGB, 1-channel binary logits	Resolution preserves burn wound details
Batch Size & Steps	Batch size = 8; 80 steps/epoch (640 samples/epoch)	Sampling with replacement based on source ratios
Maximum Epochs	18 epochs	Sufficient cycles to complete curriculum schedule
Loss Function	0.5 × BCEWithLogits + 0.5 × SoftDice	Balances pixel-level loss and shape similarity

Source Sampling in Train	25% manual, 55% pseudo, 20% None	Prevents distribution shift and controls false positives
Checkpoint Criterion	SelectionScore = ValidDice - 0.15 × NegativeFPR	Penalizes over-prediction across healthy skin
Best Checkpoint	best_random_curriculum_seed2026.pt (Epoch 6)	Achieves Validation Dice = 71.29% and Score = 0.7099

During training, in addition to evaluating Dice and IoU on validation images with manual polygons, the team measured the Negative False Positive Rate (NegativeFPR) on the 98 held-out None images. The checkpoint selection score was calculated as: SelectionScore = ValidationDice - 0.15 × NegativeFPR. This formula penalizes models that achieve high positive Dice but over-predict burn masks across healthy skin images. For the best checkpoint of the Random Curriculum strategy (seed 2026, epoch 6), metadata records a validation Dice of approximately 0.7129.

For comparison, the team evaluated Curriculum U-Net++ v5 against the manual-only U-Net++ model from the v6 pipeline using the same preprocessing and metrics. On the 21-image manual validation set, the results were: mean Dice of 0.7130 vs. 0.6478; mean IoU of 0.6067 vs. 0.5307. On the 13-image manual test set, the results were: mean Dice of 0.7215 vs. 0.6592; mean IoU of 0.6014 vs. 0.5464. The Curriculum model outperformed the baseline on 9 out of 13 test images. This comparison is against the v6 manual-only checkpoint, not a verified v8 checkpoint.

Table 7.6. Performance comparison between Manual-only U-Net++ and Curriculum U-Net++

Dataset	Metric	Manual-only v6	Curriculum v5	Difference
Validation (n=21)	Dice	0.6478	0.7130	+0.0652
Validation (n=21)	IoU	0.5307	0.6067	+0.0760
Test (n=13)	Dice	0.6592	0.7215	+0.0623
Test (n=13)	IoU	0.5464	0.6014	+0.0550

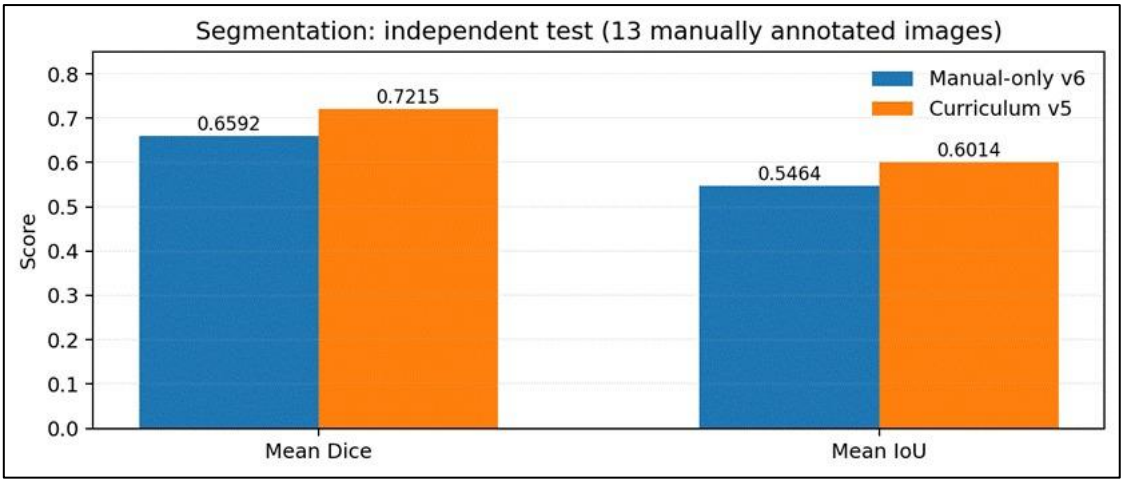


Figure 7.3. Comparison chart of Mean Dice and Mean IoU on 13 independent test images

An exploratory 95% bootstrap confidence interval for the mean Dice difference on the 13 test images is [-0.0716, 0.2306], which includes 0. Therefore, while the results favor Curriculum learning, they are not sufficient to claim a statistically significant advantage across all image sources. An additional test on 115 images labeled None in the classification validation set showed non-empty mask predictions on 99.13% of images for the manual-only model, compared to only 6.09% for Curriculum. This indicates a clear reduction in false positive masks on image-level None samples, though it does not represent pixel-level medical ground-truth accuracy.

The team added three training logs for Random Curriculum corresponding to seeds 42, 123, and 2026, each covering 18 epochs. These logs record training loss, validation Dice/IoU, NegativeFPR on the holdout None images, and SelectionScore for each epoch. This allows direct visualization of the training process (Figures 7.4-7.6) rather than only comparing final metrics. Note that the CSV logs record training loss and validation metrics, but not validation loss.

Table 7.7 shows the checkpoint with the highest SelectionScore for each seed; this is not necessarily the epoch with the lowest training loss. For seed 2026, the checkpoint was selected at epoch 6, which is also the checkpoint used in the benchmark above.

Table 7.7. Best Random Curriculum checkpoint results based on SelectionScore across three seeds

Random Seed	Selected Epoch	Validation Dice	Validation IoU	False Positive Rate (FPR)	Selection Score
Seed 42	Epoch 15	0.6924	0.5843	6.12%	0.6832
Seed 123	Epoch 12	0.7000	0.5893	3.06%	0.6954
Seed 2026	Epoch 6	0.7129	0.6066	2.04%	0.7099

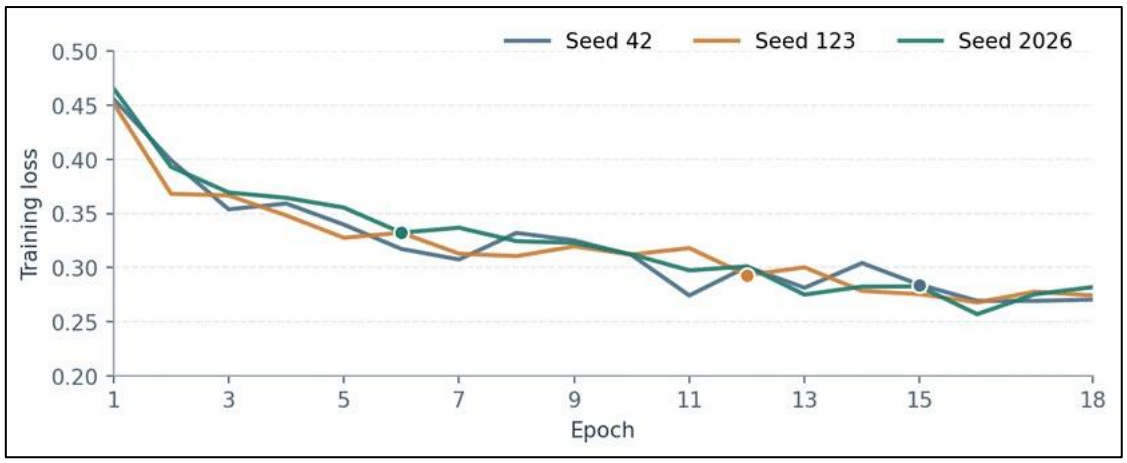


Figure 7.4. Training Loss curves by epoch for Random Curriculum across three seeds (42, 123, 2026)

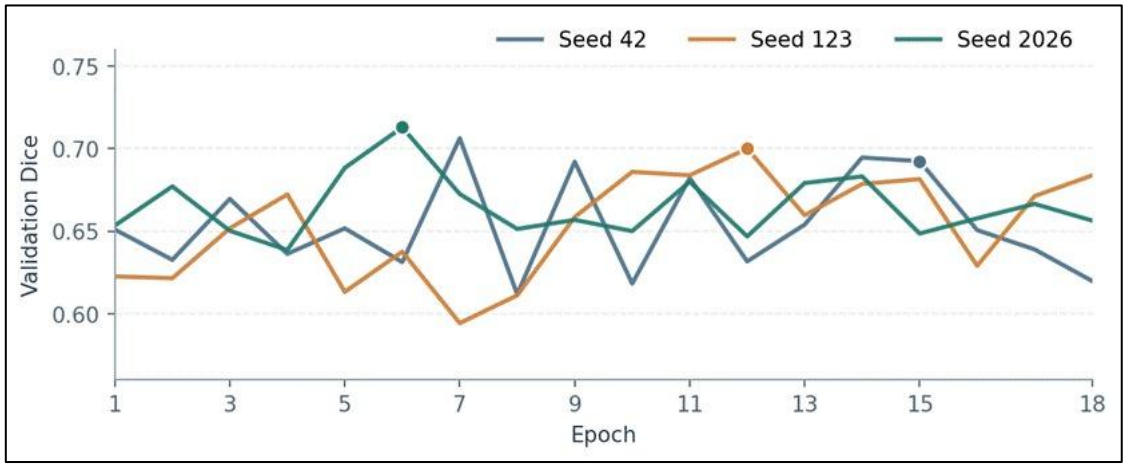


Figure 7.5. Validation Dice variation by epoch across three seeds, with dots marking selected checkpoints

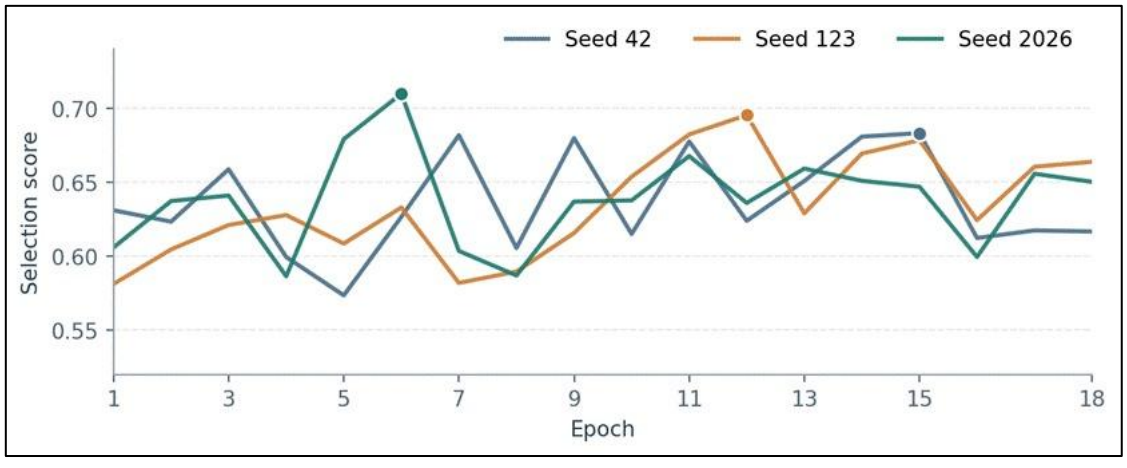


Figure 7.6. SelectionScore curves incorporating false positive penalties across epochs

For seed 2026, training loss decreased from 0.4656 to 0.2819 between epoch 1 and epoch 18, but Validation Dice fluctuated across epochs (0.6534 at epoch 1 and 0.6564 at epoch 18). Epoch 6 achieved Validation Dice = 0.7129, Validation IoU = 0.6066, NegativeFPR =

2.04%, and SelectionScore = 0.7099, the highest among all three seeds. Therefore, the final model uses the epoch 6 checkpoint even though training loss continued to fall afterward. This confirms that validation metrics and controlling false positives are more important than simply minimizing training loss.

Results from the three seeds reflect expected random variation during training. For seed 42, epoch 7 reached a Validation Dice of 0.7063, but the SelectionScore checkpoint was chosen at epoch 15 (Dice 0.6924, NegativeFPR 6.12%) due to the false positive penalty. These curves provide evidence for the optimization process and checkpoint selection criteria; they do not replace the independent evaluation on the 13 test images in Table 7.6.

The main limitations of the segmentation branch are the small number of manual validation/test masks, uneven boundary precision, and binary masks that do not differentiate burn degrees (1, 2, or 3) per pixel. Sample images outside the dataset serve only as qualitative illustrations and lack suitable ground truth for calculating out-of-distribution Dice or Macro-F1. Thus, the model should be presented as a promising deep learning experiment, not a clinically certified diagnostic tool.

7.4. EXPERIMENTAL MODEL SELECTION & ARCHITECTURE DECISIONS

7.4.1. Objectives and Experimental Setup on NVIDIA A100 GPU

To choose models based on evidence rather than architectural novelty, the team trained 5 classification candidates and 4 segmentation candidates on the same processed dataset. Each model was run across three random seeds (42, 123, 2026), resulting in 27 full training runs on an NVIDIA A100-SXM4-40GB GPU. Each run saved the best checkpoint based on the validation set. Aggregated results report the sample mean and standard deviation across the three seeds, rather than relying on a single run (Table 7.8).

Table 7.8. Model Selection experimental setup; configurations taken from the Full notebook

Criterion	Classification	Segmentation
Candidates	5 CNN/Transformer architectures	4 decoder/encoder architectures
Evaluation Set for Selection	462 validation images, 4 classes	21 manual validation masks + 98 holdout None images
Input, Batch Size	224 × 224, 32	320 × 320, 8
Training Budget	15 epochs per seed	18 epochs × 80 steps/epoch per seed
Seeds	42, 123, 2026	42, 123, 2026
Selection Metric	Validation Macro-F1	Validation Dice - 0.15 × NegativeFPR
Initialization	Pretrained (timm)	Pretrained encoder, random curriculum

Architectures within the same task used identical data splits, checkpoint criteria, and epoch budgets. Nevertheless, differences in pretrained checkpoints, parameter counts, and hyperparameter sensitivity still affect outcomes. This is an empirical comparison under specified settings, not a proof of absolute superiority. The test set was not used for checkpoint selection in the Model Selection notebook.

7.4.2. Classification Architecture Selection

The five candidate models were ConvNeXtV2 Nano, DeiT III Small, Swin Transformer Tiny, ViT Base/16, and EfficientNetV2-S. For each seed, the best epoch was selected by validation Macro-F1. Table 7.9 reports the mean ± sample standard deviation of the best Macro-F1 scores across the three seeds.

Table 7.9. Classification Model Selection results; each model trained across three seeds

Model	Parameters (M)	Validation Macro-F1 (Mean ± SD)	Best Seed Macro-F1
ConvNeXtV2 Nano	14.99	0.8551 ± 0.0068	0.8616
DeiT III Small	21.68	0.8520 ± 0.0056	0.8584
Swin Transformer Tiny	27.52	0.8490 ± 0.0082	0.8584
ViT Base/16	85.80	0.8411 ± 0.0038	0.8452
EfficientNetV2-S	22.16	0.8254 ± 0.0088	0.8319

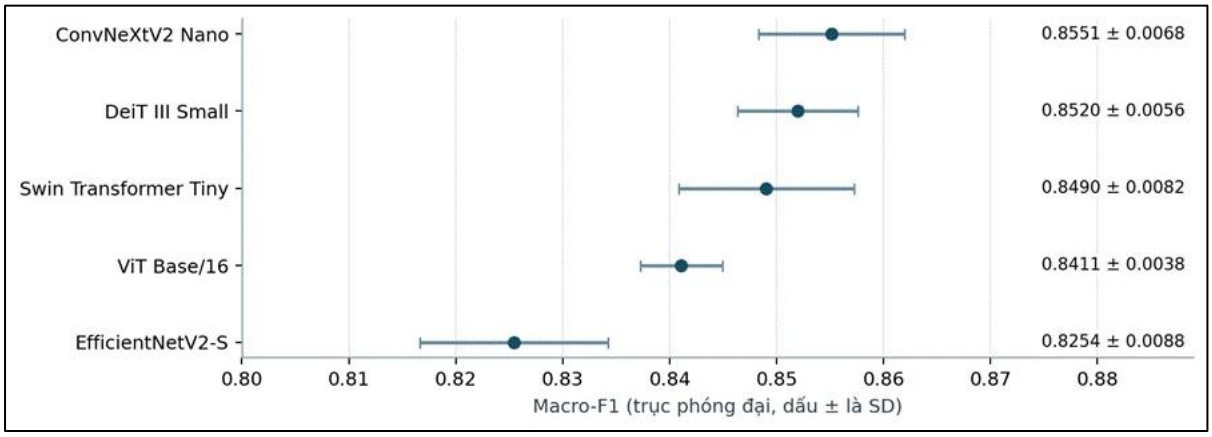


Figure 7.7. Average Macro-F1 and standard deviation across three seeds; the horizontal axis is magnified to show small differences

ConvNeXtV2 Nano led with an average Macro-F1 of 0.8551 ± 0.0068; DeiT III Small achieved 0.8520 ± 0.0056, and Swin Tiny reached 0.8490 ± 0.0082. For seed 42 alone, Swin ranked highest (0.8584), but this advantage did not hold across all three seeds. ConvNeXtV2 also has the smallest parameter count among the five candidates (~14.99 million). Therefore, the team selected ConvNeXtV2 Nano for the classification branch. The seed 2026 checkpoint achieved the highest validation score among the new runs (0.8616), though this should not be interpreted as the test score for that checkpoint.

7.4.3. Segmentation Architecture Selection

The team compared U-Net++ ResNet-34 retrained in the full round against SegFormer MiT-B2, SegFormer MiT-B3, and UPerNet MiT-B2. All four used the random_curriculum sampling strategy and selected checkpoints using SelectionScore = ValidationDice - 0.15 × NegativeFPR. NegativeFPR measures the proportion of None images predicted to have a mask after the 0.5 threshold. Table 7.10 presents the scores at the selected checkpoints for each seed.

Table 7.10. New Segmentation Model Selection results on validation: Average Dice, Selection Score, and NegativeFPR across three seeds

Architecture	Parameters (M)	Mean Dice	Score (Mean ± SD)	Mean NegativeFPR
UPerNet MiT-B2	32.53	0.6989	0.6989 ± 0.0025	0.00%
SegFormer MiT-B3	44.60	0.6963	0.6963 ± 0.0202	0.00%
SegFormer MiT-B2	24.72	0.7002	0.6961 ± 0.0080	2.72%
U-Net++ ResNet34 (new)	26.08	0.6896	0.6774 ± 0.0087	8.16%

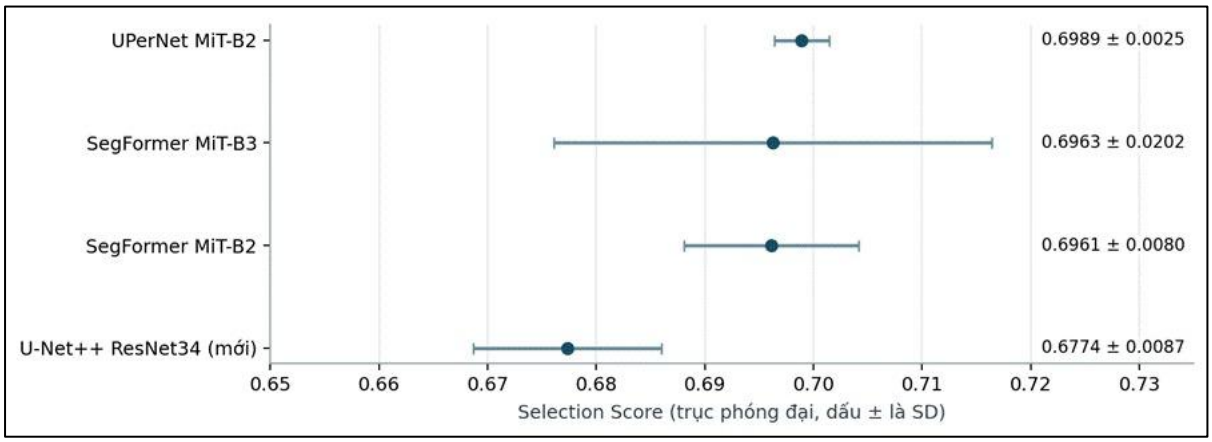


Figure 7.8. Average validation Selection Score and standard deviation across three seeds; horizontal axis is magnified to distinguish architectures

UPerNet MiT-B2 achieved the highest average Selection Score among the new candidates (0.6989 ± 0.0025) and maintained a 0% NegativeFPR across the 98 None images at all three checkpoints. SegFormer MiT-B3 reached 0.6963 ± 0.0202, showing higher variation across seeds. In the earlier Curriculum U-Net++ v5 logs, the Mean Dice at selected checkpoints was 0.7018, with an average Selection Score of 0.6962 ± 0.0133. Although both experiment rounds used related data and metrics, their training runs were not completely identical. The small score difference between UPerNet and Curriculum U-Net++ should not be considered statistical proof of superiority.

7.4.4. Direct Comparison of Two Segmentation Models on Independent Test Set

To assess performance beyond the checkpoint selection set, the team evaluated UPerNet MiT-B2 (seed 42, best epoch 15) and Curriculum U-Net++ v5 (seed 2026, best epoch 6) on the same 13 test images with manual ground-truth masks, using the same 320 × 320 resolution and 0.5 threshold. No retraining or threshold tuning was performed on the test set. Mean Dice/IoU is calculated on a per-image basis, while Pooled Dice/IoU is computed across all combined pixels.

Table 7.11. Direct comparison between two checkpoints on the segmentation test set [12]

Metric on 13 Test Images	Curriculum U-Net++	UPerNet MiT-B2
Mean Dice	0.7215	0.7170
Mean IoU	0.6014	0.6196
Pooled Dice	0.7330	0.8100
Pooled IoU	0.5785	0.6807
Images with Higher Dice	4/13	9/13
Empty Masks on Burn Images	0/13	1/13

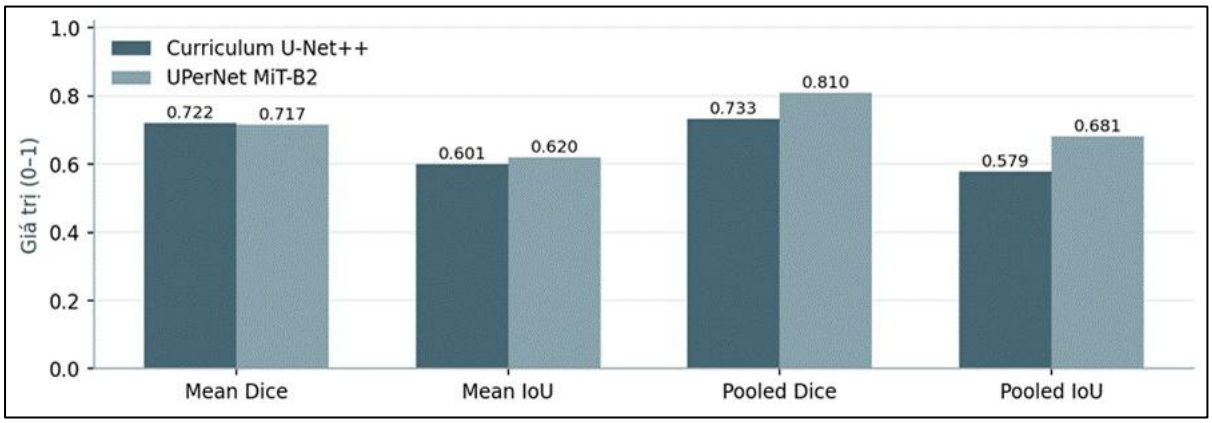


Figure 7.9. Test results by image-level averages and pooled pixel metrics; metrics summarize different aspects of the same predictions

UPerNet achieved a better Dice score on 9 of the 13 images and led in Mean IoU and both pooled metrics, showing solid prediction on larger areas. However, UPerNet produced an

empty mask on one image that contained an annotated burn; on that same image, Curriculum U-Net++ achieved a Dice of about 0.8798. As a result, the Mean Dice for Curriculum remained slightly higher (0.7215 vs. 0.7170). An exploratory paired-bootstrap 95% confidence interval for the Mean Dice difference (UPerNet - Curriculum) is [-0.1803, 0.1231], which includes 0. Because the test set contains only 13 images and was observed during development, it cannot establish general superiority for either model.

7.4.5. Final Model Decision and Deployment Limitations

The team finalized ConvNeXtV2 Nano for classification due to its highest average validation Macro-F1 among the five architectures and its compact parameter count. For segmentation, the team retained Curriculum U-Net++ ResNet-34 (random_curriculum checkpoint, seed 2026) in the current model pipeline. This decision was based on its slightly higher test Mean Dice, absence of empty masks on the 13 test images, and thoroughly verified demo pipeline. UPerNet MiT-B2 is preserved as a strong alternative candidate rather than dismissed, as the Mean Dice difference is minimal and it leads in several other metrics.

The two final branches run in parallel: ConvNeXtV2 outputs an image-level label, while U-Net++ provides a binary mask for visual display and contradiction checking. An empty mask does not automatically turn a burn prediction into None. This choice balances performance, stability, and academic demonstration needs, without claiming clinical-grade safety or diagnostic accuracy. Before broader deployment, further checks using pHash for near-duplicate images across splits, out-of-distribution evaluation, and additional specialist-verified masks are necessary.

7.5. Large Language Model & RAG

Within the system, a Large Language Model (LLM) combined with Retrieval-Augmented Generation (RAG) serves as an intelligent medical assistant. By verifying and synthesizing information from an indexed, reliable knowledge base rather than relying on unconstrained text generation, this subsystem prevents hallucinations, explains visual classification results clearly, and guides users through standard thermal burn first aid during the crucial initial minutes after an injury.

The system connects to the Groq API to run the open-source LLaMA 3.3 70B Versatile model, delivering response times in milliseconds to meet rapid emergency needs. The knowledge base is extracted from 18 reputable textbooks and professional guidelines (including Military Medical Academy and Ministry of Health documents), divided into 2,274 text chunks stored in knowledge_base_burns_full.json. BM25 retrieval and vector embeddings quickly fetch the 3 to 5 most relevant chunks to build the prompt context for the language model.

In terms of medical safety, the system prioritizes emergency 115 call warnings whenever a Third-Degree burn or signs of inhalation injury (such as stridor or hoarseness from closed-space fires) are detected. First-aid instructions strictly follow Vietnam Ministry of Health Decision 635/QD-BYT and NHS standards: continuous cooling under clean, running

tap water (15-20°C) for 20 minutes; strictly avoiding ice, moisturizers, lard, or toothpaste during the acute stage; and keeping blisters intact. Unverified professional titles such as 'board-certified' were also removed from system prompts to maintain strict medical accuracy.

CHAPTER 8. SYSTEM ARCHITECTURE & USER INTERFACE DESIGN

8.1. THREE-TIER ARCHITECTURE DIAGRAM FOR PROTOTYPE

8.1.1. Overview

After completing the modeling of the system's functions, data, and operational flows, the group proceeded to design the system's three-tier architecture diagram. This architecture divides the system into three distinct functional layers based on their respective responsibilities. The Presentation Layer handles user interaction, the Application Layer processes business logic, and the Data Layer manages data required for system operations. It should be noted that the group designed the architecture and interface exclusively for the Patient and Family Members user group; therefore, functions for IT Staff and Medical Staff are not represented. To validate the core AI logic without heavy cloud infrastructure costs during this project phase, the target mobile-first architecture was implemented as a local single-page web prototype hosted on a Python/FastAPI local server.

The overall three-tier structure is interconnected through communication paths. The Presentation Layer communicates with the Application Layer through HTTPS/REST APIs, using JSON to transmit requests and responses. Between the latter two layers, SQL queries are executed through database connections, as illustrated in the diagram. In addition, the system communicates with an external service, Groq Cloud LLM, to support the AI Chatbot feature.

To align with the implementation scope of this project phase, Figure 8.1 illustrates the 3-tier architecture specifically configured for the Patient-Facing Prototype. While the comprehensive system design encompasses administrative (IT staff) and clinical review (medical advisor) modules as modeled in Chapter 4, the active prototype architecture focuses on the presentation, business logic, and database layers serving the primary patient user group.

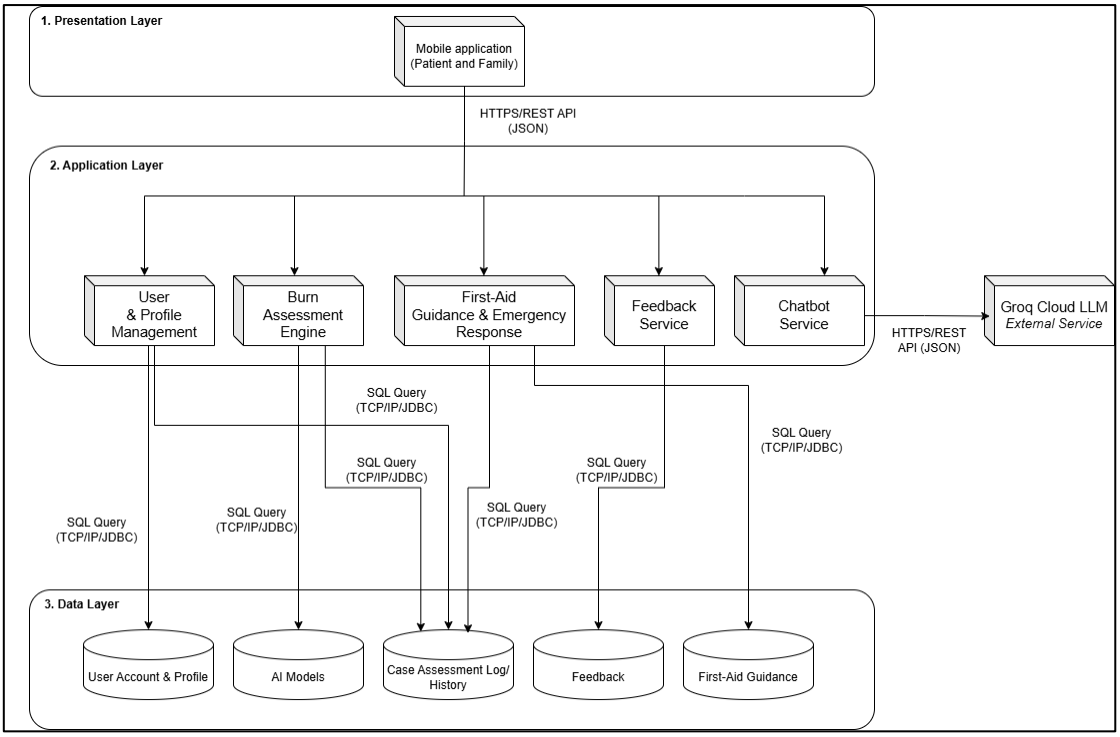


Figure 8.1. Three-tier Architecture Diagram (Patient-Facing Prototype)

8.1.2. Architecture components

The Presentation Layer is the interface layer responsible for providing the means of interaction between users and the system. In the proposed architecture, this layer consists of a Mobile Application designed for patients and their family members, serving as the primary access point for system functions. It focuses on receiving input data and displaying output information rather than directly performing assessment operations or managing data.

The Application Layer is the central business logic layer responsible for coordinating the system's functions. Since the system is designed exclusively for the Patient and Family Members user group, this layer contains five modules: User & Profile Management, which manages user account and profile information; Burn Assessment Engine, which coordinates image-based burn assessment using AI classification and segmentation models; First-Aid Guidance & Emergency Response, which provides first-aid guidance and handles situations requiring emergency responses; Feedback Service, which receives and processes user feedback; and Chatbot Service, which handles conversational requests and integrates with Groq Cloud LLM. Dividing the layer into modules helps define the responsibilities of each system function clearly.

The Data Layer is responsible for organizing and providing data to support the operations of the business logic modules. This layer contains five data stores corresponding to those in the Data Flow Diagram (D1 to D5). Each data store serves a corresponding group of functions, distinguishing different types of information according to their purposes. However, the actual organization and storage of data depend on the system's deployment approach.

8.1.3. External Service - Groq Cloud LLM

Groq Cloud LLM is represented in the diagram as an external service integrated with the Application Layer through the Chatbot Service. Unlike the internal business logic components, this service provides language processing capabilities that the system can leverage through API communication.

When a user submits a question, the Chatbot Service receives the request and prepares the necessary data to send to Groq Cloud LLM. The request is transmitted as JSON through an HTTPS/REST API. After processing, the external service returns a response, which the Chatbot Service receives and forwards to the user interface. By positioning the Chatbot Service as an intermediary, it separates the system's conversational logic from the service providing the language model. This allows chatbot requests and integration to be managed through a dedicated business logic component.

8.1.4. Communication paths

The system architecture uses different communication mechanisms depending on the relationships between components. These mechanisms define how requests are transmitted, data is exchanged, and results are received during system operation.

The Presentation Layer and Application Layer communicate through HTTPS/REST APIs, using JSON to transmit request and response data. This mechanism allows the mobile application to send input information to the business logic components and receive the corresponding processing results.

Between the Application Layer and Data Layer, business logic components access data through SQL queries over TCP/IP/JDBC connections. This supports the retrieval or updating of information in the relevant data stores.

8.2. NAVIGATION FLOW CHART FOR PROTOTYPE

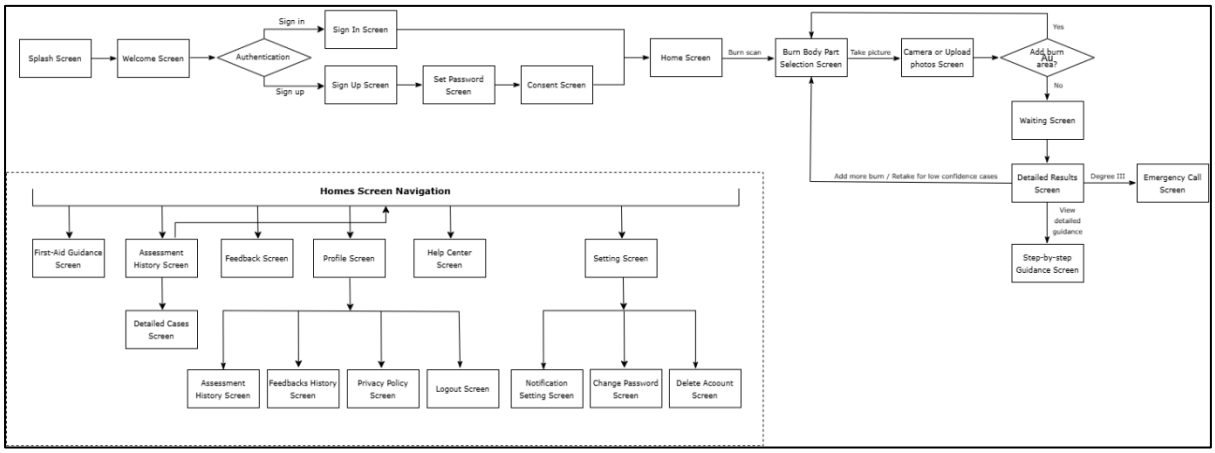


Figure 8.2. Navigation Flowchart

The navigation flowchart illustrates the navigation paths between screens of the burn assessment system, focusing on the interface designed for Patient and Family Members. The navigation paths are divided into three main groups: user authentication, burn assessment, and navigation from the home screen.

The Authentication Flow begins at the Splash Screen and proceeds to the Welcome Screen, where users choose to sign in or sign up. The sign-in branch leads to the Sign In Screen, while the sign-up branch proceeds through the Sign Up Screen, Set Password Screen, and Consent Screen. After completion, both flows lead to the Home Screen.

The Burn Assessment Flow begins at the Home Screen when users navigate to the Burn Body Part Selection Screen to select the affected body area, then proceed to the Camera or Upload Photos Screen to capture or upload images. Users can choose to add another burn area or proceed to the Waiting Screen and then the Detailed Results Screen. From the results screen, users can view step-by-step first-aid guidance and are directed to the Emergency Call Screen for third-degree burns, or return to add another burn area and retake photos when the confidence level of the result is low.

In the Home Screen Navigation flow, the Home Screen serves as the entry point to First-Aid Guidance, Assessment History, Feedback, Profile, Help Center, and Settings. From Assessment History, users can access the Detailed Cases Screen. The Profile branch leads to screens for managing assessment history, feedback history, the privacy policy, and logging out. Meanwhile, Settings provides options for managing notifications, changing passwords, and deleting accounts.

8.3. INTERFACE PROTOTYPE

All 28 functional screens and diagnostic dialogs of the BurnAid AI application were developed and captured directly from the live operational software environment. Below is the detailed presentation organized into specific functional groups:

8.3.1. Launch, Introduction, and User Authentication

This interface group establishes application identity and provides secure access for users. It includes the Splash screen (screen-splash), the Welcome screen introducing preliminary assessment and first-aid features (screen-welcome), the medical account Sign-In screen (screen-signin), the user Sign-Up screen (screen-signup), the secure Password Setup screen (screen-setpass), and the digital Medical Privacy Policy screen (screen-privacy).



Figure 8.3. Splash screen



Figure 8.4. Welcome screen

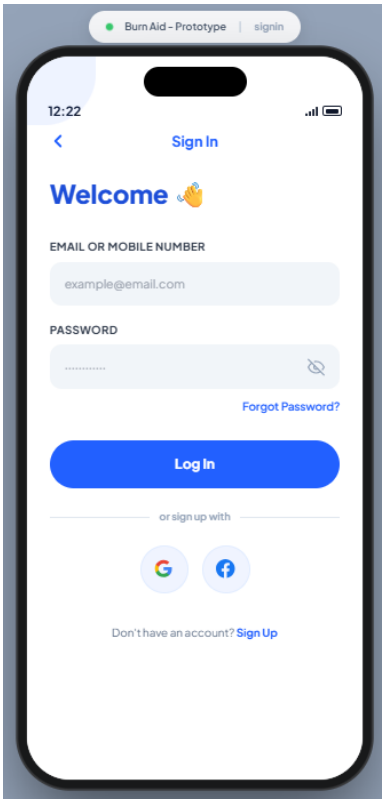


Figure 8.5. Sign-in screen

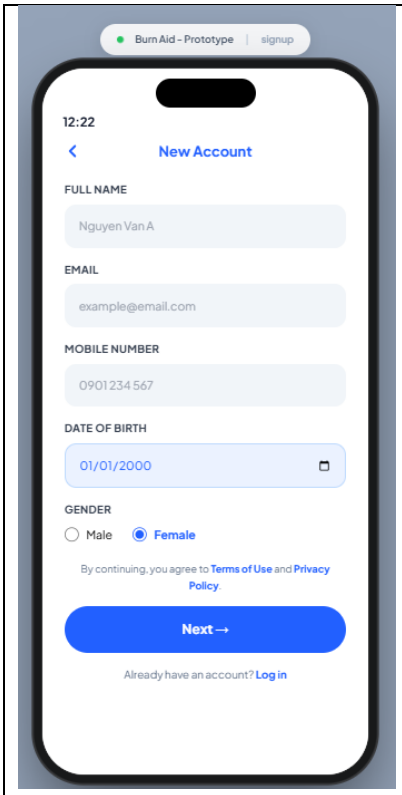


Figure 8.6. Sign-up screen

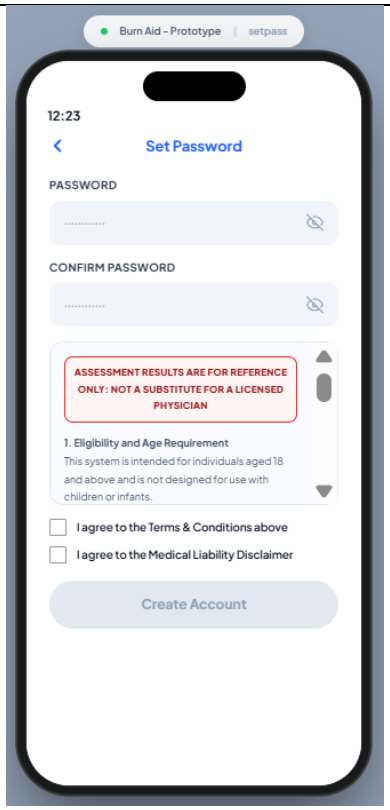


Figure 8.7. Set password screen

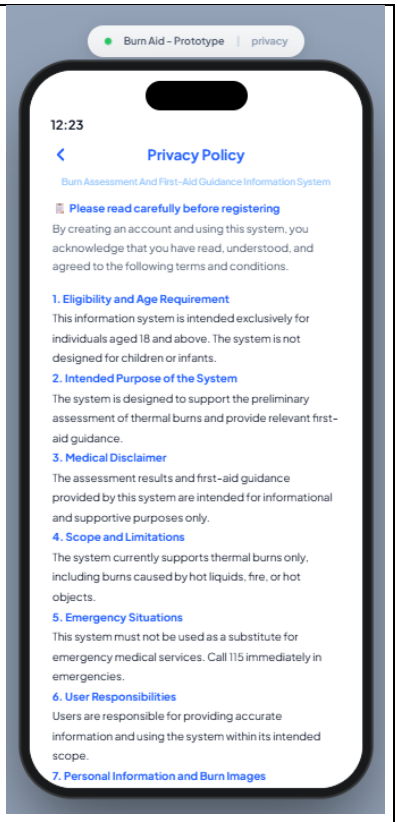


Figure 8.8. Privacy policy screen

8.3.2. Home Navigation and Burn Wound Data Collection

This functional group guides users through clinical data collection clearly and efficiently. Starting from the Home screen (screen-home), users proceed to the Add Burn Area screen (screen-add-burn) to view or manage affected regions. Next, the 2D interactive Human Body Map screen (screen-body-select) allows users to select front and back views based on Prof. Le The Trung's TBSA rule and adjust the affected percentage slider. The Photo Acquisition screen (screen-take-pic) supports capturing photos directly via live camera or uploading image files from device storage. The Quick Clinical Survey screen (screen-survey) screens for inhalation injury and systemic symptoms before transitioning to the AI Waiting Screen (screen-waiting) while deep learning models execute inference on the backend server.

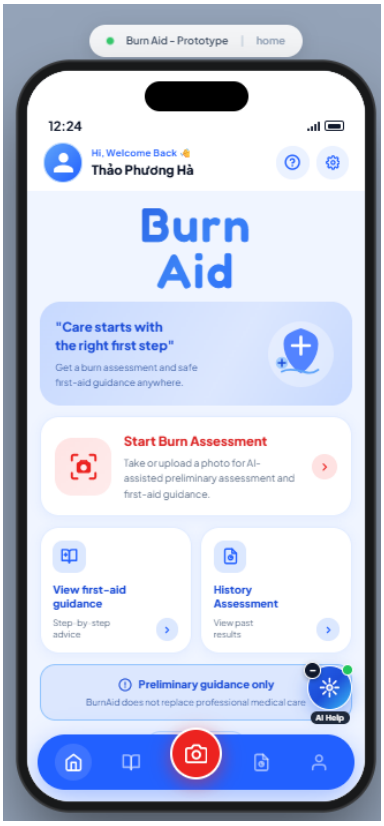


Figure 8.9. Home screen

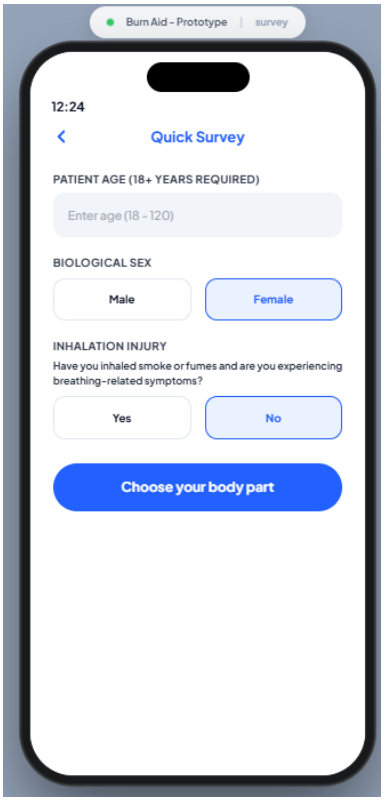


Figure 8.10. Clinical survey screen

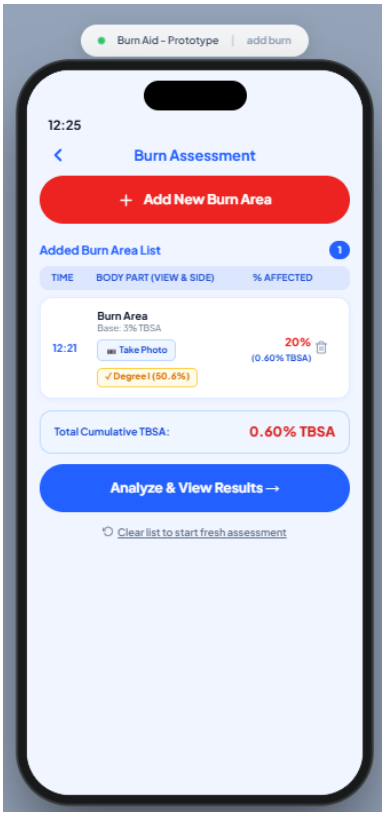


Figure 8.11. Add burn area screen

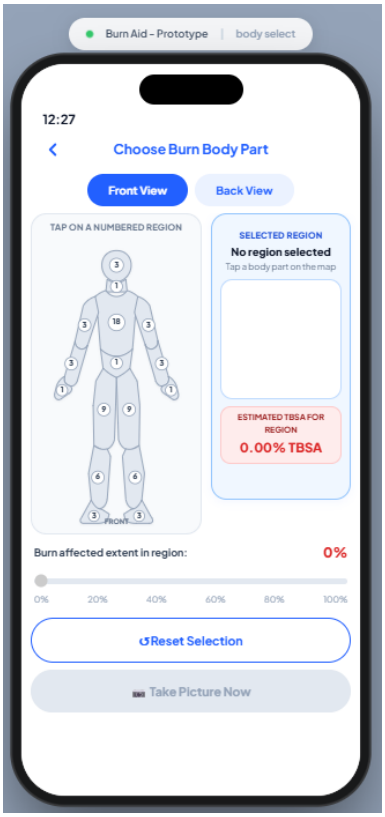


Figure 8.12. Body region selection screen

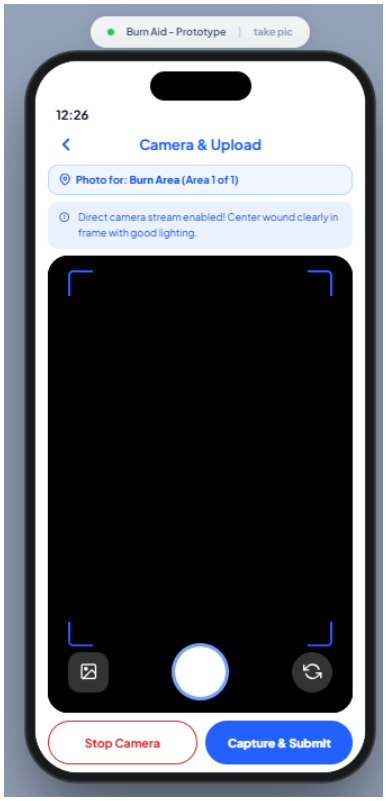


Figure 8.13. Photo capture screen

8.3.3. Diagnostic Results Triage, Safety Warnings, and First Aid

The assessment results module flexibly handles three computer vision scenarios: When an image contains healthy skin with no burn signs, the system displays the No Burn Detected screen (screen-no-burn). If the input image is blurry, poorly lit, or yields low confidence (< 70%), the Low Confidence Inconclusive Advisory screen (screen-results-low) informs the user that burn depth cannot be confirmed safely, provides universal first-aid precautions, and recommends retaking a clearer photo or consulting a doctor. When a thermal burn is identified with sufficient confidence, the Detailed Results screen (screen-results) presents the burn degree, confidence score, total cumulative TBSA, and regional first-aid guidance grounded in Decision 635/QD-BYT. If a full-thickness Degree III burn or airway compromise is detected, the Red Alert Emergency 115 Dialog (dlg-ambulance-115) is triggered immediately. Users can also access the AI Assistant Chat modal (modal-ai-chat) for interactive medical guidance or consult the First-Aid Handbook screen (screen-guidance).

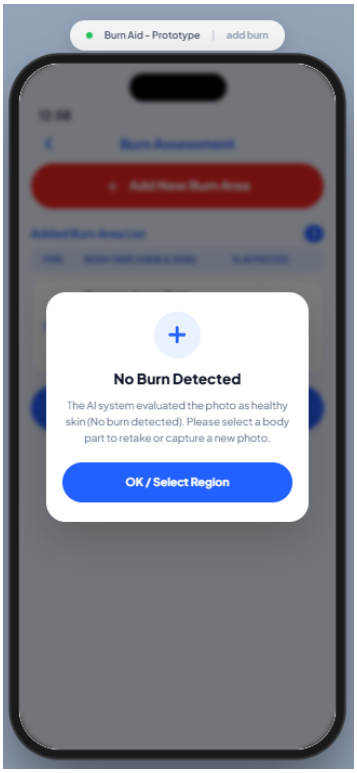


Figure 8.14 Healthy skin screen (No Burn)

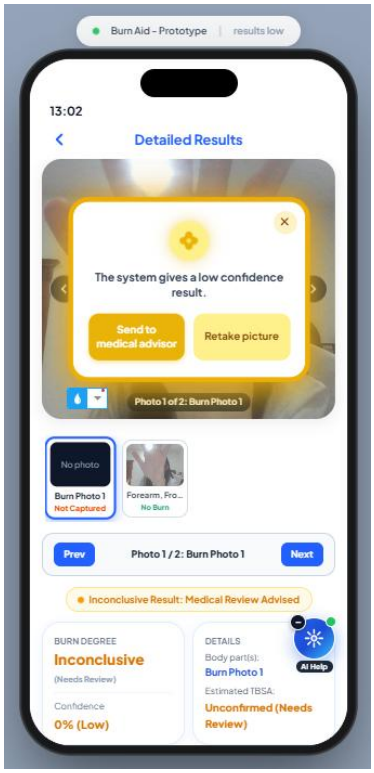


Figure 8.15. Low confidence advisory screen

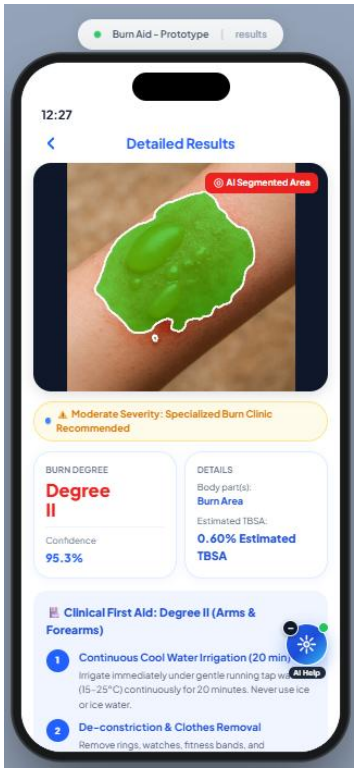
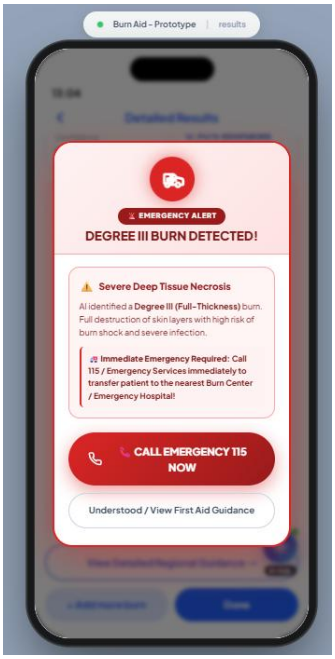
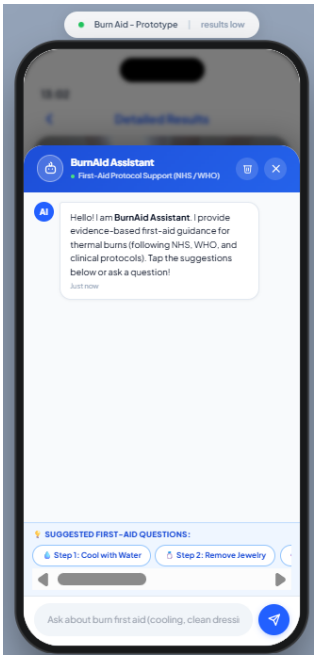


Figure 8.16. Detailed results screen





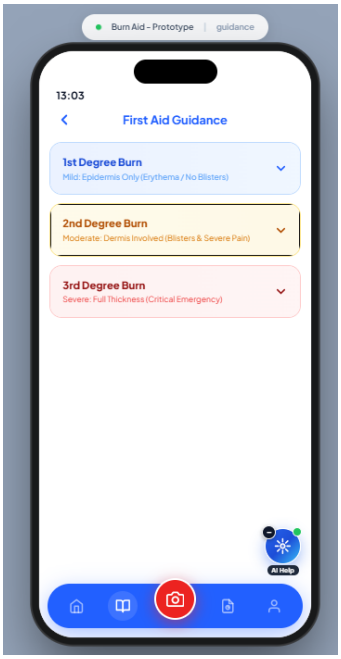


Figure 8.17. Emergency 115 alert popup	Figure 8.18. AI assistant chat modal	Figure 8.19. First-aid handbook screen

8.3.4. Profile Management, History, and System Settings

This supporting group manages user profile information and application configurations. It includes the User Profile screen (screen-profile), the Edit Profile screen (screen-edit-profile), the Assessment History screen (screen-history) displaying summary cards, the Case Details screen (screen-result-list) listing individual evaluated areas, the Feedback History screen (screen-feedback-history), the Submit Feedback screen (screen-feedback), the System Settings dashboard (screen-settings), Notification Management (screen-notifications), the Password Manager (screen-password-mgr), and the Medical Help Center (screen-help).

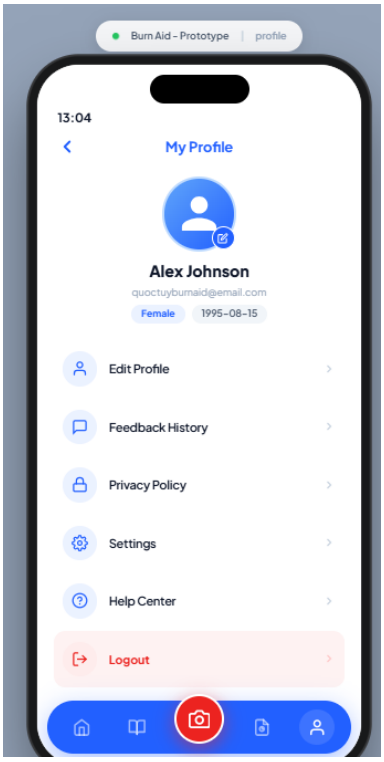


Figure 8.20. User profile screen

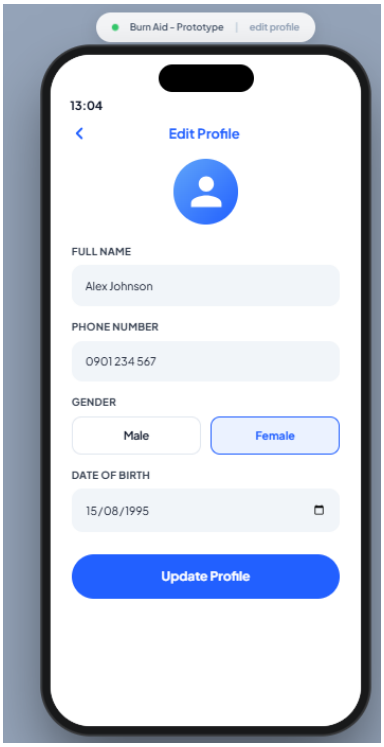


Figure 8.21. Edit profile screen

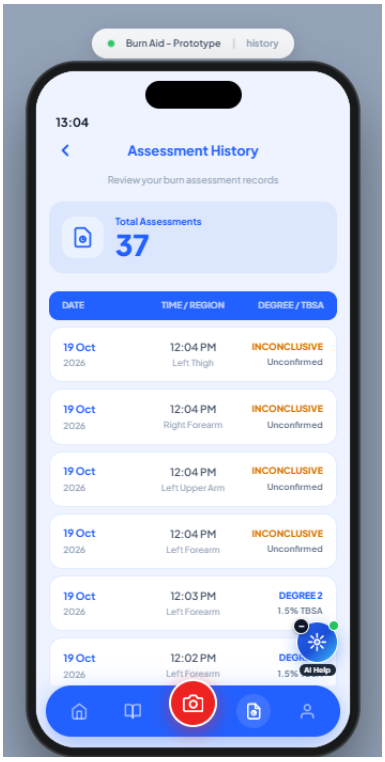


Figure 8.22. Assessment history screen

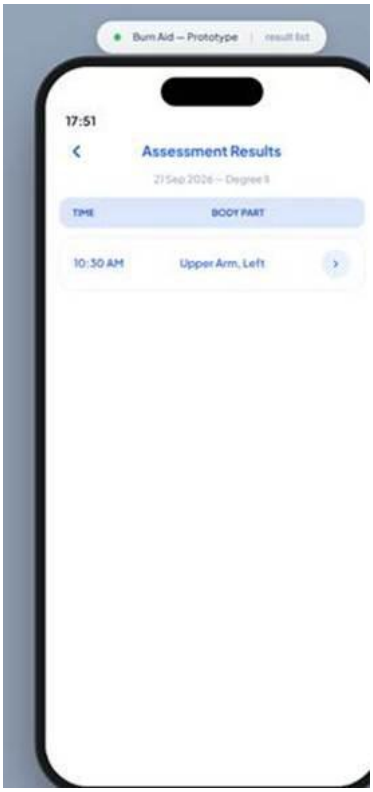


Figure 8.23. Case details list screen

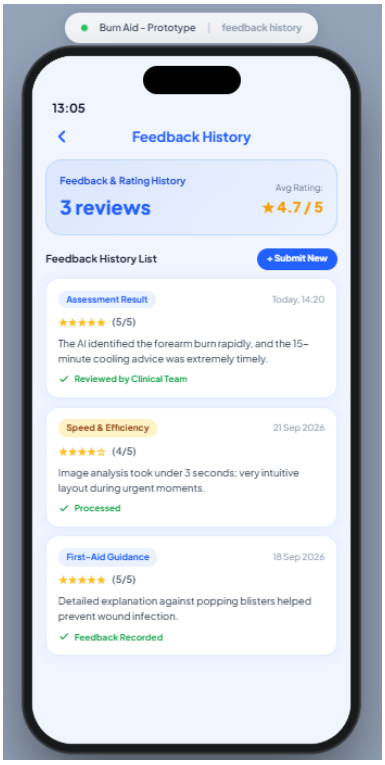


Figure 8.24. Feedback history screen

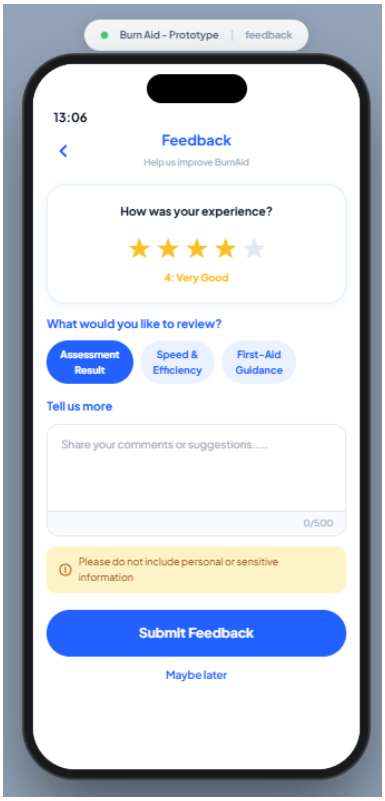
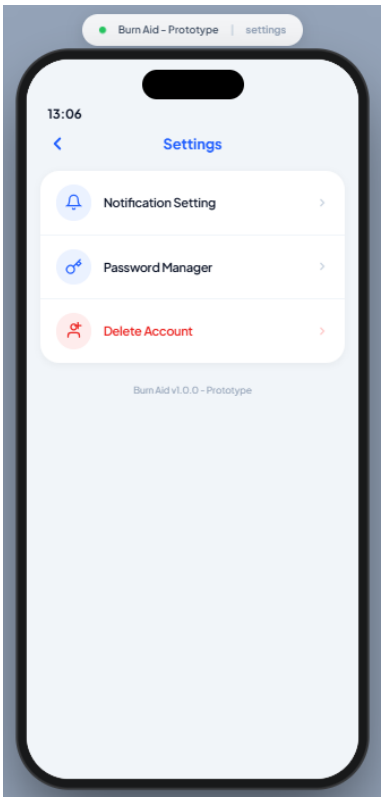
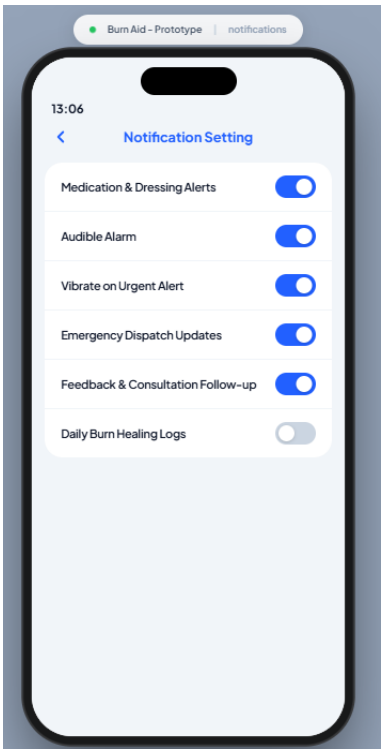


Figure 8.25. Submit feedback screen





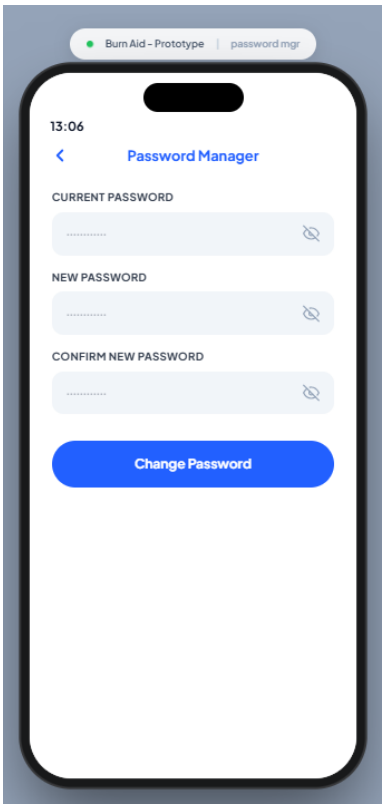


Figure 8.26. System settings screen	*Figure 8.27. Notification settings screen*	*Figure 8.28. Password manager screen*

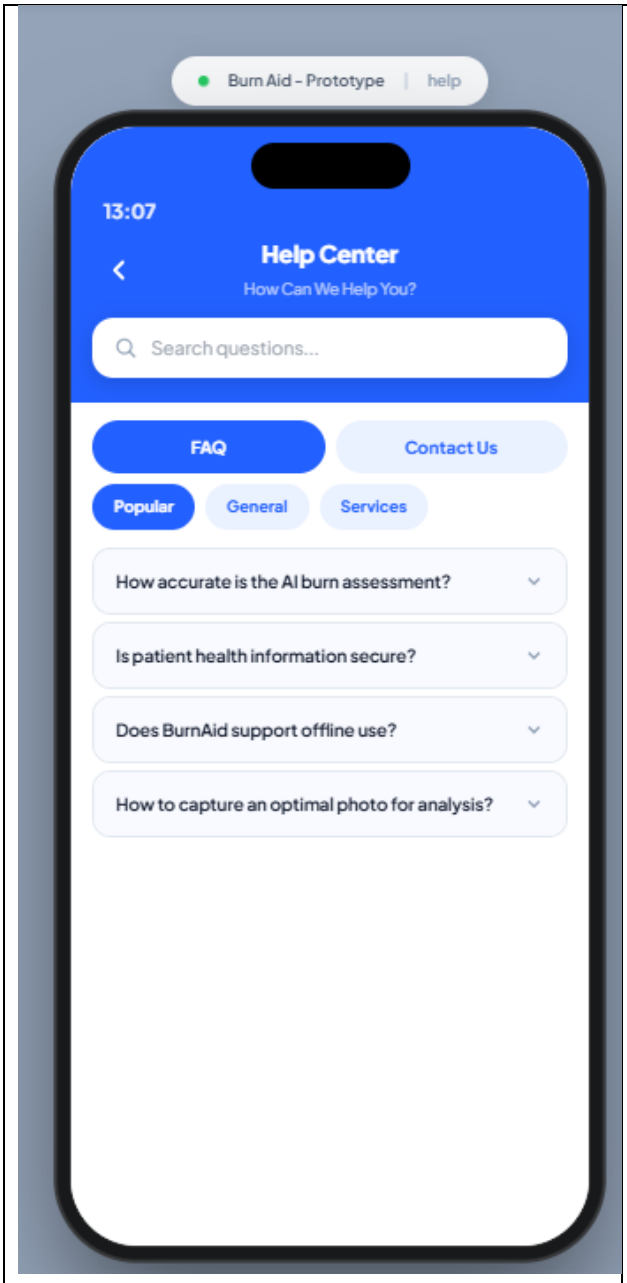


Figure 8.29. Medical help center screen

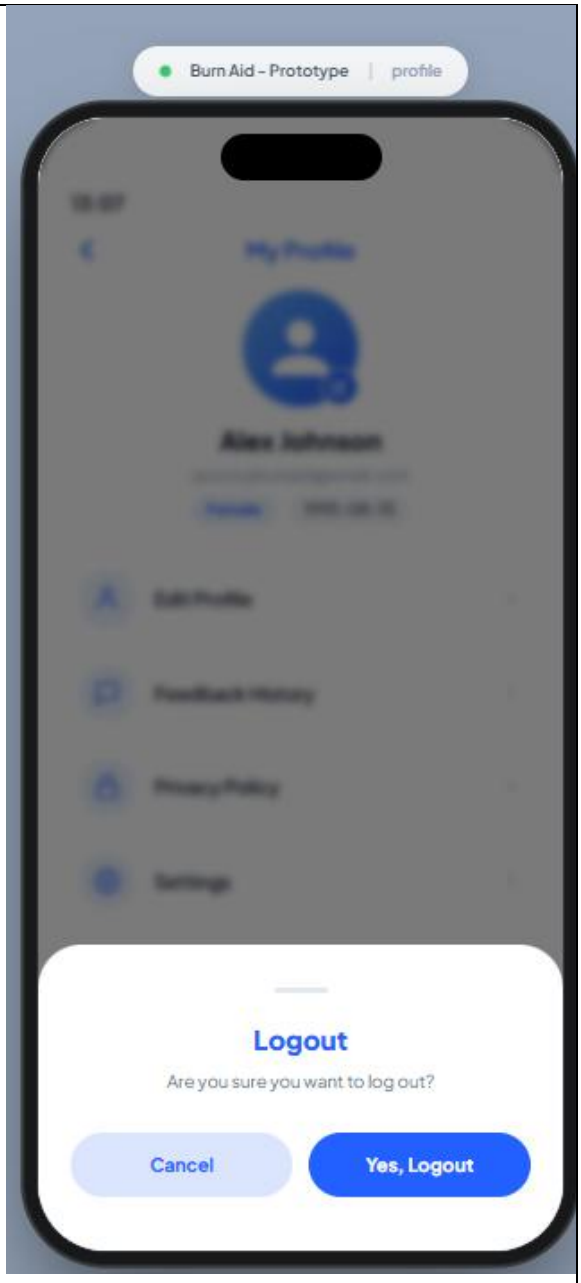


Figure 8.30. Log out screen

8.4. LOCAL SYSTEM EXPERIMENTAL DEPLOYMENT

8.4.1. Local Deployment Rationale

The BurnAid AI system is designed for local deployment on personal computers running Windows. This architecture processes patient medical images locally in system memory without sending private photos to external cloud servers, which preserves medical data privacy and ensures reliable operation even with limited internet connectivity.

To facilitate seamless local execution and offline testing during the prototype phase, the enterprise-grade physical database schema designed in SQL Server (Section 6.5) was mapped directly to a lightweight SQLite database (*burnaid.db*). This implementation preserves all core table structures, data types, primary/foreign key relationships, and validation constraints while eliminating the overhead of configuring a heavy relational database server.

The system startup workflow is packaged into a single operation using the batch script Chay_He_Thong.bat. When executed, the script verifies the Python 3.10+ runtime, installs any missing dependencies listed in requirements.txt, releases local network port 8000 if occupied, and launches the Starlette and Uvicorn ASGI server at http://127.0.0.1:8000. It then automatically opens the default web browser to display the Single-Page Application (SPA) without requiring technical setup from the user. Assessment records are stored locally in the SQLite database via database.py using production Schema v9.1.

8.4.2. Local Project Directory Tree

The directory tree structure of the local system after codebase optimization and refactoring:

```
BurnAid_AI_System/
  ├── Chay_He_Thong.bat
  ├── requirements.txt
  ├── database.py
  ├── burnaid.db
  ├── BurnAid_Database_v9.1_Full_Production.sql
  ├── api_server.py
  ├── clinical_rules.py
  ├── Burn_rag.py
  ├── Burn_aid_prototype.html
  ├── knowledge_base_burns.txt
  ├── models/
  │    ├── convnextv2_Burn_4class_best.pth
  │    └── best_random_curriculum_seed2026.pt
  └── assets/
```

Table 8.1. Core System Files and Technical Functional Roles

File	Architectural Component	Role and Functionality
Chay_He_Thong.bat	Automation Script	Validates Python 3.10+ runtime, installs dependencies, releases local port 8000, starts the ASGI server, and opens the default browser.
api_server.py	ASGI API Server	Routes,predict,,chat,,assess ment, and,history; validates image quality; coordinates deep learning inference.
clinical_rules.py	Clinical Rules Engine	Calculates % TBSA using Prof. Le The Trung's 1-3-6-9-18 rule, stratifies risk levels, and triggers 115 emergency protocols.
Burn_rag.py	RAG Clinical Assistant	Performs BM25 retrieval across 2,274 clinical knowledge chunks, constructs medical prompts, and interfaces with Groq Cloud LLM.
database.py	Database Manager	Manages SQLite database connections, executes Schema v9.1 migrations, and persists patient records and assessment histories.
burnaid.db	Local Database	Local SQLite database file storing patient profiles, assessment records, affected regions, TBSA calculations, and feedback under Schema v9.1.

BurnAid_Database_v9.1_Full_Production.sql	Database DDL Schema	Production SQL script containing complete table definitions, foreign keys, triggers, and indices for real-time assessment data.
Burn_aid_prototype.html	SPA User Interface	Unified responsive Single-Page Application interface including body map selection, camera feed, TBSA slider, results, and emergency modals.
knowledge_base_burns.txt	Medical Knowledge Base	Curated medical text repository containing 2,274 chunks extracted from Decision 635/QD-BYT and clinical burn guidelines.
convnextv2_Burn_4class_best.pth	Classification Weights	Optimal binary weights of the ConvNeXtV2 Nano model (14.99M parameters) for 4-class burn depth classification (epoch 14).
best_random_curriculum_seed2026.pt	Segmentation Weights	Optimal binary weights of the Curriculum U-Net++ ResNet-34 model for burn wound boundary segmentation (epoch 6).

CHAPTER 9. TESTING, EVALUATION & CONCLUSION

9.1. REQUIREMENTS ACHIEVED

The BurnAid AI system has completed functional testing and practical evaluation on a web prototype directly connected to a PyTorch backend server and an SQLite database. The tests confirmed that the system operates stably and meets the main technical goals set for the project.

9.1.1. Burn Severity Classification and Injury Segmentation Using Computer Vision

One of the project's main achievements is the successful development of a two-stage computer vision process that can identify different levels of injury. The ConvNeXtV2 model classifies images into four groups: first-degree burns, second-degree burns, third-degree burns, and healthy skin, achieving a Macro F1-score of 85.37%. For cases identified as burns, the U-Net++ model automatically segments the injury boundaries, with a Dice score of 71.29%. It then creates a visual overlay of the damaged tissue. When the image is identified as healthy skin, the system skips the segmentation step to prevent an incorrect mask from being created on normal skin. The model pipeline takes between 0.82 seconds on a GPU and 1.45 seconds on a CPU, meeting the need for a quick response during emergency first aid.

9.1.2. Digital Clinical Rules, Burn Area Calculation, and Risk Classification

The system digitizes Professor Lê Thế Trung's 1–3–6–9–18 rule for estimating burn area in adults aged 18 and above. Users can select body regions and adjust the estimated percentage of the affected area, allowing the system to calculate the total body surface area burned (TBSA). Based on TBSA, age, burn depth, and important areas such as the face, hands, feet, and perineal region, the system classifies risk into 3 levels. When these signs are detected, the system immediately provides guidance to call emergency services at 115. For children under 18, the system records their age but does not apply the adult formula; instead, it warns users that the child should be referred to a specialized pediatric facility.

9.1.3. Medical Question-and-Answer Assistant and Technical Safety Checks

The medical advice module uses retrieval-augmented generation (RAG), drawing on 2,274 text passages extracted from 18 specialist references and Decision No. 635/QĐ-BYT issued by Vietnam's Ministry of Health. These sources are combined with a large language model to provide guidance on first-aid steps based on medical knowledge. The main first-aid instruction is to cool the burn under clean, cool running water for 20 minutes, in line with recommendations from the Ministry of Health and international health authorities. The system also warns against unsafe folk remedies, such as applying ice, python fat, or toothpaste. Technically, the system has five safety checks: filtering dark or blurry images, controlling image-processing status, reporting server status when the model has not finished loading, saving uncertain results in the database, and using token-based authentication to help protect users' privacy.

9.2. SYSTEM LIMITATIONS

Although BurnAid AI operated stably in simulated test cases, practical evaluation showed that some limitations still need to be addressed:

9.2.1. Limitations in Clinical Scope and Types of Injury

The system was designed and trained specifically for thermal burns caused by fire, boiling water, steam, or contact with hot objects. Electrical and chemical burns are outside the scope of its image-based assessment and TBSA calculation. In addition, the current TBSA formula is standardized only for adults aged 18 and above. There is not yet an age-adjusted chart for children under 18. The system is designed only to support first aid before hospital care, so advanced fluid-resuscitation methods such as the Parkland and Evans formulas are not included. The computer vision model can currently classify burns from first to third degree, but it cannot identify deeper injuries affecting muscles and bones, classified as fourth- and fifth-degree burns.

9.2.2. Limitations in Features and System Architecture

In the current version, each assessment accepts only one image representing one body area. When a person has several burns in different parts of the body, the system cannot yet record them together and automatically add up the total affected area in a single case file. Users still select body regions from a map and adjust a slider to estimate the affected percentage. This means they must estimate the area themselves rather than mark it directly on a visual body model. In terms of infrastructure, the system runs locally, uses an SQLite database, and has basic token-based authentication. It does not yet have a separate interface for doctors to review cases with low confidence, and the 115 emergency-call button cannot share the injured person's location coordinates.

9.2.3. Limitations of the AI Model and Image Processing

The performance of the deep learning models is affected by the quality of images taken in real situations. Challenges include images taken from an angle, flash reflections on fluid-filled blisters, and inconsistent backgrounds. The training data mainly comes from international open datasets and does not fully represent the range of skin tones found among Vietnamese people. The segmentation model works well on concentrated areas of injury but may miss tiny, scattered burn spots because they can be mistaken for image noise. In addition, advanced conversations with the medical assistant still require an Internet connection to access the large language model. The service may be interrupted if the network connection fails at the scene.

9.3. IMPROVEMENT DIRECTIONS AND DEVELOPMENT PLAN

Based on the limitations found during testing, the main goals for improvement are to increase clinical accuracy, support multiple injuries, and make the technology more independent when used in the field. The development plan has two stages:

9.3.1. Short-Term Plan: Improving Features and the User Interface

In the short term, the system will focus on improving practical features for users. First, a multi-injury assessment module will be added, allowing users to take or upload several photos of different body areas during one assessment. The system will then calculate the total affected body surface area. The project will also digitize the Lund and Browder chart to calculate burn areas based on the age of children from birth to under 18. In addition, a separate professional interface will be developed so that clinical doctors can review and check cases where the analysis has low confidence. Finally, the first-aid decision tree will be expanded to include specific procedures for chemical accidents, including longer washing times, and instructions for performing cardiopulmonary resuscitation (CPR) for victims of electric shock.

9.3.2. Medium- and Long-Term Plan: Improving Technology and Expanding Healthcare Connections

In the medium and long term, the project aims to upgrade its technology and connect with community healthcare services. This would help provide emergency support when mobile signals are unavailable or during fire-related incidents. The project also plans to work with hospitals and specialist burn centers to build a local clinical dataset that better reflects Vietnamese skin tones and common household burns. The system is also planned to include a small language model that runs locally to protect personal health data in accordance with legal requirements. In addition, geographic information system (GIS) features will help users locate the nearest healthcare facility and send pre-hospital information to the responding emergency team in time.

9.4. CONCLUSION

Burns are sudden and complex injuries. They can seriously damage the skin and the tissues beneath it, and may also leave lasting physical and emotional scars. In the first moments after an accident, a lack of medical knowledge and feelings of panic can lead people to use unsafe folk remedies, such as applying ice, python fat, egg yolk, fish sauce, or other unfamiliar substances. These actions may make a minor injury worse, increase the risk of infection, and delay proper treatment. The first few minutes after an injury are especially important. Following evidence-based first-aid steps, particularly cooling the burn with clean, cool water for 20 minutes, can help remove heat, limit further tissue damage, ease pain, and protect living skin cells needed for later skin repair.

Motivated by an understanding of the difficulties faced by injured people and their families at such moments, the BurnAid AI project was developed to use advances in artificial intelligence to provide timely, appropriate, and compassionate support before the patient reaches a hospital. The system is not intended to replace diagnosis or treatment by specialist doctors. Instead, it serves as a useful support tool at the scene, helping prevent harmful first-aid mistakes and connecting patients with suitable healthcare facilities. BurnAid AI aims to help people with burns feel more reassured and make a small contribution to reducing the serious long-term effects of burn injuries.

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APPENDIX 1: SOURCE CODE REPOSITORY, DATASET

Repository URL: https://github.com/ha-thao/burnaid-ai.
Dataset URL:
[1] https://www.kaggle.com/code/faresabbasai2022/skin-burn-detection-acc-97/input
[2] https://www.kaggle.com/competitions/itobos-2024-detection/data

APPENDIX 2: TRAINING LOGS

URL:
https://docs.google.com/spreadsheets/d/1uPh1ULTeUV8c4frVGuJ50BP8p82K40Jc/edit?usp=drive_link&ouid=109821446003507263100&rtpof=true&sd=true

APPENDIX 3: MEDICAL KNOWLEDGE

Table A.1. Total body surface area by the method of Professor Le The Trung (1999)

Body Region	Reference Total %TBSA	Front	Back
Perineum/Genitalia	1%	1%	—
Neck/Nape	1%	0.5%	0.5%
Palm or dorsum of hand (one side)	1%	0.5%	0.5%
Forearm (one side)	3%	1.5%	1.5%
Upper arm (one side)	3%	1.5%	1.5%
Foot (one side)	3%	1.5%	1.5%
Face	3%	3%	—
Scalp (hairy area)	3%	1.5%	1.5%
Lower leg (one side)	6%	3%	3%
Buttocks (both)	6%	—	6%
Thigh (one side)	9%	4.5%	4.5%
Upper limb (one side)	9%	4.5%	4.5%
Lower limb (one side)	18%	9%	9%
Back (including both buttocks)	18%	—	18%
Chest & Abdomen	18%	18%	—

Table A.2. ABSI calculation

Parameters	Finding	Score
Sex	Male	0
	Female	1
Age	0-20	1
	21-40	2
	41-60	3
	61-80	4
	81-100	5
Inhalation injury	Yes	1
	No	0
Presence of full-thickness burn (degree III)	Yes	1
	No	0
TBSA (%)	1-10	1
	11-20	2
	21-30	3
	31-40	4
	41-50	5
	51-60	6
	61-70	7
	71-80	8
	81-90	9
	91-100	10

Table A.3. The ABSI explanation

ABSI	Treat to life	Probability of Survival (%)
2-3	Very low	≥ 99
4-5	Moderate	98
6-7	Moderately severe	80-90
8-9	Serious	50-70
10-11	Severe	20-40
≥ 12	Maximum	≤ 10

APPENDIX 4: TABLES

Table 2.1: Key Studies in Deep Learning Burn Classification 10
Table 2.2. Summary of the existing studies reviewed 11
Table 3.1. System Stakeholders 13
Table 3.2. User roles and requirements 14
Table 3.3. Functional requirements 15
Table 3.4. Non-functional requirements 19
Table 5.1. List of Use Cases 27
Table 5.2. UC-01 Manage Account and Profile 28
Table 5.3. UC-02 View Assessment History 30
Table 5.4. UC-03 Perform Burn Assessment 31
Table 5.5. UC-04 View Assessment Result 33
Table 5.6. UC-05 View First-Aid Guidance 35
Table 5.7. UC-06 Receive Emergency Warning 36
Table 5.8. UC-07 Open Device Dialer 37
Table 5.9. UC-08 Submit Feedback 38
Table 5.10. UC-09 Review User Feedback 39
Table 5.11. UC-10 Manage First-Aid Guidelines 40
Table 5.12. UC-11 Review Low-Confidence Cases 42
Table 5.13. UC-12 Manage AI Models 43
Table 5.14. UC-13 Manage User Account Status 44
Table 5.15. UC-14 Ask First-Aid Assistant 45
Table 6.1. Entity Account 62
Table 6.2. Entity Burn_case 62
Table 6.3. Entity Burn_area_assessment 64
Table 6.4. Entity Body_region_reference 67
Table 6.5. Entity First_aid_guideline 67
Table 6.6. Entity Assessment_guideline 68
Table 6.7. Entity AI_model 69
Table 6.8. Entity Feedback_review_log 70
Table 6.9. Entity Case_referral 71
Table 6.10. Entity Relationship 72
Table 6.11. Logical entities of the prototype 77
Table 7.1. Class distribution of the classification dataset after preprocessing 80
Table 7.2. Training parameter settings for the ConvNeXtV2 Nano model 81

Table 7.3. Detailed class-by-class evaluation results on the independent test set (n = 316) ...83
Table 7.4. Segmentation dataset composition and the 98 holdout None images.....84
Table 7.5. Training settings for the Curriculum U-Net++ ResNet-34 model.....85
Table 7.6. Performance comparison between Manual-only U-Net++ and Curriculum U-Net++ 86
Table 7.7. Best Random Curriculum checkpoint results based on SelectionScore across three seeds.....87
Table 7.8. Model Selection experimental setup; configurations taken from the Full notebook 89
Table 7.9. Classification Model Selection results; each model trained across three seeds 90
Table 7.10. New Segmentation Model Selection results on validation: Average Dice, Selection Score, and NegativeFPR across three seeds 91
Table 7.11. Direct comparison between two checkpoints on the segmentation test set [12] ..92
Table 8.1. Core System Files and Technical Functional Roles 108
Table A.1. Total body surface area by the method of Professor Le The Trung (1999).....115
Table A.2. ABSI calculation.....115
Table A.3. The ABSI explanation.....116

APPENDIX 5: FIGURES

Figure 4.1. Context Diagram 21
Figure 4.2. DFD Level 1 22
Figure 4.3. DFD Level 2 of Process 2.0 (Burn Assessment Engine) and Process 3.0 (Provide First-Aid Guidance and Emergency Response).....24
Figure 4.4. DFD Level 2 of Process 5.0 (Manage medical content, review cases, and feedback) 25
Figure 5.1. Detecting and assessing burn severity and providing first-aid guidance for Individuals System Use Case diagram.....27
Figure 5.2. Sign In/Create Account Activity Diagram 48
Figure 5.3. View Assessment History Activity Diagram 49
Figure 5.4. Receive Emergency Warning Activity Diagram.....50
Figure 5.5. View Assessment Result Activity Diagram 51
Figure 5.6. Submit Feedback Activity Diagram 52
Figure 5.7. Perform Burn Assessment Activity Diagram 53
Figure 5.8. Review User Feedback Activity Diagram.....54
Figure 5.9. Manage First-Aid Guidelines Activity Diagram 55

Figure 5.10. Manage AI Models Activity Diagram 56
Figure 5.11. Review Low-confidence Cases Activity Diagram 57
Figure 5.12. Manage User Account Status Activity Diagram 58
Figure 5.13. Manage Personal Profile Activity Diagram 59
Figure 5.14. Ask First-Aid Assistant Activity Diagram 60
Figure 6.1. Entity Relationship Diagram (each box shows the keys and main attributes; the complete list is in Table 1 to Table 9) 74
Figure 7.1. Training Loss and Validation Loss curves of the ConvNeXtV2 Nano model over 15 epochs 83
Figure 7.2. Validation Accuracy and Macro F1-score across epochs 83
Figure 7.3. Comparison chart of Mean Dice and Mean IoU on 13 independent test images 87
Figure 7.4. Training Loss curves by epoch for Random Curriculum across three seeds (42, 123, 2026) 88
Figure 7.5. Validation Dice variation by epoch across three seeds, with dots marking selected checkpoints 88
Figure 7.6. SelectionScore curves incorporating false positive penalties across epochs 88
Figure 7.7. Average Macro-F1 and standard deviation across three seeds; the horizontal axis is magnified to show small differences 90
Figure 7.8. Average validation Selection Score and standard deviation across three seeds; horizontal axis is magnified to distinguish architectures 91
Figure 7.9. Test results by image-level averages and pooled pixel metrics; metrics summarize different aspects of the same predictions 92
Figure 8.1. Three-tier Architecture Diagram (Patient-Facing Prototype) 96
Figure 8.2. Navigation Flowchart 97
Figure 8.3. Splash screen 99
Figure 8.4. Welcome screen 99
Figure 8.5. Sign-in screen 99
Figure 8.6. Sign-up screen 100
Figure 8.7. Set password screen 100
Figure 8.8. Privacy policy screen 100
Figure 8.9. Home screen 101
Figure 8.10. Clinical survey screen 101
Figure 8.11. Add burn area screen 101
Figure 8.12. Body region selection screen 102
Figure 8.13. Photo capture screen 102
Figure 8.14 Healthy skin screen (No Burn) 103
Figure 8.15. Low confidence advisory screen 103

Figure 8.16. Detailed results screen 103
Figure 8.17. Emergency 115 alert popup 104
Figure 8.18. AI assistant chat modal 104
Figure 8.19. First-aid handbook screen 104
Figure 8.20. User profile screen 104
Figure 8.21. Edit profile screen 104
Figure 8.22. Assessment history screen 104
Figure 8.23. Case details list screen 105
Figure 8.24. Feedback history screen 105
Figure 8.25. Submit feedback screen 105
Figure 8.26. System settings screen 106
Figure 8.27. Notification settings screen 106
Figure 8.28. Password manager screen 106
Figure 8.29. Medical help center screen 106
Figure 8.30. Log out screen 106

THE END